



Analysis, Design and Simulation of analog/RF Digital Circuits with Open-Source EDA Tools and Process Design Kit



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"Scan me"



Part 1

Advanced Compact MOSFET Model: ACM2



"Scan me"



Carlos Galup-Montoro

https://github.com/ACMmodel/MOSFET_model

Outline

- **Introduction: ACM timeline**
- **ACM2 model**
- **Long-channel: I_D and g_m/I_D models**
- **Long-channel: MOS basic amplifier design**

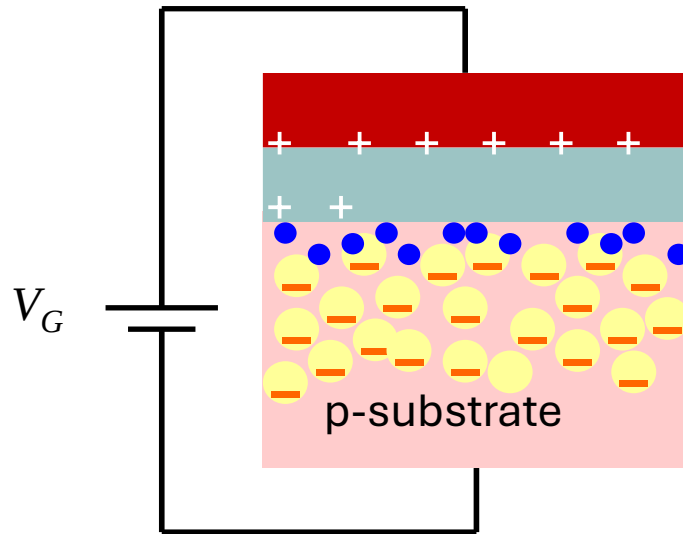
ACM timeline

- 1993 ϕs -based model (SBMICRO, Campinas, Br)
- 1995 Long-channel charge-based model (SSE, Nov.)
- 1996 gm/ID model (SBMICRO, Aguas de Lindoia, Br)
- 1997 Unified current control model (ISCAS, Hong Kong)
- 1998 Most referenced ACM paper (JSSC, Oct.)
- 2000 ACM model in SMASH simulator (CICC, Orlando)
- 2021 4-parameter single-piece model (NorCAS, Oslo)
- 2023 ACM2 in VERILOG-AMS (NEWCAS, Edinburgh)

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The capacitive model of the MOSFET



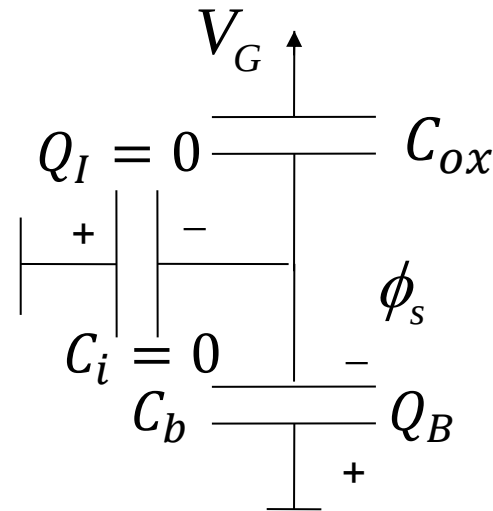
ϕ_s surface potential

C_{ox} oxide capacitance per unit area

C_b depletion capacitance per unit area

Q_I carrier charge density

V_{T0} threshold voltage



$$\frac{\Delta\phi_s}{\Delta V_G} = \frac{C_{ox}}{C_{ox} + C_b} = \frac{1}{n}$$

$$\phi_s = 2\phi_F + \frac{V_G - V_{T0}}{n} = 2\phi_F + V_P$$

Drain current model: main simplifications

First approximation ACM2

$$dQ_I = nC_{ox}d\phi_s$$

Second approximation ACM2

$$I_D = \mu W \left(-Q_I \frac{d\phi_s}{dy} + \phi_t \frac{dQ_I}{dy} \right)$$

drift

Diffusion

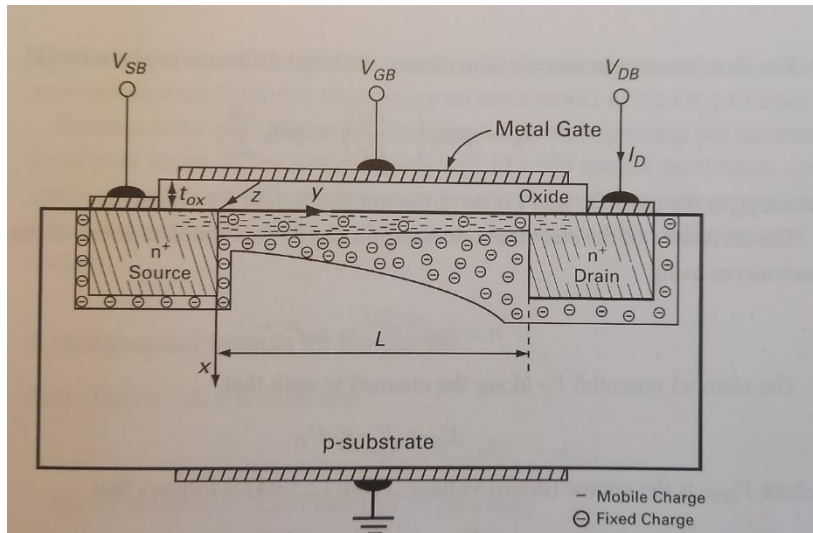
Q_I carrier charge density

W transistor width

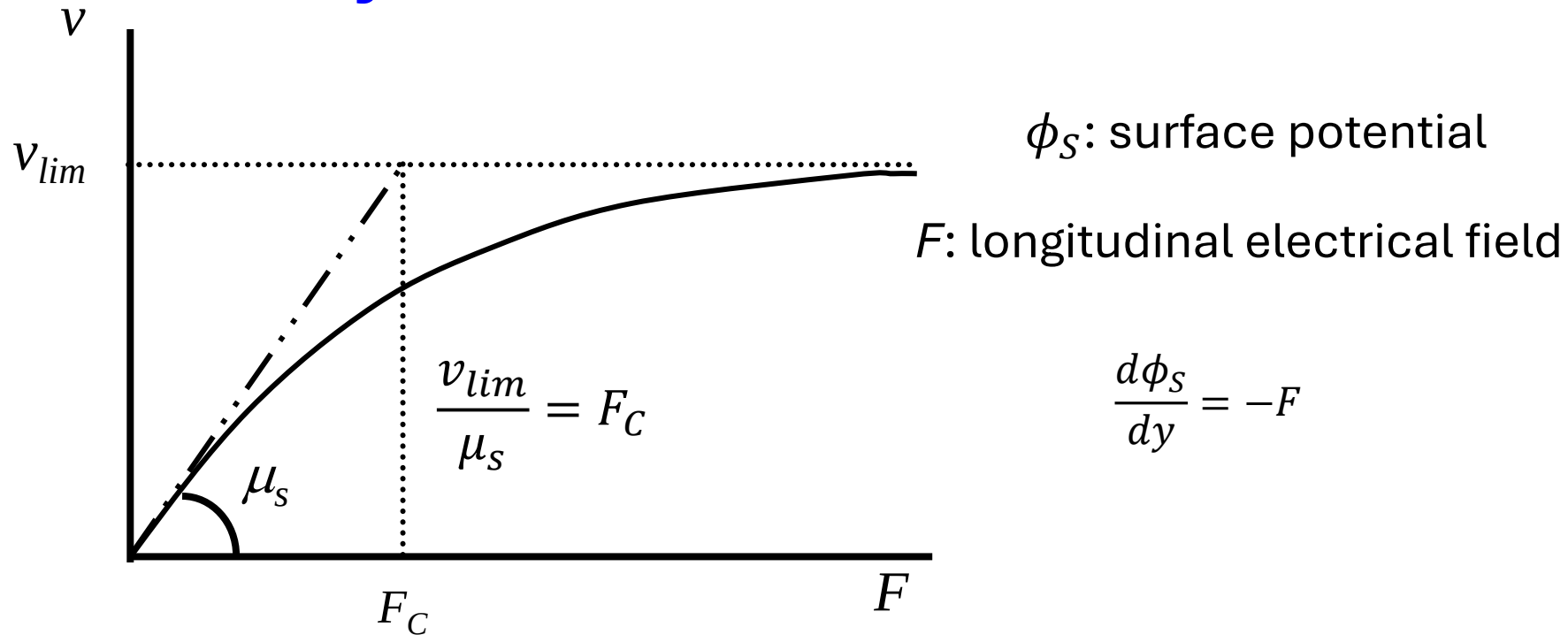
μ carrier mobility

ϕ_t thermal voltage

26 mV @ 300K



Velocity saturation



Third approximation ACM2

$$\mu = \frac{\mu_s}{1 + \frac{\mu_s}{v_{lim}} \frac{d\phi_s}{dy}}$$

Allows analytical integration for I_D

ACM2 current law

From the 3 approximations:
normalized current vs. normalized
charge densities at source and drain



$$i_D = \frac{(q_S + q_D + 2)}{1 + \zeta(q_S - q_D)} (q_S - q_D)$$

$$i_D = I_D/I_S \quad I_S = \frac{W}{L} \mu_s n C_{ox} \frac{\phi_t^2}{2}$$

normalization (specific) current

$$q_{S(D)} = Q_{S(D)} / (-nC_{ox}\phi_t)$$

$-nC_{ox}\phi_t$ thermal charge

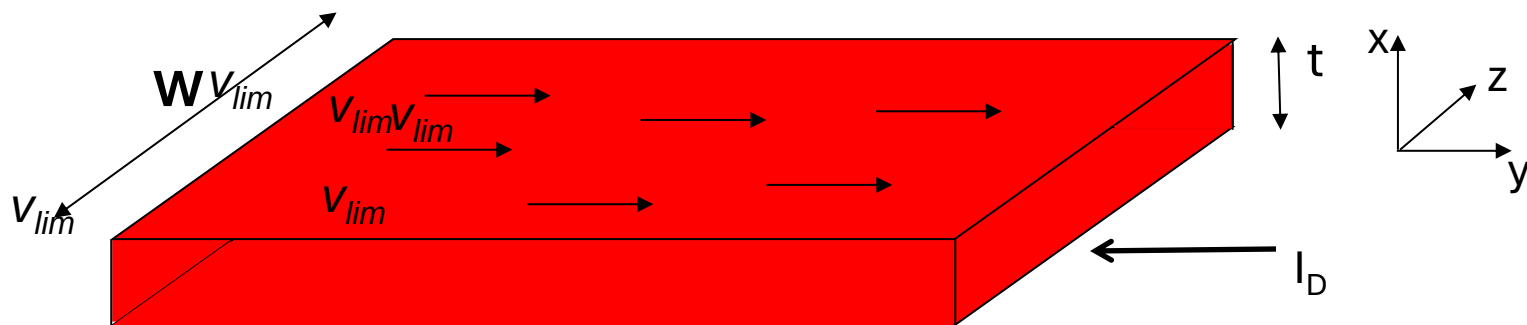
$$SI : q_{S(D)} \gg 1 \quad WI : q_{S(D)} \ll 1$$

Short-channel parameter ζ :

$$\zeta = \frac{(\mu_s \phi_t / L)}{v_{lim}}$$

ratio of diffusion-related velocity to saturation velocity

Physics-based saturation



Saturation current due to saturation velocity of the carriers

$$I_{Dsat} = -W Q_{Dsat} v_{lim}$$

Q_{Dsat} is the saturation inversion charge per unit area

or, using normalized variables

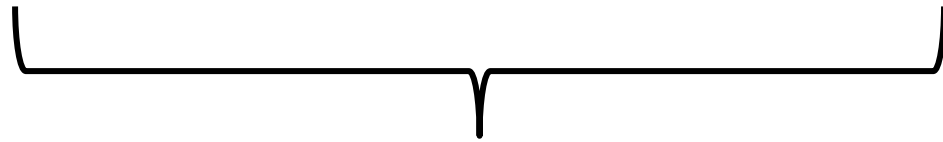
$$i_{Dsat} = \frac{2}{\zeta} q_{dsat}$$

“Carrier velocity approaches v_{sat} , but never reaches v_{sat} ”

Y.Taur TED March 2019

Physics-based saturation: design model

$$i_{Dsat} = \frac{2}{\zeta} q_{dsat} \qquad i_{Dsat} = \frac{(q_s + q_{Dsat} + 2)}{1 + \zeta(q_s - q_{Dsat})} (q_s - q_{Dsat})$$



$$q_{Dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

or equivalently

$$q_s = \sqrt{1 + \frac{2}{\zeta} q_{dsat}} - 1 + q_{dsat}$$

The Unified Charge Control Model (UCCM)

$$\frac{V_P - V_C}{\phi_t} = q_I - 1 + \ln q_I$$

$$V_S \leq V_C \leq V_D$$

$$q_S \leq q_I \leq q_D$$



$$\frac{V_{DS}}{\phi_t} = q_S - q_D + \ln \frac{q_S}{q_D}$$

The “Regional” Weak (WI) and Strong Inversion (SI) Approximations

WI

SI

$$q_I \ll 1 \rightarrow V_P - V_C \ll -\phi_t$$

$$q_I \gg 1 \rightarrow V_P - V_C \gg \phi_t$$

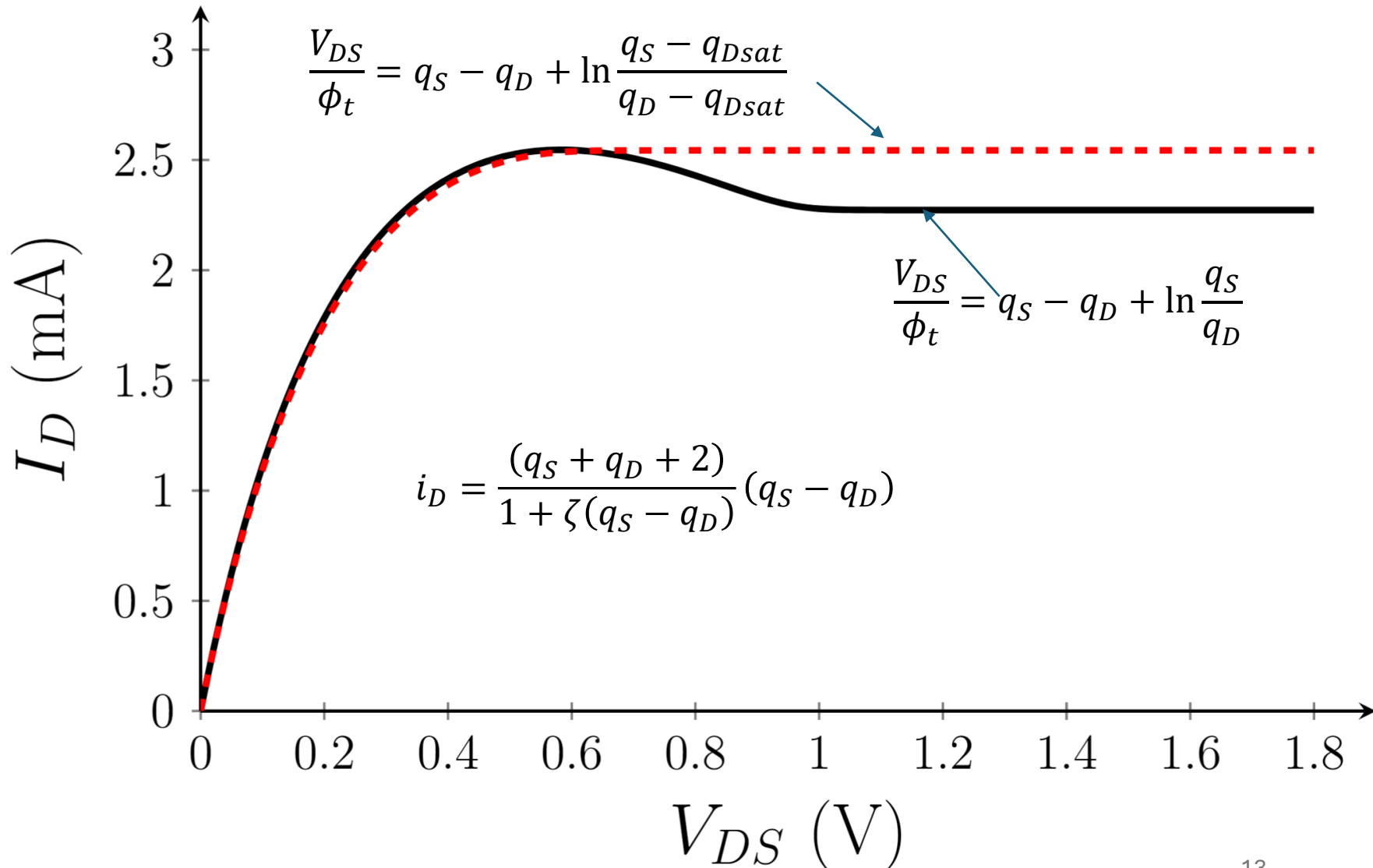
$$q_I \cong e^{\frac{V_P - V_C + \phi_t}{\phi_t}}$$

$$q_I \cong \frac{V_P - V_C + \phi_t}{\phi_t}$$

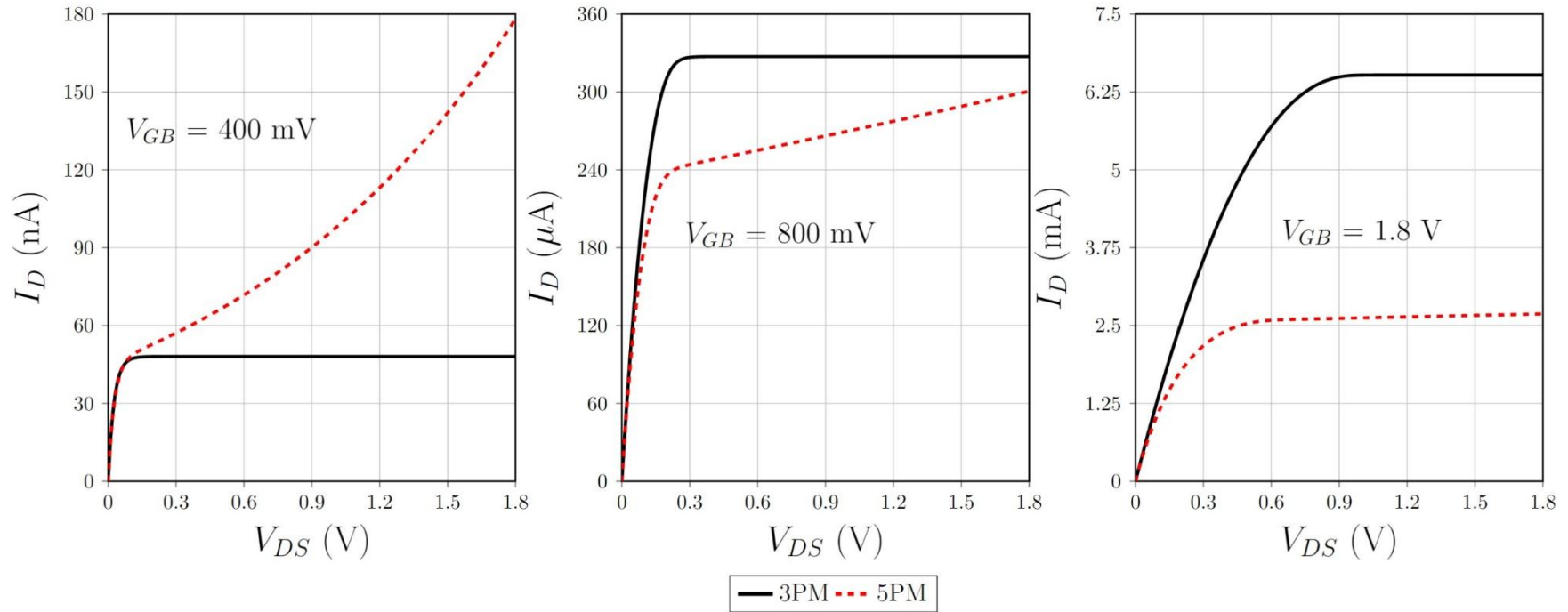
Error <10% for $q_I < 0.22$

Error <10% for $q_I > 20$

UCCM including the effect of velocity saturation



Output curves including DIBL and v_{sat}



DIBL model: $V_T = V_{T0} - \sigma(V_S + V_D)$

Transistor	W/L ($\mu m/\mu m$)	V_{T0} (mV)	I_S (μA)	n	σ	ζ
NMOS2V	5/0.18	528	5.52	1.37	0.025	0.056

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- Long-channel: MOS basic amplifier design

Long-channel charge-based model

For short channel MOS

$$I_D = I_S \frac{(q_S + q_D + 2)}{1 + \zeta(q_S - q_D)} (q_S - q_D)$$

$$\frac{V_P - V_S}{\phi_t} = q_S - 1 + \ln q_S \quad \frac{V_{DS}}{\phi_t} = q_S - q_D + \ln \frac{q_S - q_{Dsat}}{q_D - q_{Dsat}}$$

For long channel MOS $\zeta=0 \rightarrow q_{Dsat}=0$

$$I_D = I_S(q_S + q_D + 2)(q_S - q_D) = I_S[(q_S^2 + 2q_S) - (q_D^2 + 2q_D)]$$

$$\frac{V_P - V_{S(D)}}{\phi_t} = q_{S(D)} - 1 + \ln q_{S(D)}$$

Obs: $\frac{dq_{S(D)}}{dV_G} = -\frac{1}{n} \frac{dq_{S(D)}}{dV_{S(D)}} \quad \frac{dq_D}{dV_D} = -\frac{1}{\phi_t} \frac{q_D}{q_D + 1} \quad \frac{dI_D}{dq_D} = -2I_S(q_D + 1)$

Small-signal transconductances

$$g_{md} = \frac{dI_D}{dV_D} = \frac{dI_D}{dq_D} \frac{dq_D}{dV_D} = 2I_S(q_D + 1) \frac{1}{\phi_t} \frac{q_D}{q_D + 1} = \frac{2I_S}{\phi_t} q_D$$

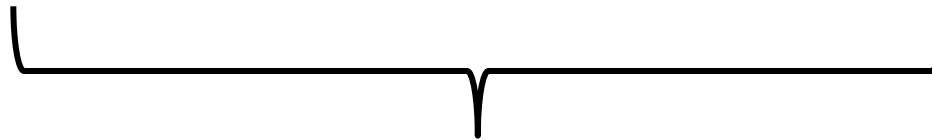
Symmetry



$$g_{ms} = \frac{2I_S}{\phi_t} q_S$$

**Amazingly simple
(see Appendix)**

$$g_m = \frac{dI_D}{dV_G} = \frac{dI_D}{dq_S} \frac{dq_S}{dV_G} + \frac{dI_D}{dq_D} \frac{dq_D}{dV_G} \qquad \frac{dq_{S(D)}}{dV_G} = -\frac{1}{n} \frac{dq_{S(D)}}{dV_{S(D)}}$$



$$g_m = \frac{g_{ms} - g_{md}}{n}$$

$$g_m = \frac{g_{ms}}{n} \longrightarrow \text{in saturation}$$

Unified Current Control Model (UICM)-I

$$I_D = I_S[(q_S^2 + 2q_S) - (q_D^2 + 2q_D)] \quad (A)$$

(A) can also be written as

$$I_D = I_F - I_R = I_S[i_f - i_r] \quad (B)$$

I_F, I_R : forward and reverse currents

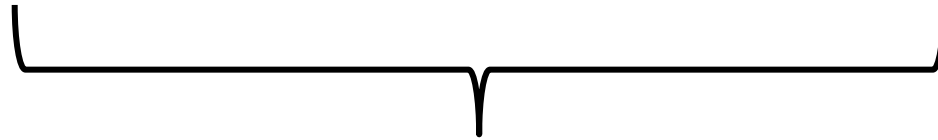
$i_{f(r)} = q_{S(D)}^2 + 2q_{S(D)}$: forward (reverse) inversion coefficients

$$\downarrow$$
$$q_{S(D)} = \sqrt{1 + i_{f(r)}} - 1$$

Unified Current Control Model (UICM)-II

$$\frac{V_P - V_{S(D)}}{\phi_t} = q_{S(D)} - 1 + \ln q_{S(D)}$$

$$q_{S(D)} = \sqrt{1 + i_{f(r)}} - 1$$



Normalized UICM

$$\frac{V_P - V_{S(D)}}{\phi_t} = \sqrt{1 + i_{f(r)}} - 2 + \ln \left(\sqrt{1 + i_{f(r)}} - 1 \right)$$



$$\frac{V_{DS}}{\phi_t} = q_S - q_D + \ln \frac{q_S}{q_D} = \sqrt{1 + i_f} - \sqrt{1 + i_r} + \ln \left(\frac{\sqrt{1 + i_f} - 1}{\sqrt{1 + i_r} - 1} \right)$$

$$g_m/I_D$$

$$g_{ms(d)} = \frac{2I_S}{\phi_t} \left(\sqrt{1 + i_{f(r)}} - 1 \right) = \frac{W}{L} \mu C_{ox} n \phi_t \left(\sqrt{1 + i_{f(r)}} - 1 \right)$$

$$g_m = \frac{g_{ms} - g_{md}}{n}$$

$$\frac{g_m}{I_D} = \frac{2}{n \phi_t (\sqrt{1 + i_f} + \sqrt{1 + i_r})}$$

For $V_{DS}/\phi_t \ll 1$ we have $i_f \cong i_r$

In saturation $i_f \gg i_r$

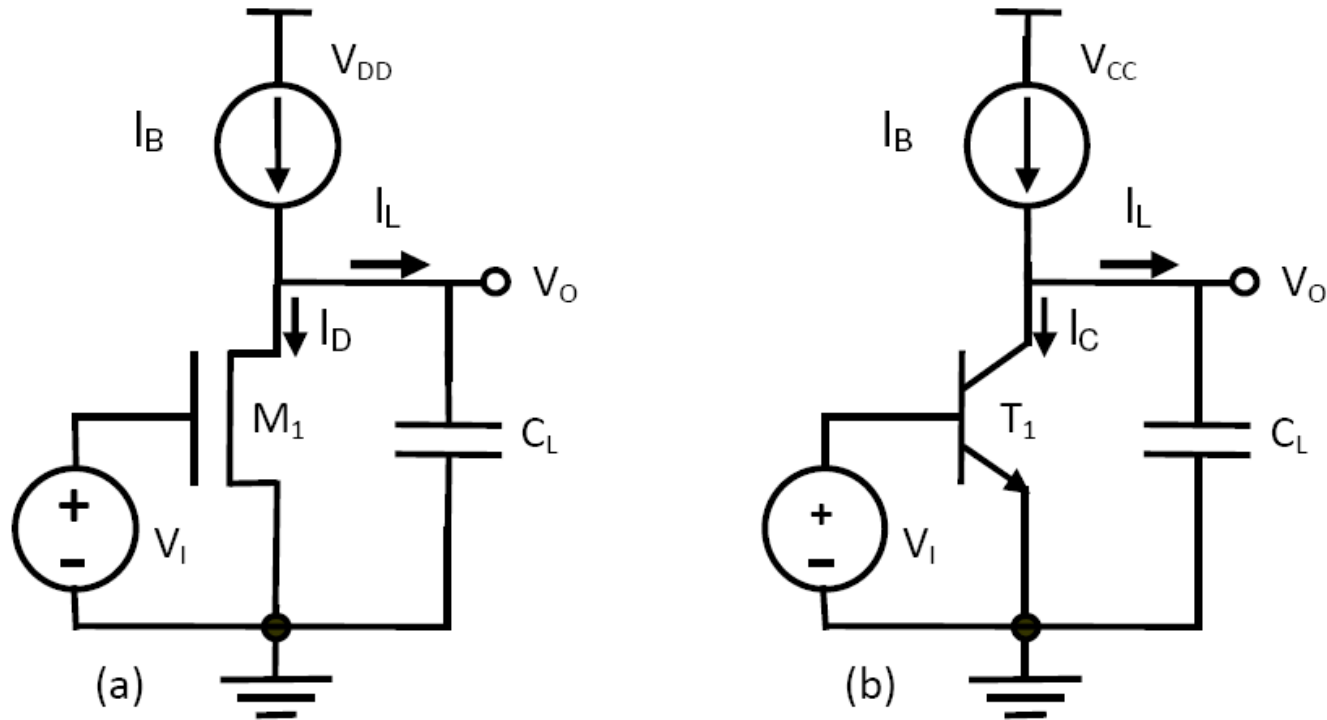
$$\frac{g_m}{I_D} \cong \frac{1}{n \phi_t \sqrt{1 + i_f}}$$

$$\frac{g_m}{I_D} \cong \frac{2}{n \phi_t (\sqrt{1 + i_f} + 1)}$$

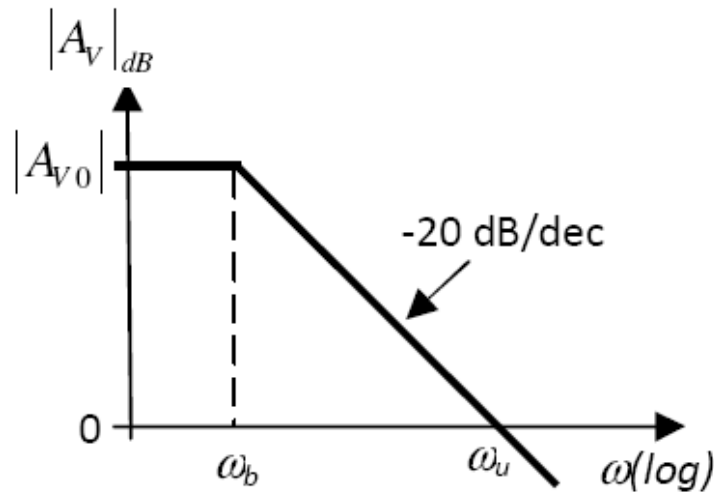
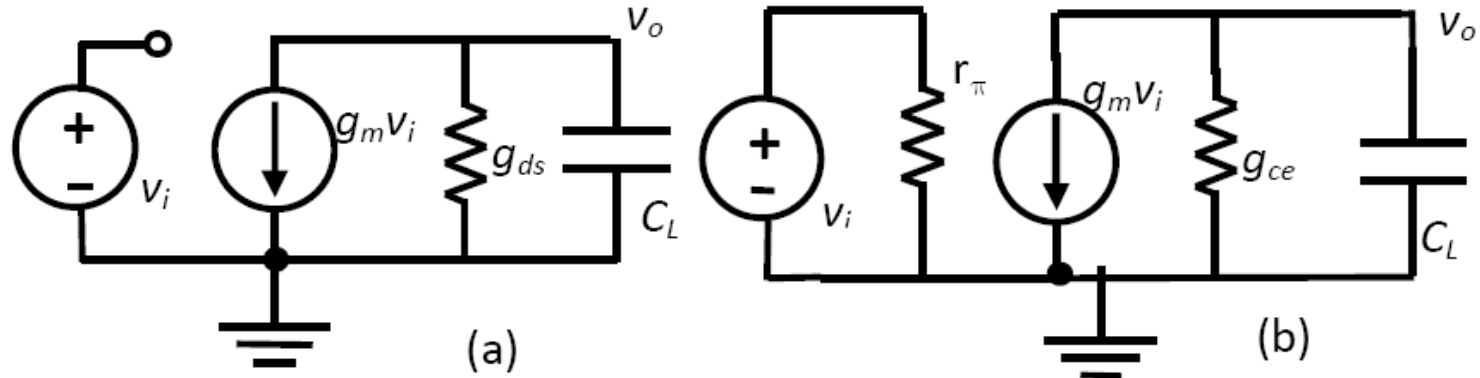
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Intrinsic gain stages: common-source and common-emitter amplifiers



Small-signal circuit and frequency response of the CS and CE amplifiers



$$v_o \cong -\frac{g_m}{j\omega C_L} v_i; \quad \omega \gg \omega_b$$

$$\omega_u = \frac{g_m}{C_L}$$

Design of the CE and CS amplifiers

$$\left| A_v(\omega_u) \right| = 1$$

$$g_m = \omega_u C_L = 2\pi \cdot GB \cdot C_L$$

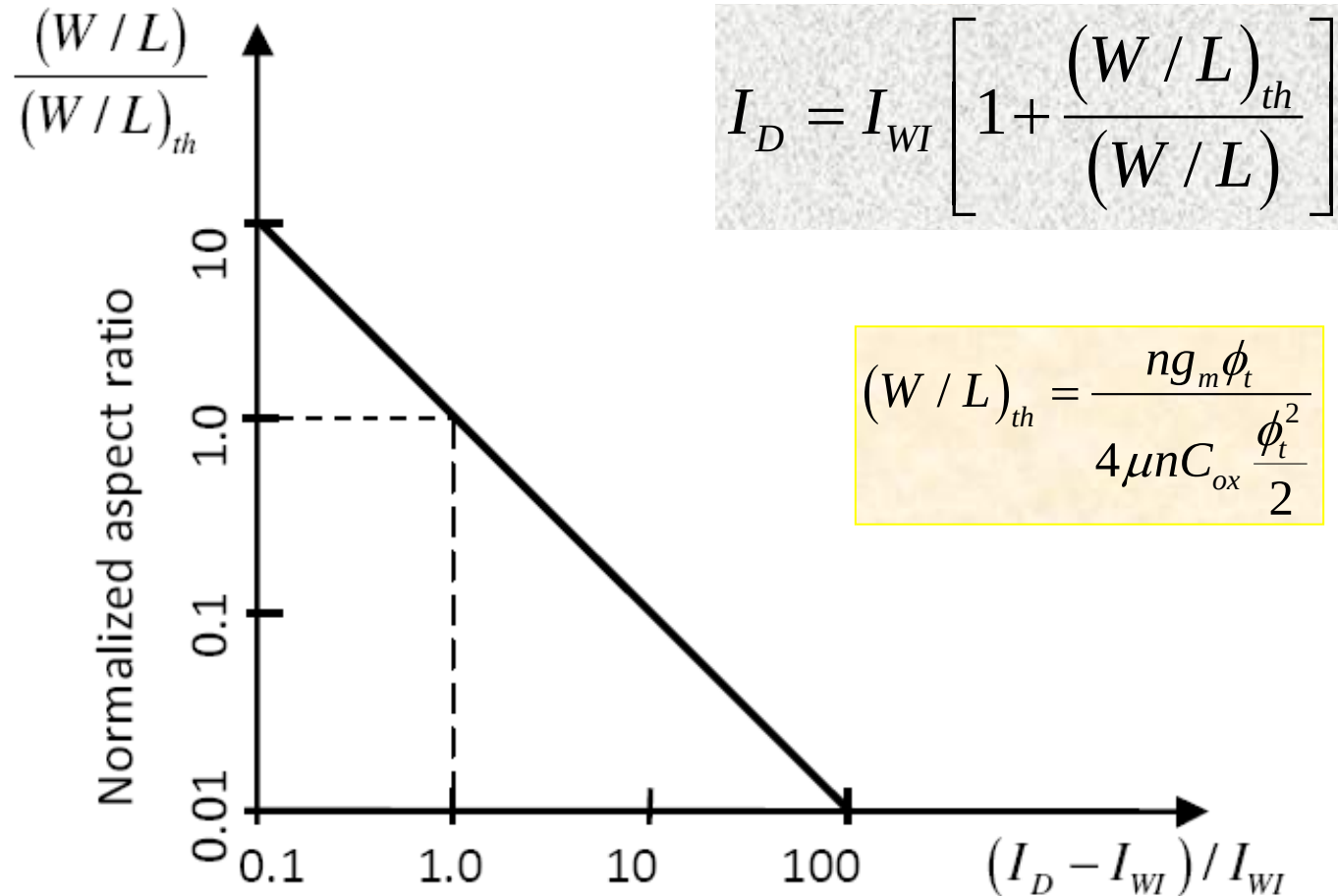
BJT in the direct active region

$$I_C = I_S e^{V_{BE}/\phi_t} \quad \Rightarrow \quad I_C = g_m \phi_t = 2\pi \cdot GB \cdot C_L \cdot \phi_t$$

MOSFET in saturation

$$I_D = n g_m \phi_t \frac{(\sqrt{1 + i_f} + 1)}{2} = n g_m \phi_t \left[1 + \frac{g_m}{2\mu C_{ox} \phi_t (W/L)} \right]$$

Aspect ratio vs. current excess in MOSFET design



REFERENCES

- A. I. A. Cunha, M. C. Schneider and C. Galup-Montoro, "Derivation of the Unified Charge Control Model and Parameter Extraction Procedure", Solid-State Electronics, March 1999.
- O. C. Gouveia Filho, A. I. A. Cunha, M. C. Schneider and C. Galup-Montoro, "Advanced compact model for short-channel MOS transistors", IEEE Custom Integrated Circuits Conference, Orlando, FL, USA, May 2000.
- C. Galup-Montoro and M. C. Schneider, *MOSFET Modeling for Circuit Analysis and Design*, World Scientific, 2007.
- C. M. Adornes, D. G. Alves Neto, M. C. Schneider and C. Galup-Montoro, "Bridging the Gap between Design and Simulation of Low-Voltage CMOS Circuits", Journal of Low Power Electronics and Applications, June 2022.
- D. G. Alves Neto *et al*, "Design-oriented single-piece 5-DC parameter MOSFET model," *IEEE Access*, 2024.

Appendix: the exact model of the long-channel MOSFET¹

$$I_D = -\frac{W}{L} \int_{V_S}^{V_D} \mu Q_I(V_C) dV_C$$

Consequently, the **exact** expressions for g_{mS} and g_{mD} are

$$g_{mD} = \frac{dI_D}{dV_D} = -\frac{W}{L} \mu Q_D = \frac{2I_S}{\phi_t} q_D$$

$$g_{mS} = -\frac{dI_D}{dV_S} = -\frac{W}{L} \mu Q_S = \frac{2I_S}{\phi_t} q_S$$

¹ H. C. Pao and C. T. Sah, 'Effects of diffusion current on characteristics of metal-oxide (insulator)-semiconductor transistors' Solid-State Electronics, Oct 1966



Part 2

Advanced Compact MOSFET Model: Parameter extraction



"Scan me"

Deni Germano Alves Neto

https://github.com/ACMmodel/MOSFET_model



Outline

- **ACM2 :**
 - **Parameter Extraction**
 - **V_{T0} , I_S and n**
 - **Sigma & Zeta**
 - **DC characteristics**
 - **Transconductances in Saturation**
 - **Example: Inverter CMOS**
- **Overview of Open-source environment**
 - **Automatic parameter extraction**

ACM2: A simple 5-DC-parameter MOSFET model

$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s)$$

Used to calculate q_s

$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

$$\frac{V_{DS}}{\phi_t} = q_s - q_d + \ln\left(\frac{q_s - q_{dsat}}{q_d - q_{dsat}}\right)$$

Used to calculate q_d

$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \zeta(q_s - q_d)} (q_s - q_d)$$

Specific
current
 I_S (W, L)

Threshold
voltage
 V_{T0} (W, L)

Slope
factor
 n (W, L)

DIBL
factor
 σ (W, L)

V_{sat}
effect
 ζ (W, L)

3PM-ACM model in a nutshell

$$I_D = I_S [i_f - i_r] \quad \text{where} \quad I_S = \mu C_{ox} n \frac{\phi_t^2}{2} \frac{W}{L} = I_{SH} \frac{W}{L}$$

$$\frac{V_P - V_{S(D)B}}{\phi_t} = \sqrt{1 + i_{f(r)}} - 2 + \ln \left(\sqrt{1 + i_{f(r)}} - 1 \right) \quad V_P \cong \frac{V_{GB} - V_{T0}}{n}$$

DC eqs

If we choose $i_f = 3 \Rightarrow RHS = 0 \Rightarrow V_{GB} = V_{T0}$

$$\frac{V_{DS}}{\phi_t} = \sqrt{1 + i_f} - \sqrt{1 + i_r} + \ln \left(\frac{\sqrt{1 + i_f} - 1}{\sqrt{1 + i_r} - 1} \right)$$

$$g_{ms(d)} = \frac{2I_S}{\phi_t} \left(\sqrt{1 + i_{f(r)}} - 1 \right) \Rightarrow \frac{W}{L} = \frac{g_{ms(d)} \phi_t}{2I_{SH} (\sqrt{1 + i_{f(r)}} - 1)}$$

Small-signal eqs

$$g_m = \frac{g_{ms} - g_{md}}{n}$$

$$\frac{g_m}{I_D} = \frac{d(\ln I_D)}{dV_G} = \frac{2}{n \phi_t (\sqrt{1 + i_f} + \sqrt{1 + i_r})}$$

I_S, V_{T0} and n extraction

The g_m/I_D method

Let us choose: $V_{DS} = \phi_t/2$ and $i_f = 3$

$$\frac{V_{DS}}{\phi_t} = \sqrt{1 + i_f} - \sqrt{1 + i_r} + \ln \left(\frac{\sqrt{1 + i_f} - 1}{\sqrt{1 + i_r} - 1} \right)$$

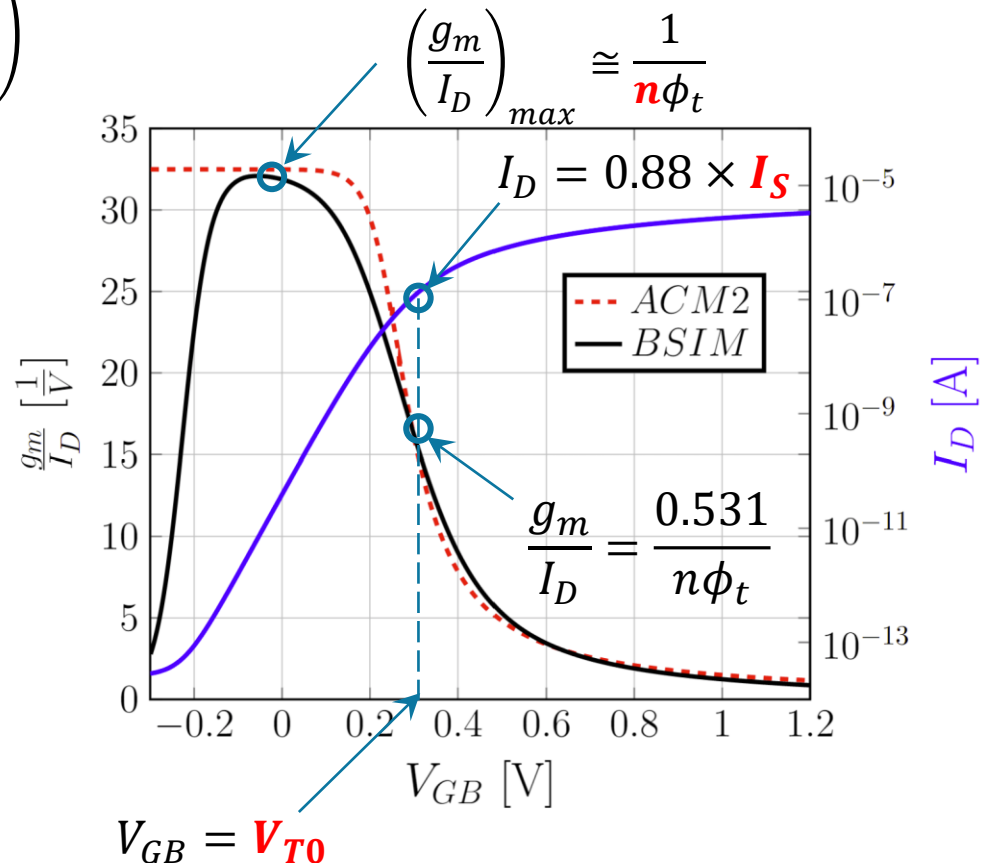
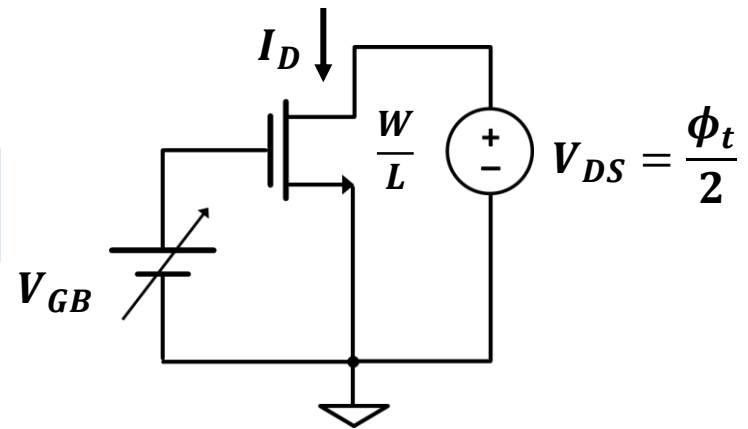
$$\text{Thus: } i_r = 2.12$$

$$\frac{g_m}{I_D} = \frac{d(\ln I_D)}{dV_G} = \frac{2}{n\phi_t(\sqrt{1 + i_f} + \sqrt{1 + i_r})}$$

$$\frac{g_m}{I_D} = \frac{0.531}{n\phi_t} = 0.531 \left(\frac{g_m}{I_D} \right)_{max}$$

$$I_D = I_S(i_f - i_r)$$

$$I_D = (3 - 2.12) I_S = 0.88 I_S$$



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ACM2 : Transconductances in saturation

$$I_{Dsat} = \frac{2I_S}{\zeta} q_{dsat} \quad \text{where} \quad \left\{ \begin{array}{l} I_S = \mu C_{ox} n \frac{\phi_t^2}{2} \frac{W}{L} = I_{SH} \frac{W}{L} \\ q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}} \end{array} \right.$$

$$\text{Definitions} \quad \left\{ \begin{array}{l} g_m \triangleq \frac{\partial I_{Dsat}}{\partial V_G} \quad g_d \triangleq \frac{\partial I_{Dsat}}{\partial V_D} \quad g_{msat3} \triangleq \frac{\partial^3 I_{Dsat}}{\partial V_G^3} \end{array} \right.$$

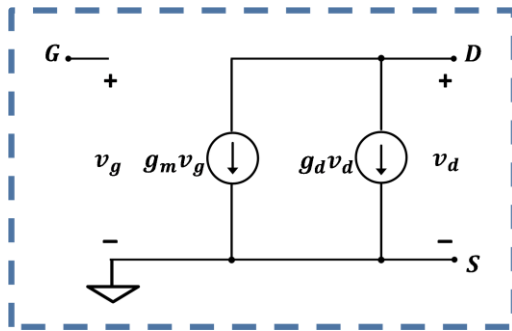
Transconductances in terms of q_s :

$$g_{msat} = \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \quad g_{dsat} = \sigma \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \quad \Rightarrow \quad g_{dsat} = \sigma g_{msat}$$

$$g_{msat3} = \frac{16I_S}{(n\phi_t)^3} \frac{q_s}{(q_s + 1)^3} \frac{2 - 2\zeta q_s - 3\zeta q_s^2}{(\zeta q_s + 2)^4}$$

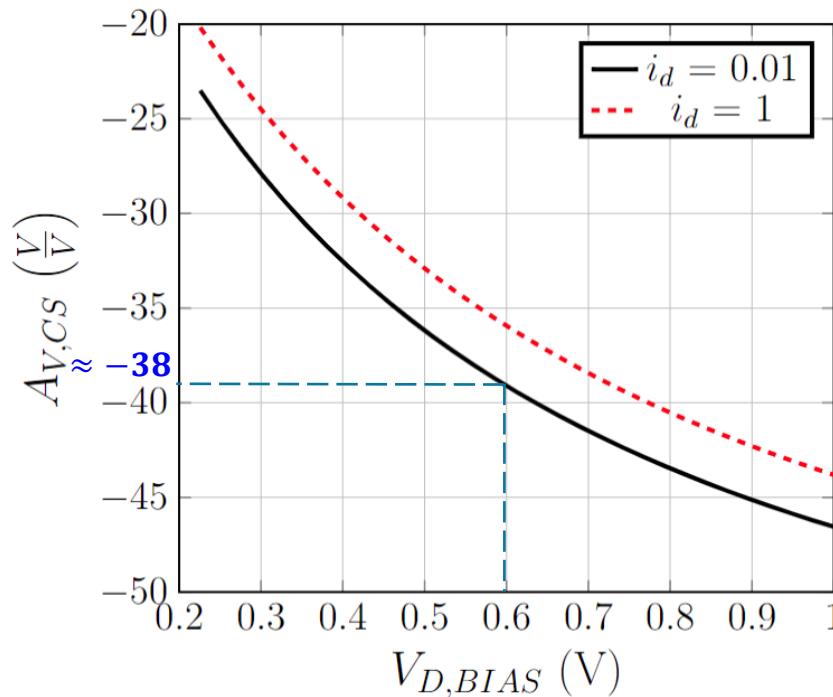
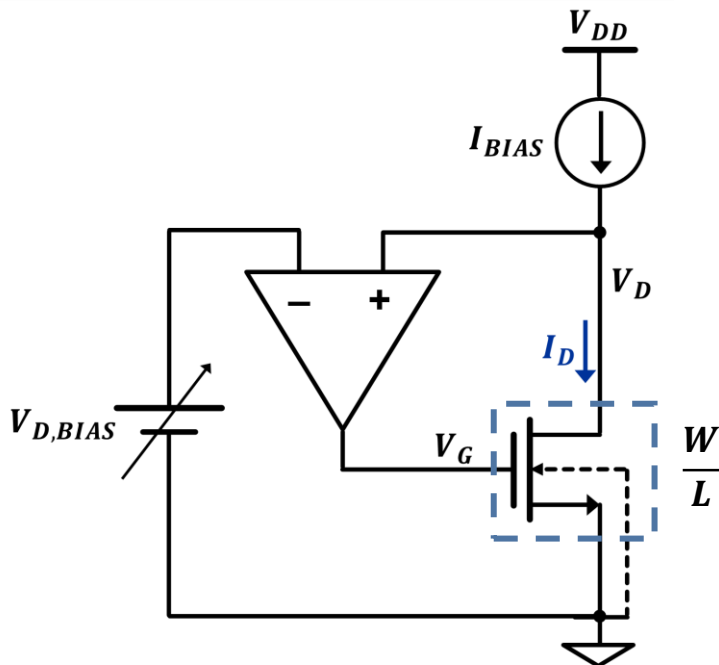
Extraction of σ

Common-Source Intrinsic-Gain method



In saturation : $g_{dsat} = \sigma g_{msat}$

$$A_{V,CS} = \frac{\Delta v_D}{\Delta v_G} = -\frac{g_{msat}}{g_{dsat}} = \frac{\frac{\partial I_{Dsat}}{\partial V_G}}{\frac{\partial I_{Dsat}}{\partial V_D}} = -\frac{\frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}}{\sigma \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}} = -\frac{1}{\sigma}$$



Example:

$$A_{V,CS} = -\frac{1}{\sigma}$$

$$\sigma = -\frac{1}{(-38)}$$

$$\sigma = 0.026$$

ζ extraction

$$i_{dsat} = \frac{2}{\zeta} q_{dsat} \quad q_s = \sqrt{1 + \frac{2}{\zeta} q_{dsat}} - 1 + q_{dsat}$$

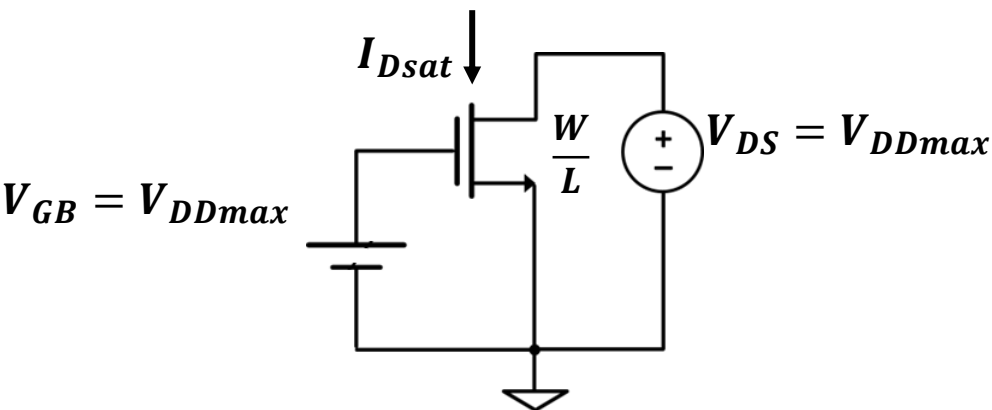
$$\zeta = \frac{2(q_s + 1 - \sqrt{1 + i_{dsat}})}{i_{dsat}}$$

- q_s calculated using parameters (V_{T0} , n , σ) and UCCM.

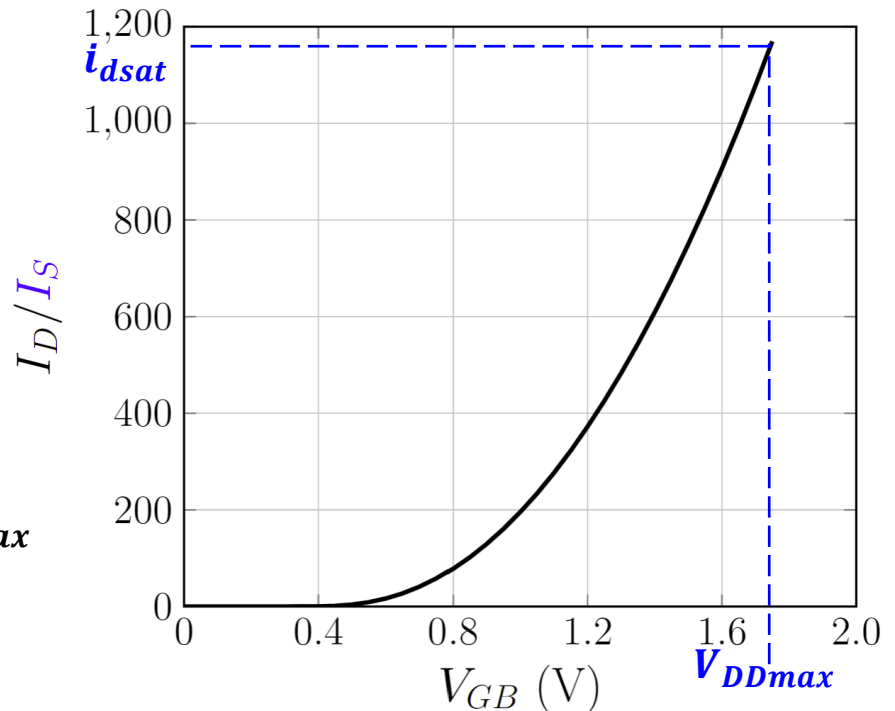


$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

$$\frac{V_P - V_{S(D)B}}{\phi_t} = q_s - 1 + \ln(q_s)$$



- Measure
 $I_{Dsat} = I_D(V_G = V_D = V_{DDmax} \text{ and } V_S = V_B)$
 $i_{dsat} = I_{Dsat}/I_S$.

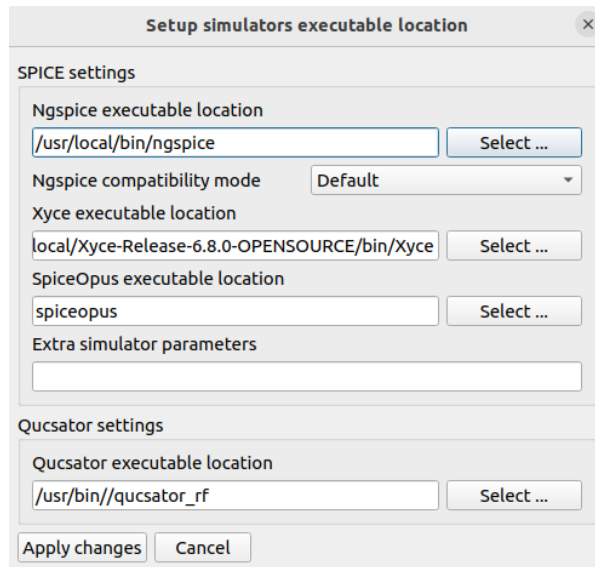


Outline

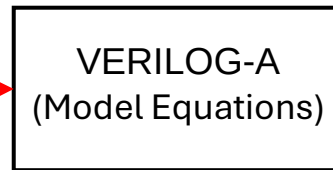
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Qucs-S: Quite universal circuit simulator with SPICE

Simulation setup:



SG13G2 PDK



Qucs-S: Quite universal circuit simulator with SPICE

Device path for the MOS model and parameters (IHP pdk)

Simulations
Control

- Ngspice simulations:

- DC
- AC
- TRAN
- S-parameters
- Noise
- Fourier

.INCLUDE SCRIPT

```
INCLSCR1
SpiceCode=.LIB /home/acm2/Desktop/IHP-Open-PDK/ihp-sg13g2/libs.tech/ngspice/models/cornerMOSlv.lib mos_tt
.control
pre_osdi /home/acm2/QucsWorkspace/psp103_nqs.osdi
.endc
```

dc simulation

DC1

Parameter
sweep

SW1
Sim=DC1
Type=lin
Param=V1
Start=0
Stop=1.5
Points=301

Input eqs.

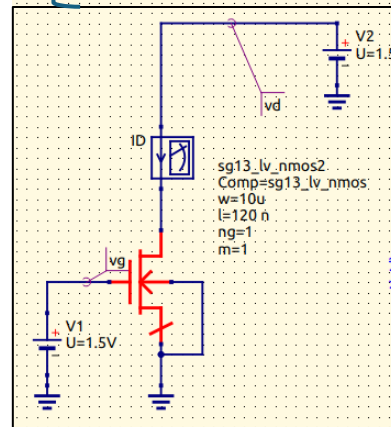
Nutmeg

NutmegEq1
Simulation=SW1
IDmax=maximum(i(vd))
IS=11.77e-6
idsat=IDmax/IS

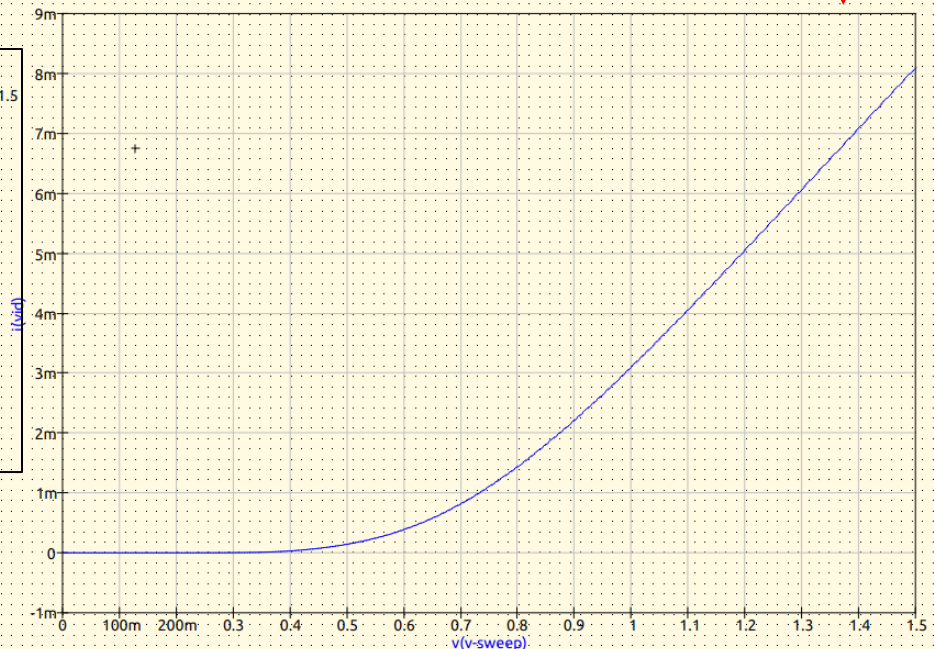
Tabular results

v(v-sweep)	i(idmax)	idsat
0	0.00809	688

Diagram results

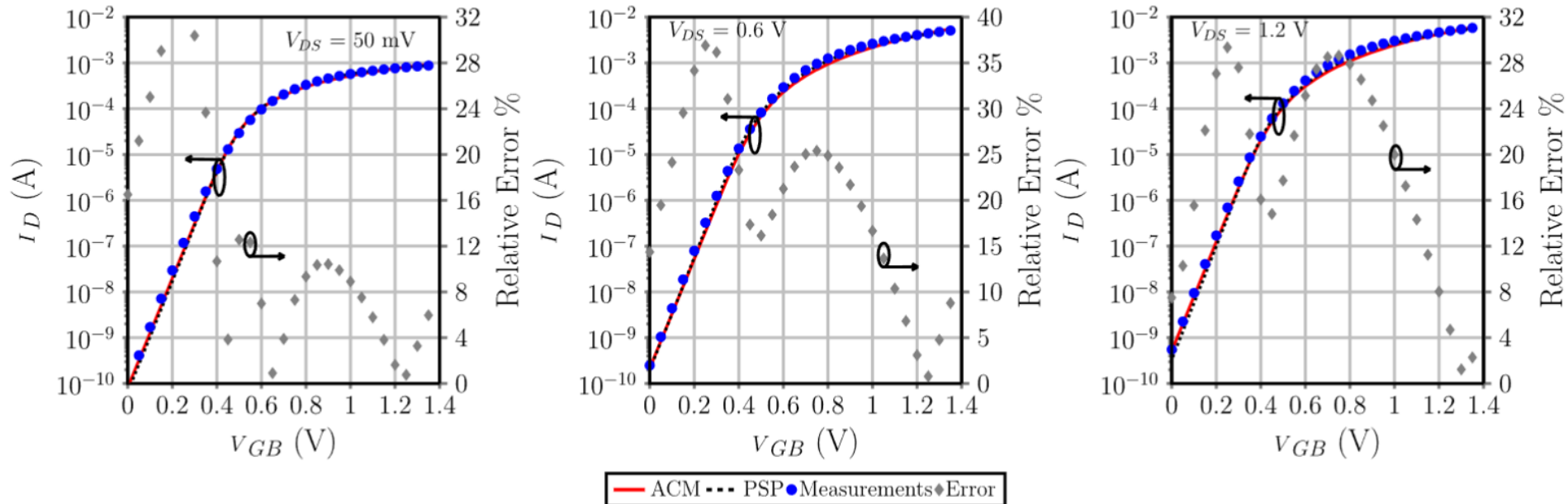


Circuit
Schematic



ACM2¹ vs PSP – 130 nm SiGe IHP²

I_D vs V_{GB}



Characteristics of a LVT NMOS bulk transistor with $W/L = 10\mu\text{m}/120\text{ nm}$.

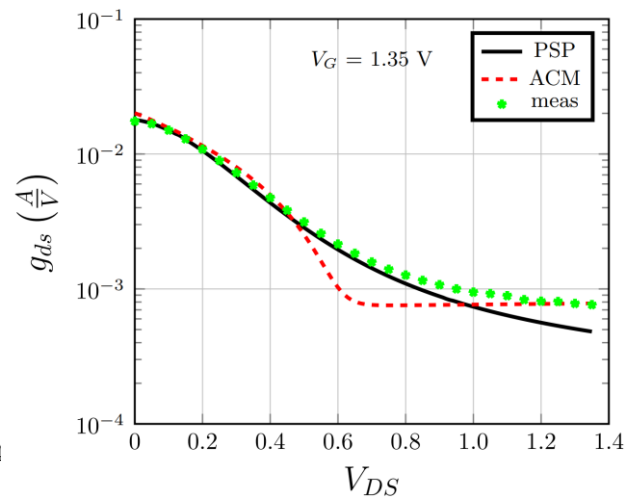
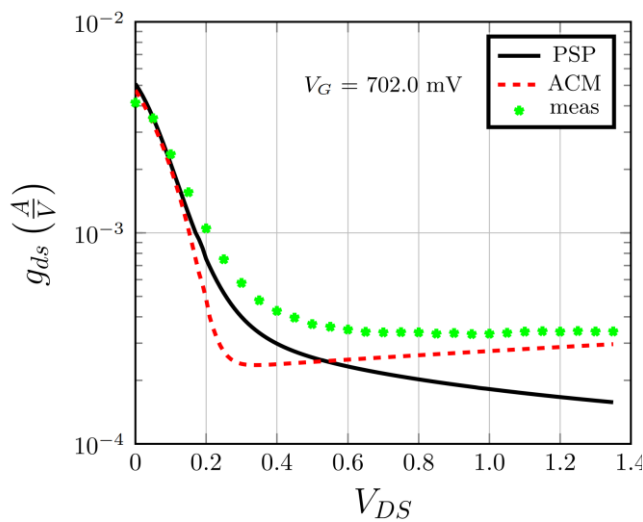
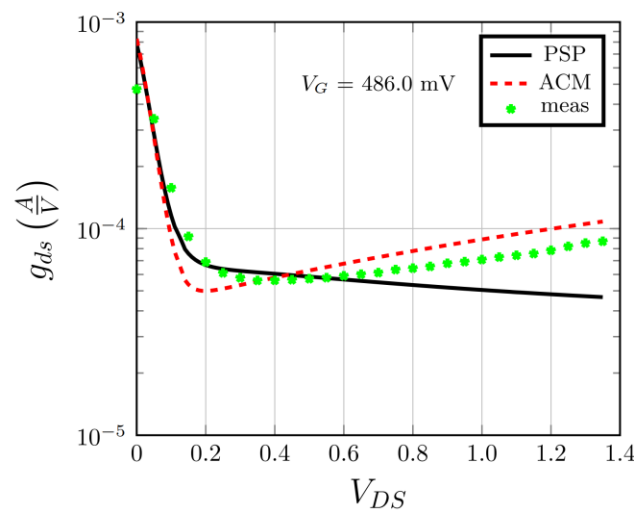
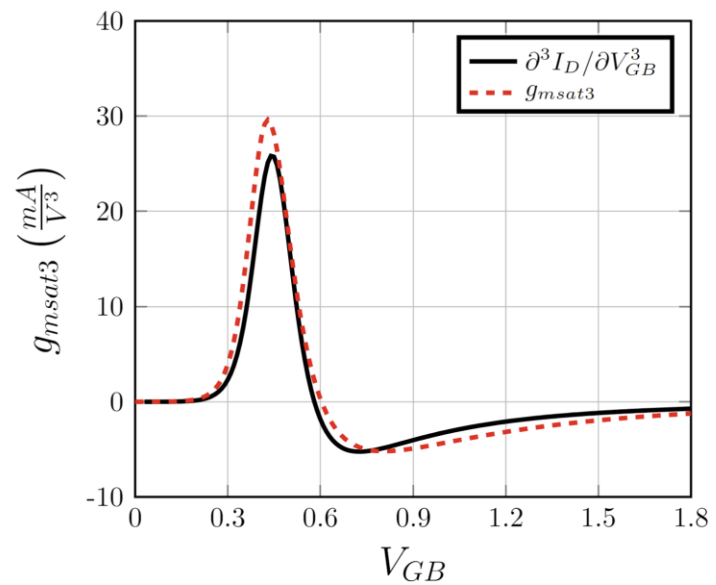
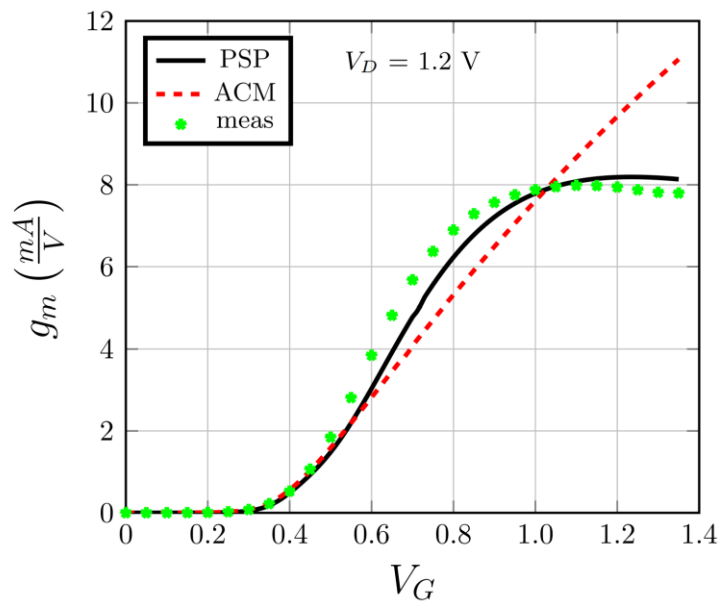
¹ ACM2 : implemented in verilog-A, compiled by OPENVAF, simulated in Ngspice

² Institut for High-Performance Microelectronics (IHP) open-source PDK

Outline

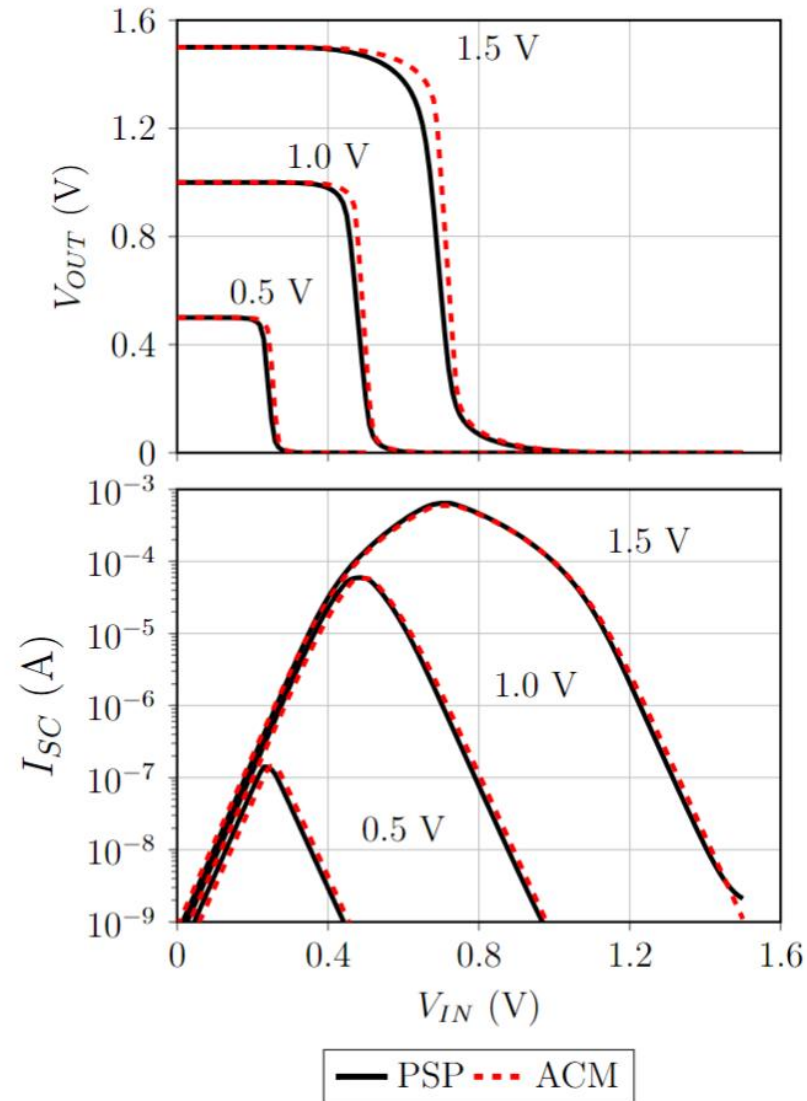
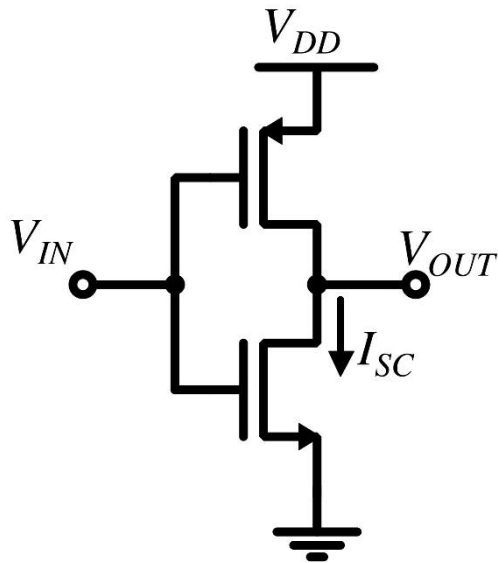
- **ACM2 :**
 - **Parameter Extraction**
 - **V_{T0} , I_S and n**
 - **Sigma & Zeta**
 - **DC characteristics**
 - **Transconductances in Saturation**
 - **Example: Inverter CMOS**
- **Overview of Open-source environment**
 - **Automatic parameter extraction**

5PM-ACM2 : Transconductance gm, gm3 and gds



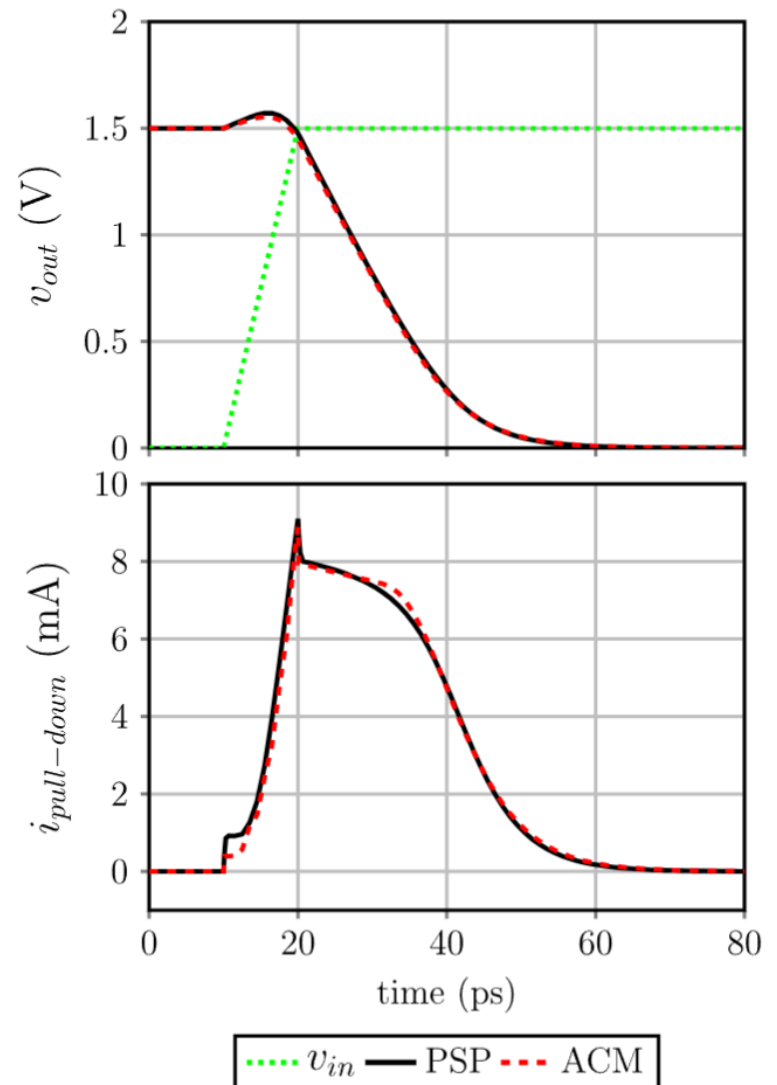
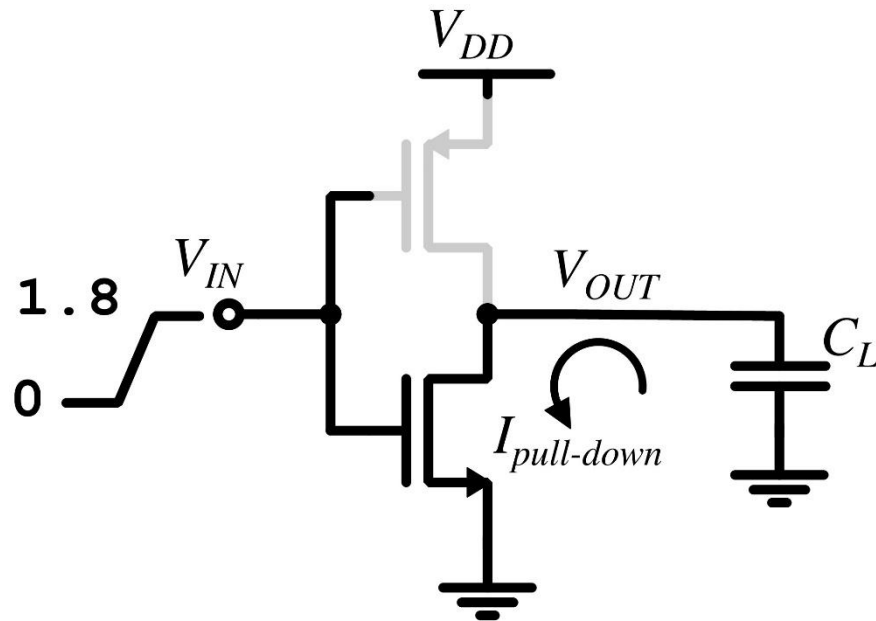
CMOS Inverter in 130 nm bulk

VTC and short-circuit current



CMOS Inverter in 130 nm bulk

Output Voltage and pull-down current



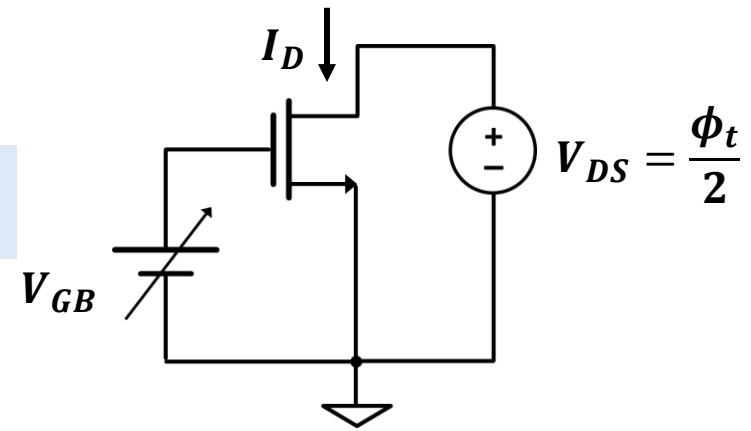
Let's go to Qucs-S!

Useful links:

- Qucs-S oficial page: <https://ra3xdh.github.io/>
- Qucs-S iterative doc: <https://qucs-s-help.readthedocs.io/>
- Ngspice oficial page : <https://ngspice.sourceforge.io/index.html>
- Google Colab: ACM2 & LNA design:
https://colab.research.google.com/drive/1s3PKF6pf3zlhTj6jc-qhCLlcGfJ_UEE?usp=sharing

I_{S0}, V_{T0} and n extraction

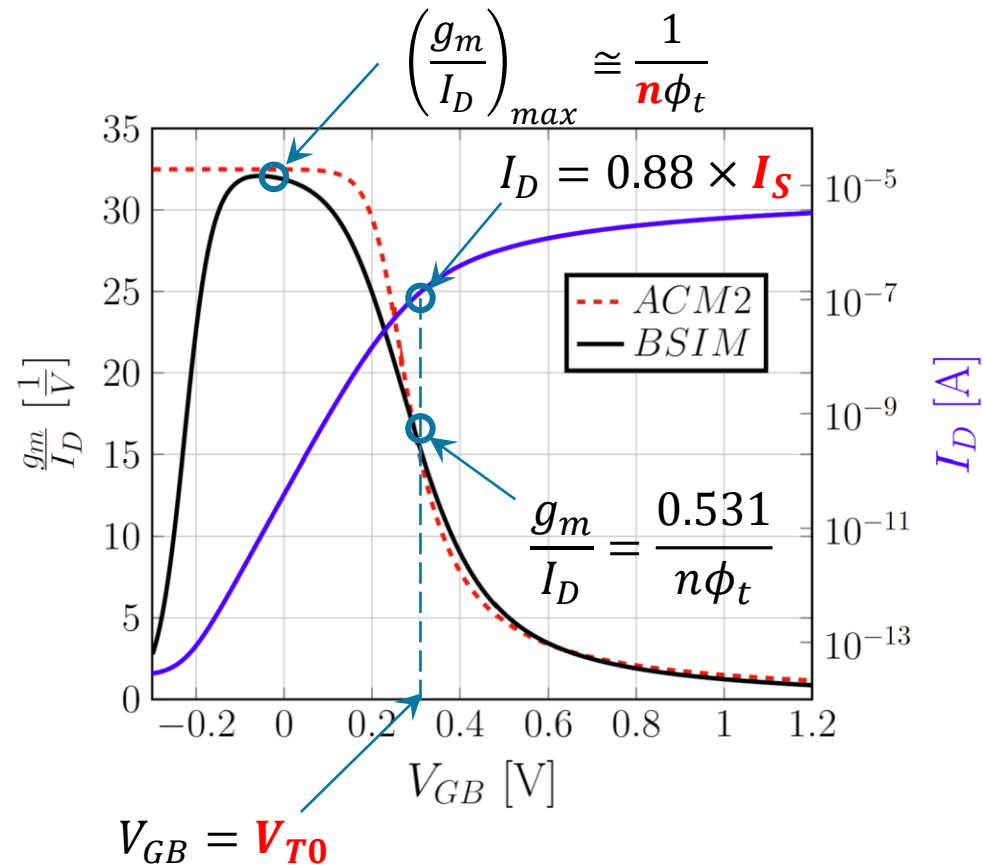
The g_m/I_D method



We identify the $\left. \frac{g_m}{I_D} \right|_{max} \rightarrow n$

We calculate $0.531 \left. \frac{g_m}{I_D} \right|_{max} \rightarrow V_{T0}$

We read the value of $I_D \rightarrow I_S$



Practical part → Go to QuCS !

Github – ACM2

Github - Content

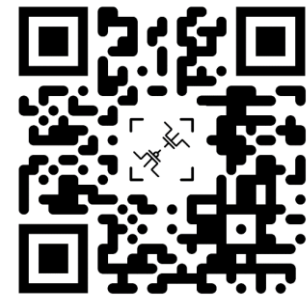
 **ACMmodel** Merge pull request #26 from gabrielmaranhao/main 71ee7cd · 6 months ago

Examples	Update SKY130 and GF180 using ACM examples on xschem
Verilog-A	Update PMOS_ACM_2V0.va
docs	Delete 5PM_NewCAS.pdf
LICENSE	Update LICENSE
README.md	Update README.md

 **README**  ECL-2.0 license

Advanced Compact MOSFET model (ACM)

ACM is a simple MOSFET model to design and simulate Analog, Mixed-Signal, and RF circuits



Verilog-A code Available!

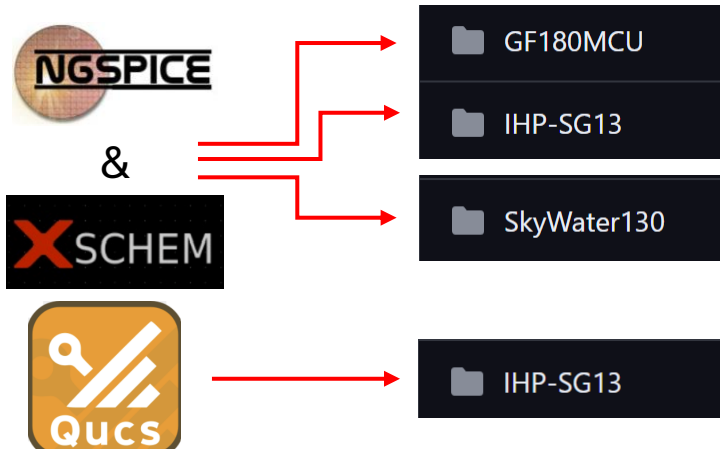
```
MOSFET_model / Verilog-A / NMOS_ACM_2V0.va

ACMmodel Update NMOS_ACM_2V0.va

Code Blame 291 lines (245 loc) · 12.3 KB

1 //*****
2 // * ACM NMOS model (Verilog-A) *
3 // * 07/2023 V2.0.0 *
4 //*****
5
6 // *****
7 // * Copyright under the ECL-2.0 license *
8 // * Universidade Federal de Santa Catarina *
9 // *
10 // * Current developers: Deni Germano Alves Neto (Doctoral student, UFSC) *
11 // * Cristina Missel Adornes (Doctoral student, UFSC) *
12 // * Gabriel Maranhao (Doctoral student, UFSC) *
13 // *
14 // * Project Supervisors: Prof. Carlos Galup-Montoro *
15 // * Prof. Marcio Chereem Schneider *
16 // *****
17
18 `include "constants.vams"
19 `include "disciplines.vams"
20
21 // function of the algorithm 443 to calculate de normalize charge densities
22 `define algo_443(Z,qn) \
23     if(Z < 0.7385) begin \
24         numeratorD = Z + (4.0/3.0)*Z*Z; \
25         denominatorD = 1.0 + (7.0/3.0)*Z+(5.0/6.0)*Z*Z; \
26         WnD = numeratorD/denominatorD; \
27     end else begin \
28         numeratorD = ln(Z)*ln(Z)+2.0*ln(Z)-3.0; \
29         denominatorD = 7.0*ln(Z)*ln(Z) + 58.0*ln(Z) +127.0; \
30         WnD = ln(Z) - 24.0*(numeratorD/denominatorD); \
```

Examples of PDKs and circuit simulators using the ACM model

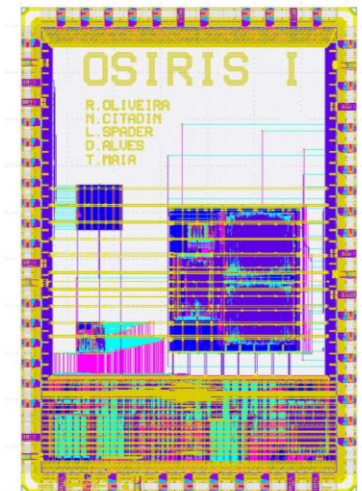
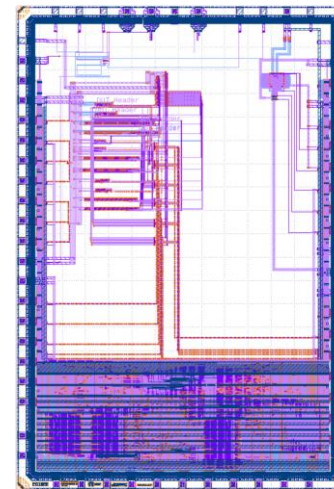
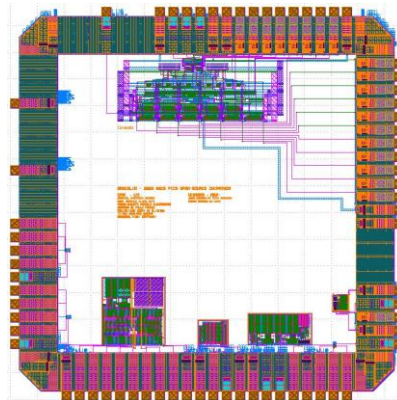


Open-Source IC design

- Chipathon - SSCS 2021 :)
- Analog-front-end for Biosignals – AFEbio
- Chipathon-SSCS 2023 & UNIC-CASS 2023
 - Analog IC design
- Mentor in **Universalization of IC Design from CASS (UNIC-CASS)** 2024



ODIN



Open-Source Design of Integrated Networks



Part 3

Analog IC design



Márcio Cherem Schneider

https://github.com/ACMmodel/MOSFET_model



Outline

Analog circuit design

1. Current mirrors
2. High-swing cascode current mirror
3. Current gain schemes
4. Current and voltage references
5. Single-stage amplifiers

ACM2 applied to weak inversion

ACM2 parameters: V_{T0} , I_S , n , σ , ζ

$$q_{S(D)} = Q_{S(D)} / (-nC_{ox}\phi_t)$$

$$q_{S(D)} = e^{(\frac{V_P - V_{S(D)B}}{\phi_t} + 1)} \quad \text{UCCM}$$

$$I_D / I_S = i_f - i_r = (I_F - I_R) / I_S$$

$$V_P = \frac{V_{GB} - (V_{T0} - \sigma(V_{DB} + V_{SB}))}{n}$$

$$i_D = I_D / I_S = 2(q_S - q_D)$$

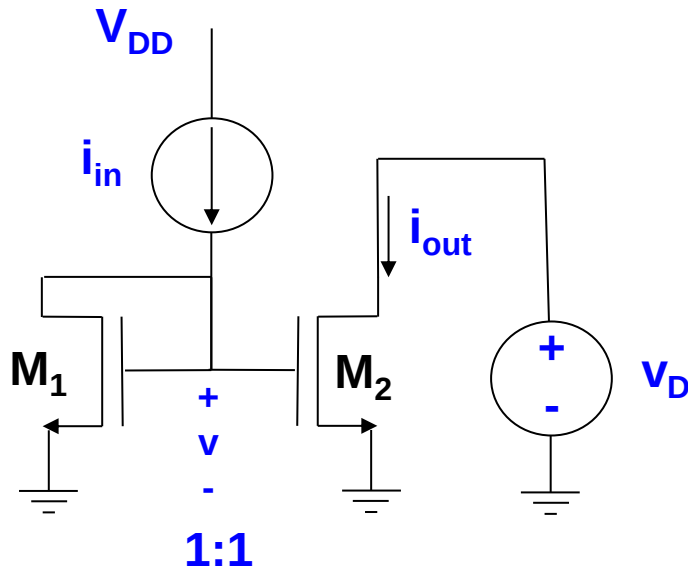
$$I_S = \frac{W}{L} \mu_s n C_{ox} \frac{\phi_t^2}{2} = S I_{SH}$$

normalization (specific) current

$$Wl : q_{S(D)} \ll 1$$

ratio of diffusion-related velocity to saturation velocity

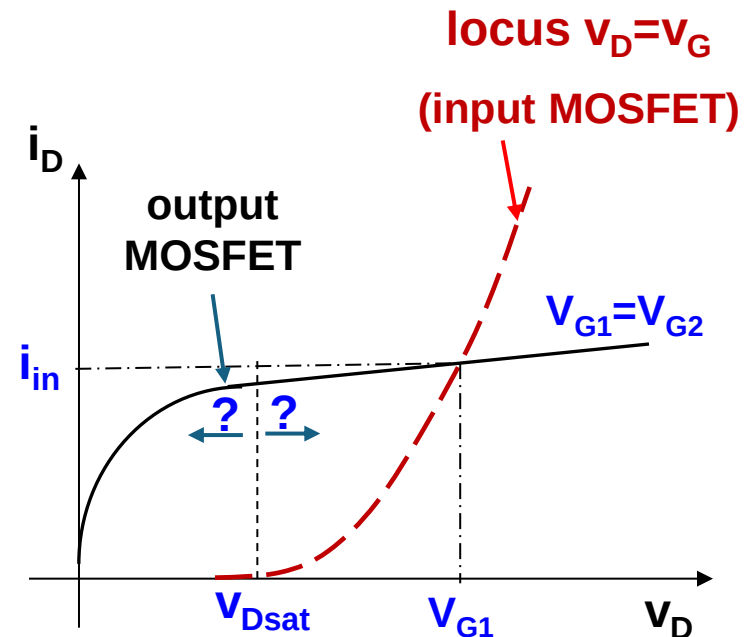
1. Current mirrors



$$M_1 = M_2$$

M_1 : $i \rightarrow v$ converter

M_2 : $v \rightarrow i$ converter



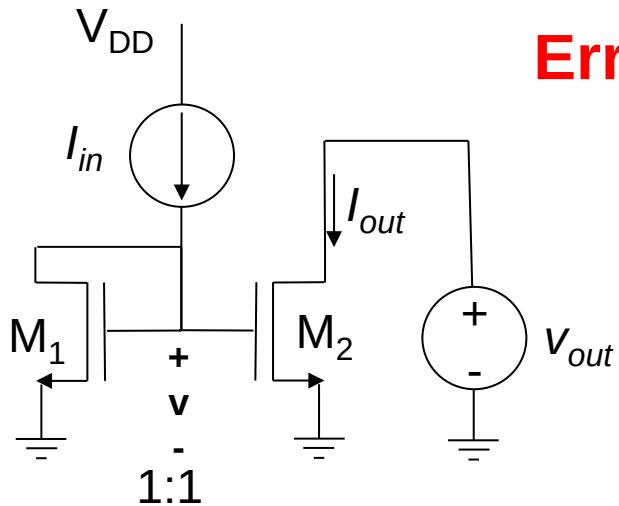
Basic principle $V_{G1} = V_{G2}$; $V_{S1} = V_{S2}$; $V_{B1} = V_{B2}$;
 $v_{out} > V_{Dsat} \rightarrow i_{out} \cong i_{in}$

What's this saturation voltage V_{Dsat} ? No clear definition.

An approximation for the saturation voltage:

$$V_{DSSat} \cong \phi_t \left(\sqrt{1 + i_f} + 3 \right)$$

1. Current mirrors



Error due to difference in drain voltages

V_A is non-physical

A simple case: error for weak inversion using ACM2

Linear approximation

$$\frac{I_{out}}{I_{in}} = \frac{I_S f(V_G - V_T, V_S)(1 + V_{out}/V_A)}{I_S f(V_G - V_T, V_S)(1 + V_{in}/V_A)}$$

$$I_{in} = I_{F1} - I_{R1} = I_{F1} [1 - e^{-V_{in}/\varphi_t}]$$

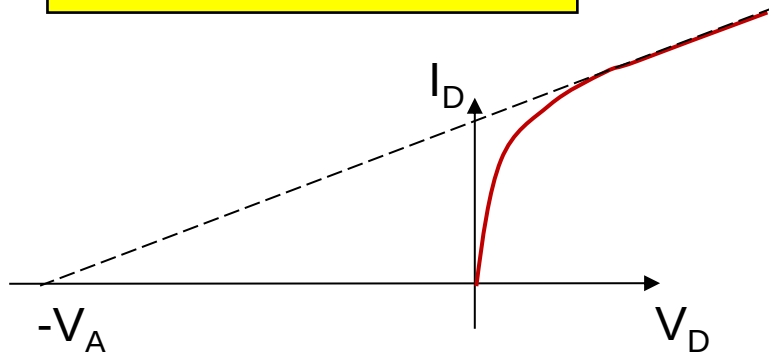
$$I_{out} = I_{F2} - I_{R2} = I_{F2} [1 - e^{-V_{out}/\varphi_t}]$$

$$I_{F2} = I_{F1} e^{\sigma(V_{out}-V_{in})/n\varphi_t}$$

$$\frac{I_{out}}{I_{in}} = e^{\frac{\sigma(V_{out}-V_{in})}{n\varphi_t}} \frac{1 - e^{-V_{out}/\varphi_t}}{1 - e^{-V_{in}/\varphi_t}}$$

$$\frac{I_{out}}{I_{in}} \cong e^{\frac{\sigma(V_{out}-V_{in})}{n\varphi_t}} \quad \text{for } V_{out} \text{ \& } V_{in} \gg \varphi_t$$

$$\frac{I_{out} - I_{in}}{I_{in}} \cong \frac{V_{out} - V_{in}}{V_A}$$



1. Current mirrors

Dependence of the output current (conductance) on the output voltage

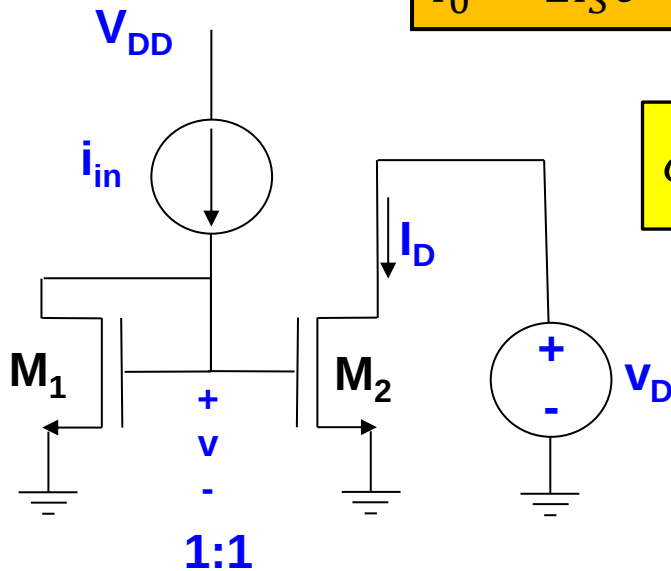
Weak inversion
 $i_f < 1$

$$I_D = I_0 e^{(V_G - V_T)/\varphi_t} [1 - e^{-V_D/\varphi_t}] = I_F \left[1 - \frac{I_R}{I_F} \right]$$

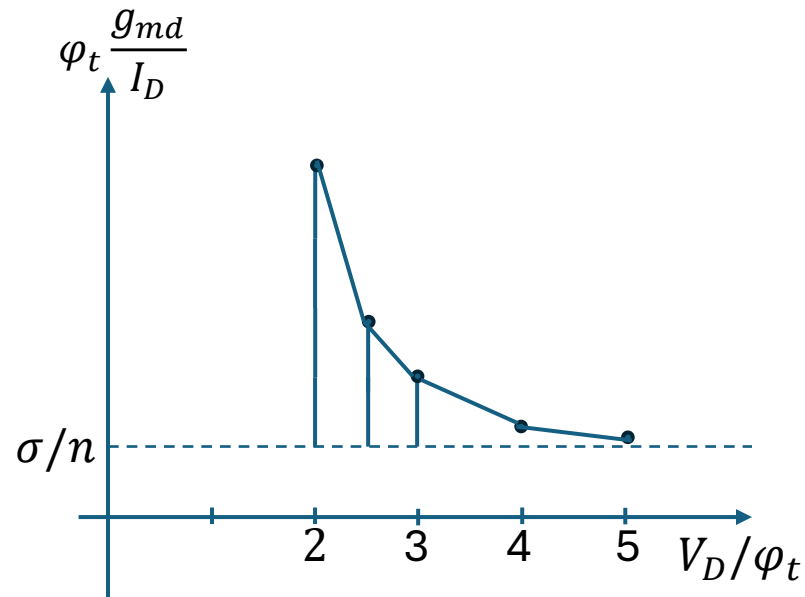
$$I_0 = 2I_S e^1$$

$$I_F$$

$$V_T \cong V_{T0} - \sigma[V_{SB} + V_{DB}]$$



$$\varphi_t \frac{g_{md}}{I_D} = \frac{1}{I_D} \frac{dI_D}{dV_D} = \frac{d \ln I_D}{dV_D} = \frac{\sigma}{n} + \frac{I_F}{I_D} e^{-V_D/\varphi_t} \quad (A)$$

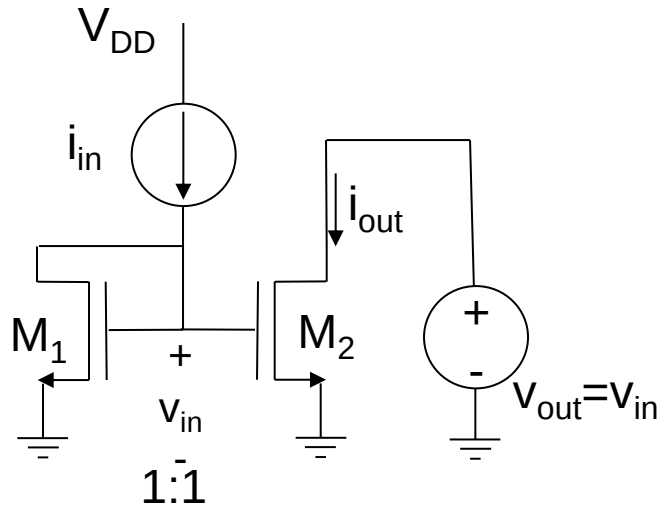


1. Current mirrors

Error due to mismatch

Main contributors to mismatch: I_S & V_T

$$I_D = I_S f(V_G - V_T) \quad (B)$$



$$\Delta I_D \cong \frac{dI_D}{dI_S} \Delta I_S + \frac{dI_D}{dV_T} \Delta V_T$$

$$\frac{dI_D}{dV_T} = -\frac{dI_D}{dV_G} = -g_m$$

$$\frac{\Delta I_D}{I_D} \approx \frac{\Delta I_S}{I_S} - \frac{g_m}{I_D} \Delta V_T \quad (C)$$

For uncorrelated mismatch sources

$$\frac{\sigma^2(I_D)}{I_D^2} \approx \frac{\sigma^2(I_S)}{I_S^2} + \left(\frac{g_m}{I_D} \right)^2 \sigma^2(V_T) \quad (D)$$

Using Pelgrom's model

$$g_m/I_D$$

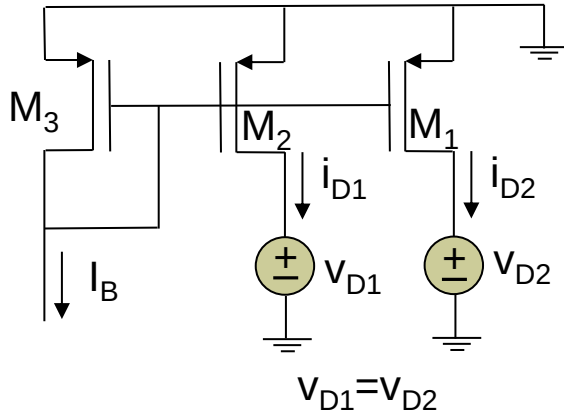
$$\frac{\sigma^2(I_D)}{I_D^2} \approx \frac{2}{WL} \left\{ A_{\beta}^2 + \left(\frac{A_{VT}}{n\phi_t} \right)^2 \left[\frac{2}{(\sqrt{1+i_f} + 1)} \right]^2 \right\}$$

In general, this is the prevailing term

1. Current mirrors

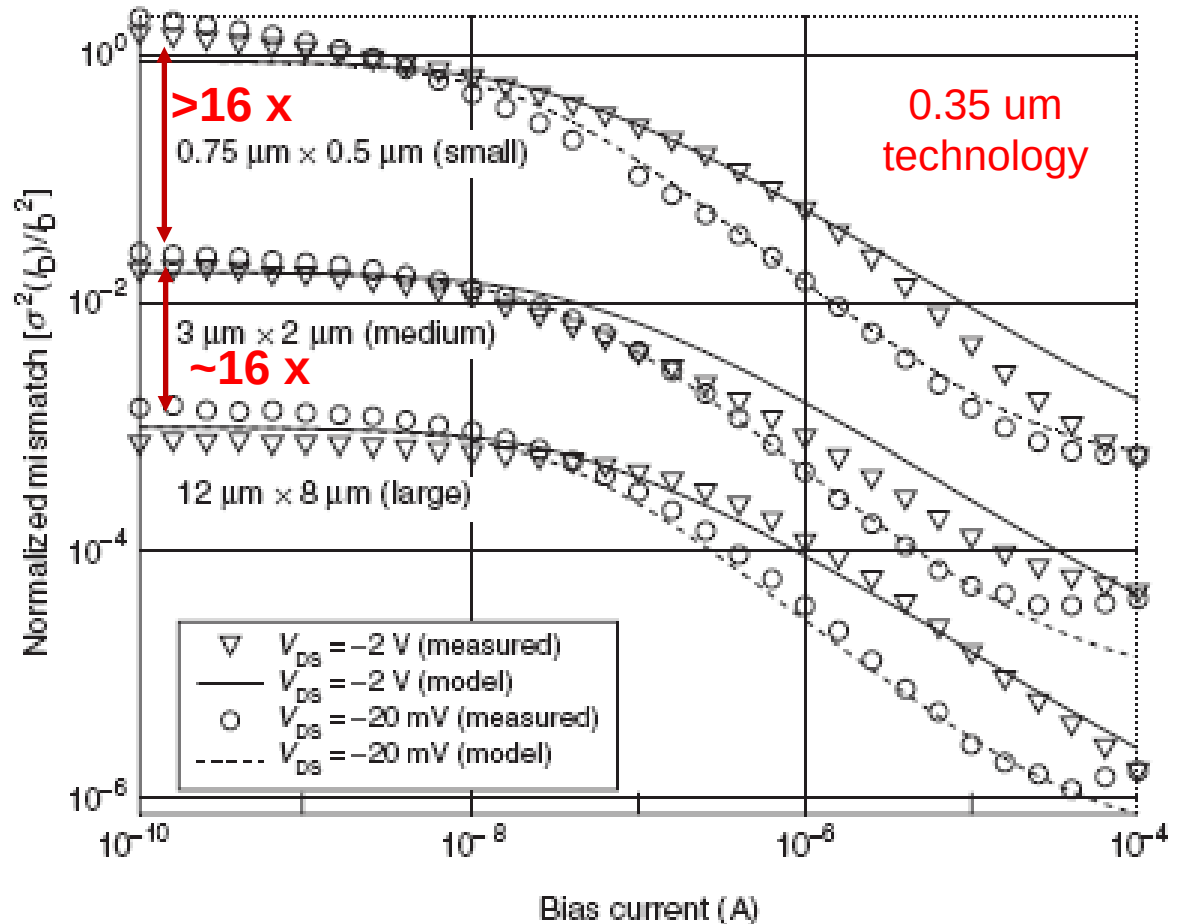
Error due to mismatch

$$\frac{\sigma^2(I_D)}{I_D^2} \approx \frac{2}{WL} \left\{ A_{ISH}^2 + \left(\frac{A_{VT}}{n\phi_t} \right)^2 \left[\frac{2}{(\sqrt{1+i_f} + 1)} \right]^2 \right\}$$



$$\epsilon_i = \frac{i_{D1} - i_{D2}}{(i_{D1} + i_{D2})/2} \Big|_i$$

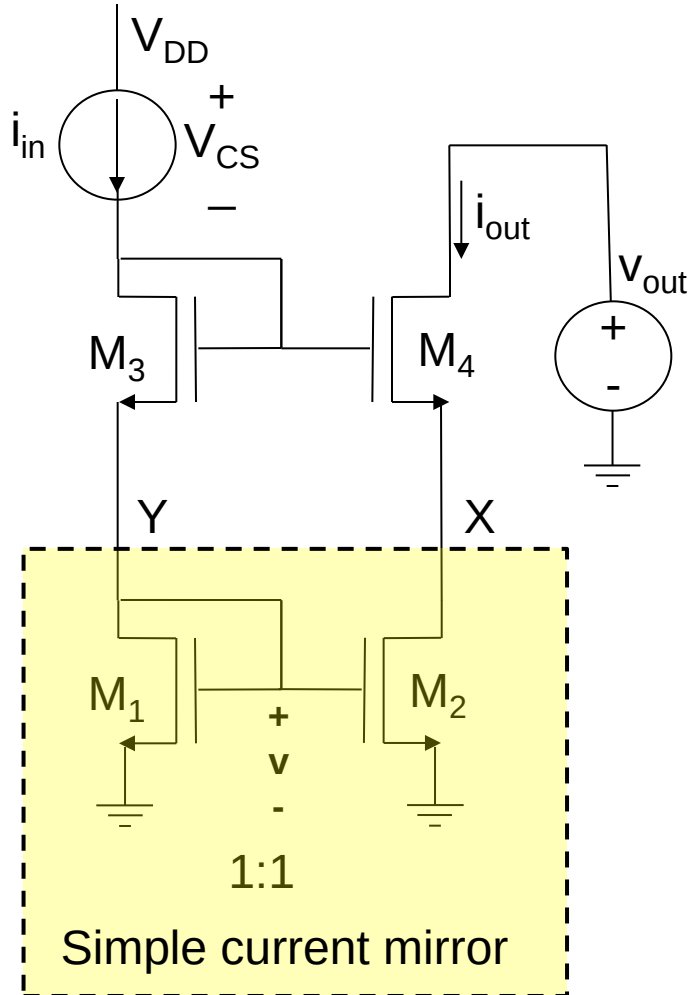
$$\frac{\sigma^2}{I_D^2} = \frac{1}{N} \sum_{i=1}^N \epsilon_i^2$$



Dependence of current matching on inversion level in linear and saturation regions $|V_{DS}| = 20 \text{ mV}$ and $|V_{DS}| = 2 \text{ V}$, respectively, for the large, medium-size, and small PMOS transistor pairs. Statistics determined for 18 pairs of transistors.

2. Cascode current mirrors

Self-biased cascode current mirror (SBCCM)



Assume $M_1 \equiv M_2 \equiv M_3 \equiv M_4$

- If M_4 operates in saturation, then $V_X \cong V_Y$ and $I_{out} \cong I_{in}$
- V_{out} is transferred to V_X with attenuation $(\frac{g_{md4}}{g_{ms4}})$ of common-gate gain
- Input constraint: $V_{DD} > V_{CS,min} + V_{GS3} + V_{GS1}$
- Output constraint: $V_{out} > V_{GS1} + V_{DS4,sat}$ for saturation of M_4 (In most cases, $V_{GS} > V_{DS,sat}$)
- The output conductance:

$$y_o = \frac{i_{out}}{v_{out}} \approx g_{md2} \frac{g_{md4}}{g_{ms4}}$$

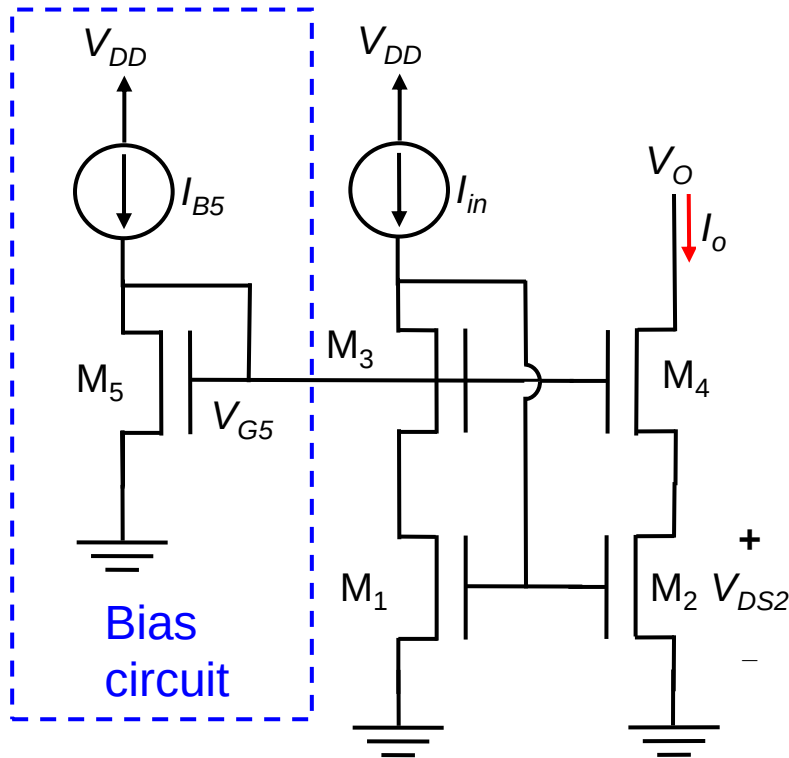
SBCCM is not a low-voltage current mirror

**Node X follows
approximately node Y**

2. Cascode current mirrors

Low-voltage cascode current mirror: lower VDD

Assume $M_1 \equiv M_2 \equiv M_3 \equiv M_4$



Goal: Design V_{G5} to bias M2 (M1) at the edge of saturation

$$\left. \begin{aligned} V_{S4} = V_{DS2} = V_{DSsat2} + \Delta V \\ f2 = i_{f4} = \frac{I_o}{I_{S4}} \approx \frac{I_{in}}{I_{S4}} \end{aligned} \right\} \begin{aligned} \Delta V &= \alpha \varphi_t \\ V_{DSsat} &= \varphi_t (\sqrt{1 + i_{f2}} + 3) \end{aligned}$$

$$V_{P4} = V_{S4} + \varphi_t \left[\sqrt{1 + i_{f4}} - 2 + \ln \left(\sqrt{1 + i_{f4}} - 1 \right) \right] \quad (I)$$

$$V_{P5} = \varphi_t \left[\sqrt{1 + i_{f5}} - 2 + \ln \left(\sqrt{1 + i_{f5}} - 1 \right) \right] \quad (II)$$

Equate (I) to (II) to find i_{f5}

Choose I_{S5} & I_{B5} to comply with $i_{f5} \approx \frac{I_{B5}}{I_{S5}}$

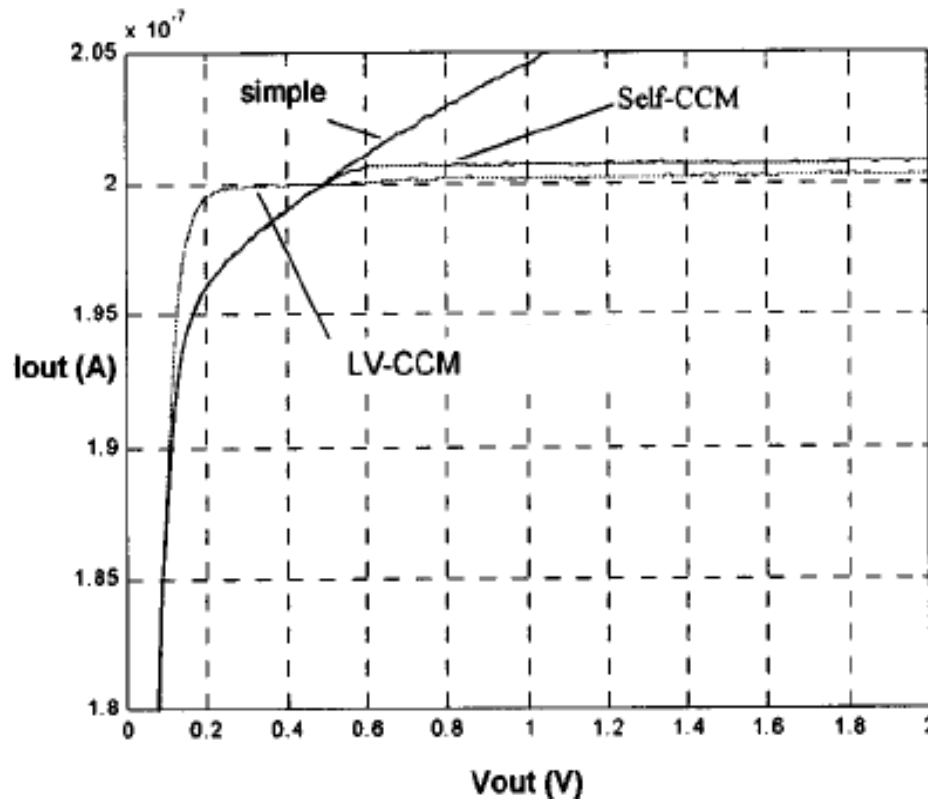
$$I_{S5} = I_{SH} \frac{W}{L}$$

$$y_o = \frac{i_{out}}{v_{out}} \approx g_{md2} \frac{g_{md4}}{g_{ms4}}$$

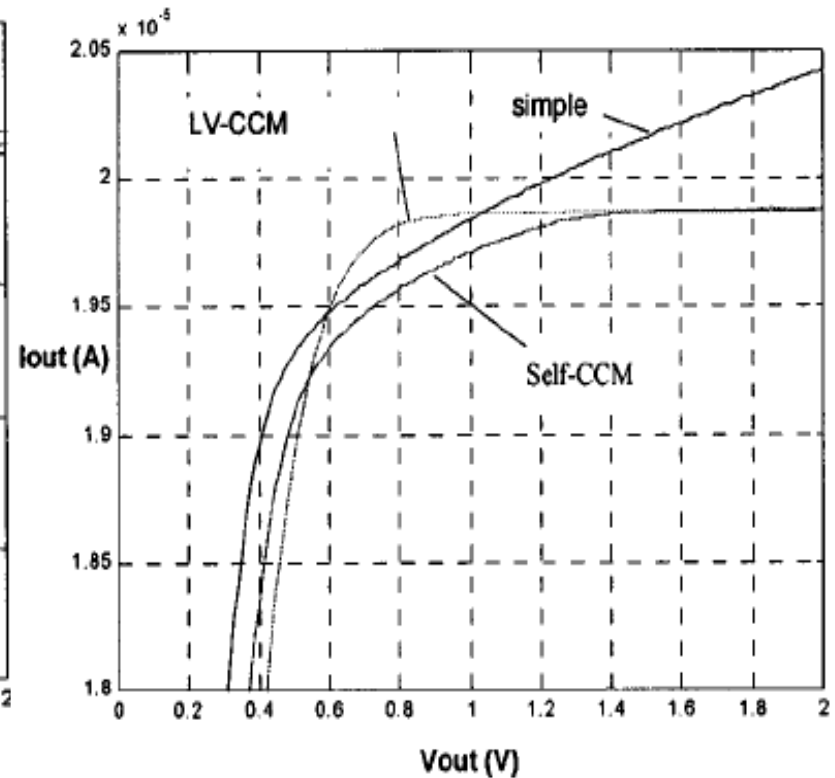
- Input constraint: $V_{DD} > V_{CS,min} + V_{DS3,sat} + V_{GS1}$
- Output constraint: $V_{out} > V_{DS2,sat} + \Delta V + V_{DS4,sat}$ for saturation of M_4

2. Cascode current mirrors

Low-voltage cascode current mirror - 3



($i_f = 1$).



($i_f = 100$).

Fig. 7. Detail of experimental output characteristics of the current mirrors

2 μ m CMOS technology

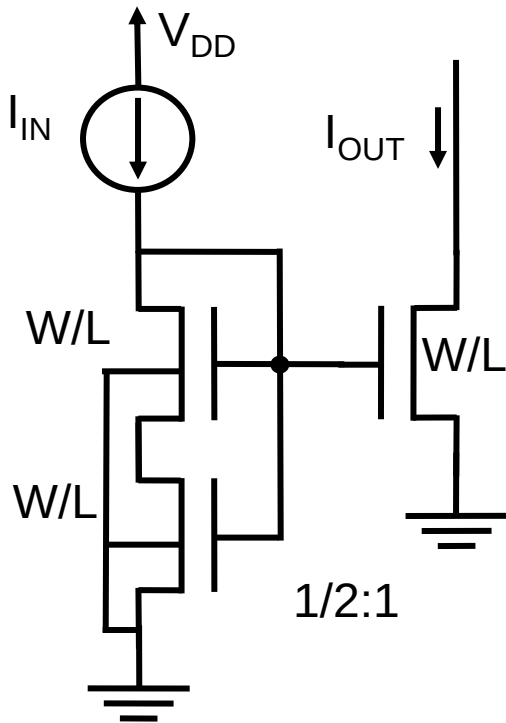
$V_{T0N} \approx 0.55$ V

• V. C. Vincence, C. Galup-Montoro and M. C. Schneider, "[A high-swing MOS cascode bias circuit](#)," *IEEE Trans. Circuits Syst. II*, vol. 47, no. 11, pp. 1325-1328, Nov. 2000.

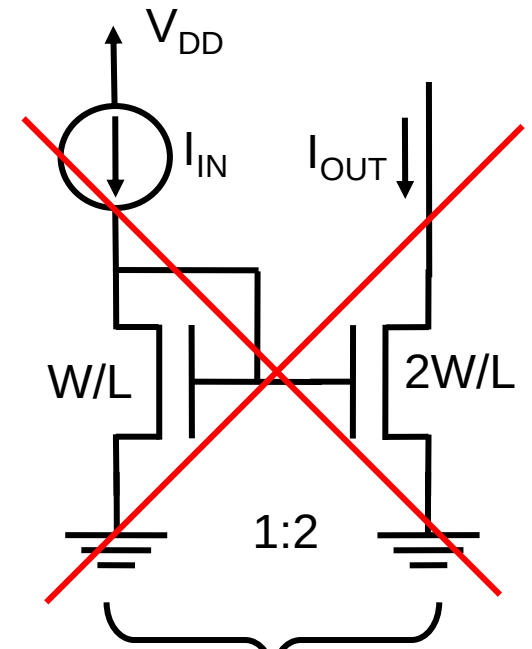
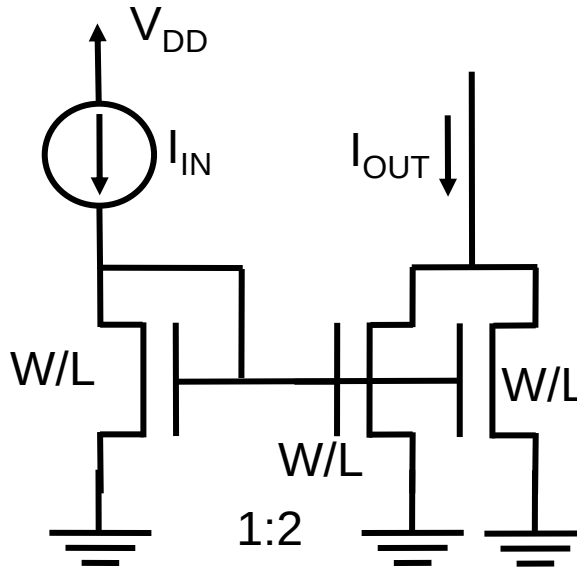
• P. Aguirre and F. Silveira, "Bias circuit design for low-voltage cascode transistors," *Proc. of SBCCI 2006*, pp. 94-98, Sep. 2006.

3. Current gain schemes

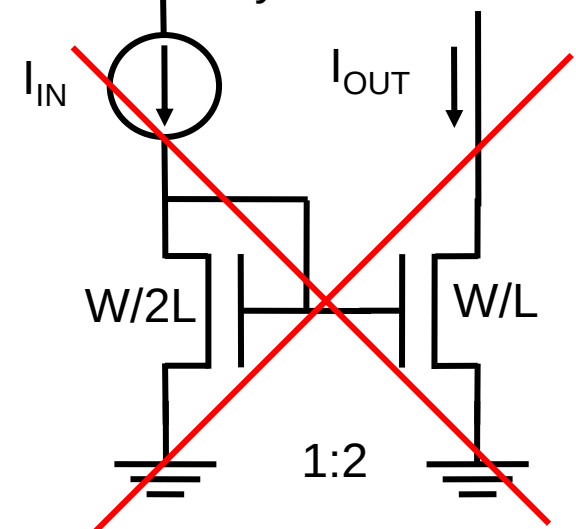
Gain-of-2 current mirror



Choose equally-sized components for good matching

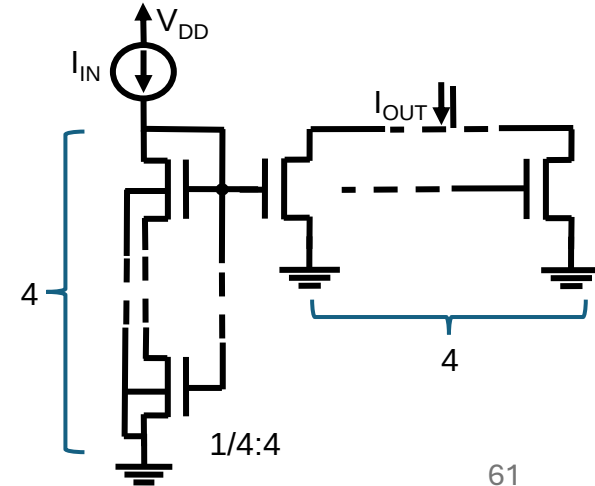
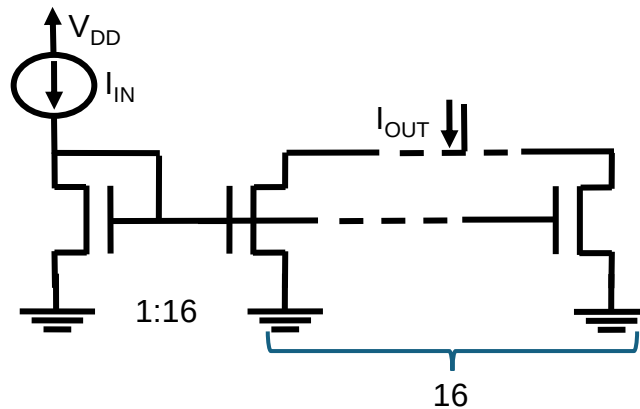
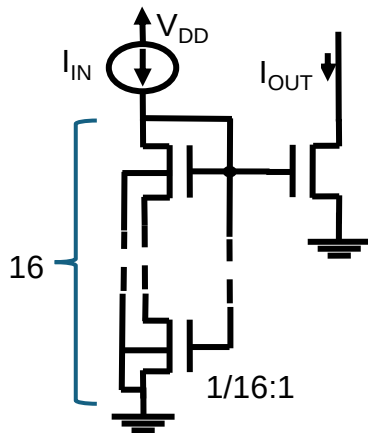
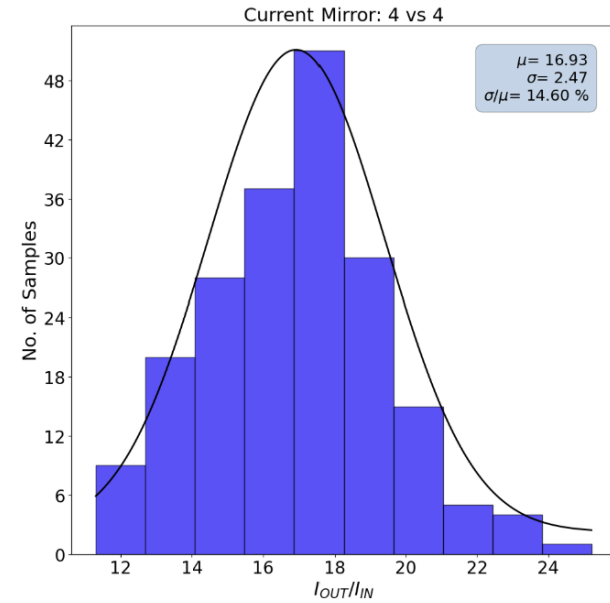
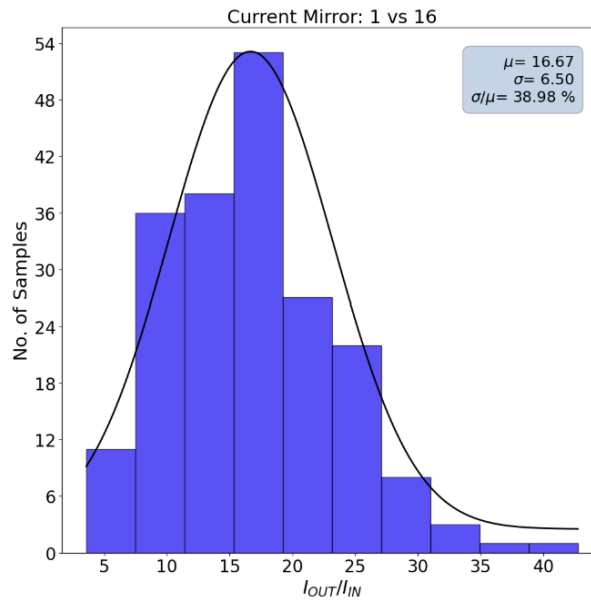
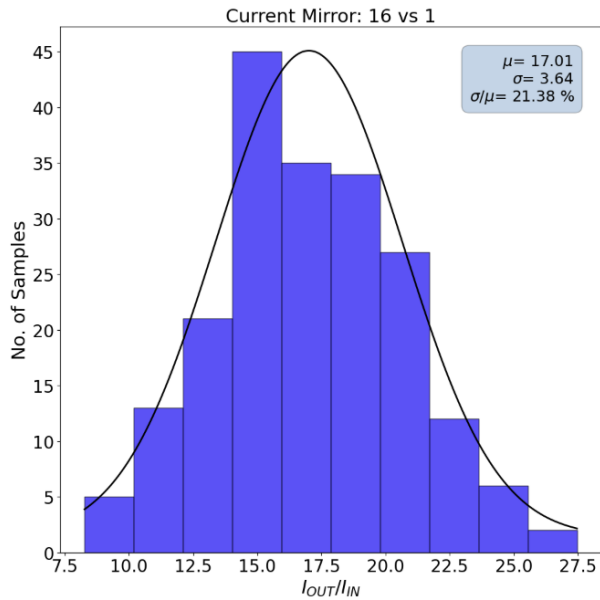


Why not?



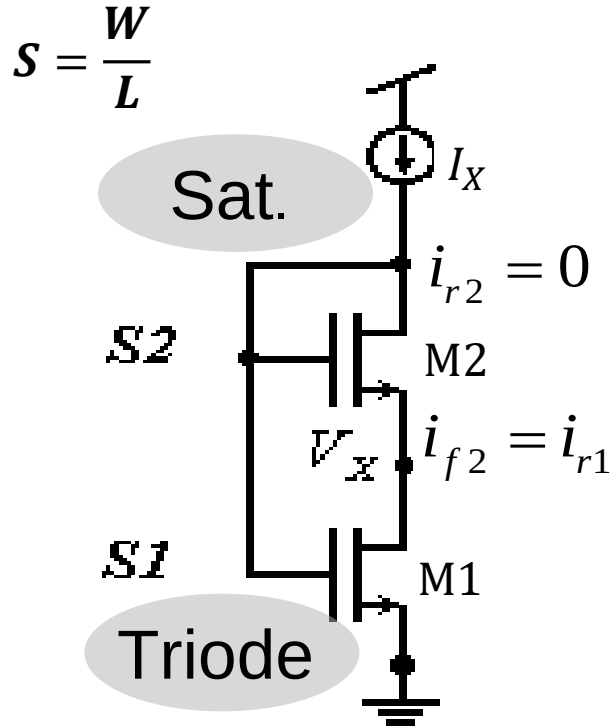
3. Current gain schemes

Example: Gain-of-16 current mirror



4. Current and voltage references

The self-cascode MOSFET (SCM)



$$I_X = I_{S2} i_{f2} = I_{S1} (i_{f1} - i_{f2})$$



$$i_{f1} = \left(1 + \frac{I_{S2}}{I_{S1}}\right) i_{f2} = \left(1 + \frac{S_2}{S_1}\right) i_{f2} = \alpha i_{f2}$$

Applying UICM to M1

$$\frac{V_X}{\phi_t} = \sqrt{1 + \alpha i_{f2}} - \sqrt{1 + i_{f2}} + \ln \left(\frac{\sqrt{1 + \alpha i_{f2}} - 1}{\sqrt{1 + i_{f2}} - 1} \right)$$

V_X is PTAT if i_{f2} is independent of temperature

In other words, V_X is PTAT if i_{f2} is a scaled replica of I_S

In weak inversion,

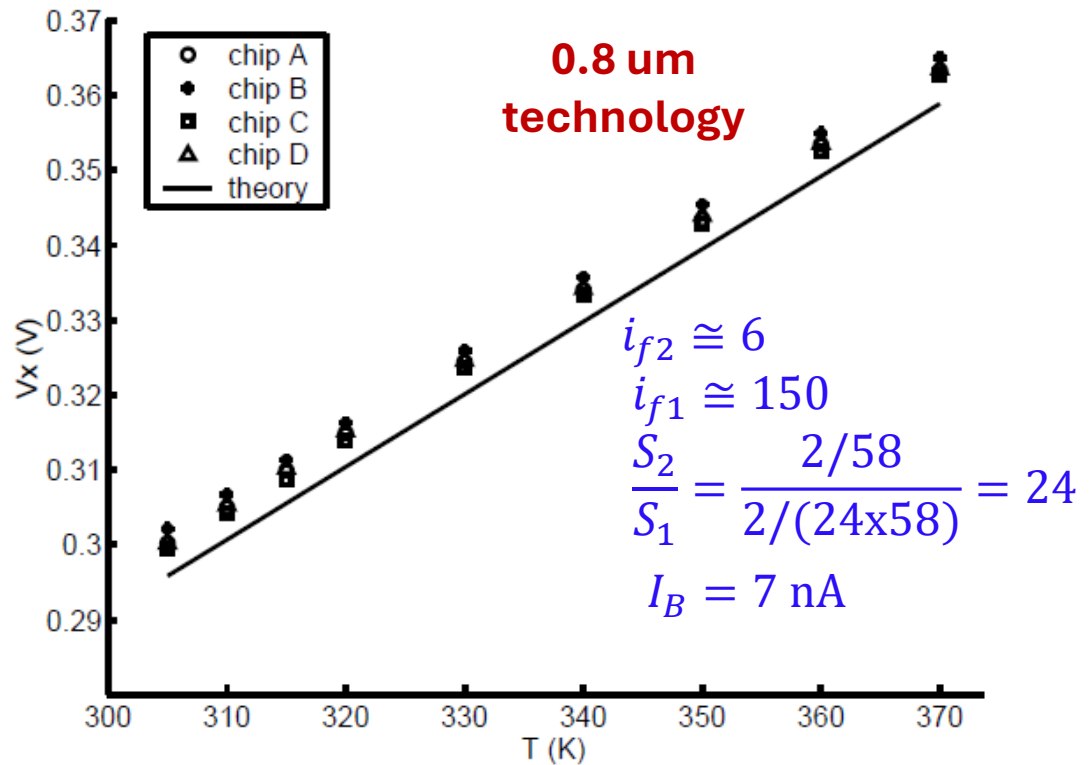
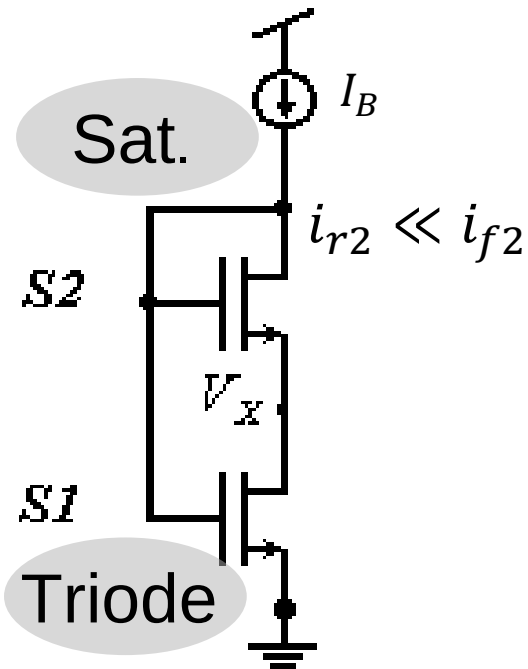
$$i_{f1} = \alpha i_{f2} \ll 1$$

$$\frac{V_X}{\phi_t} \rightarrow \ln \alpha$$

$\Rightarrow V_X$ is PTAT and independent of the current coefficient

4. Current and voltage references

The self-cascode MOSFET (SCM)



$$I = I_{S2} i_{f2} = I_{S1} (i_{f1} - i_{f2})$$

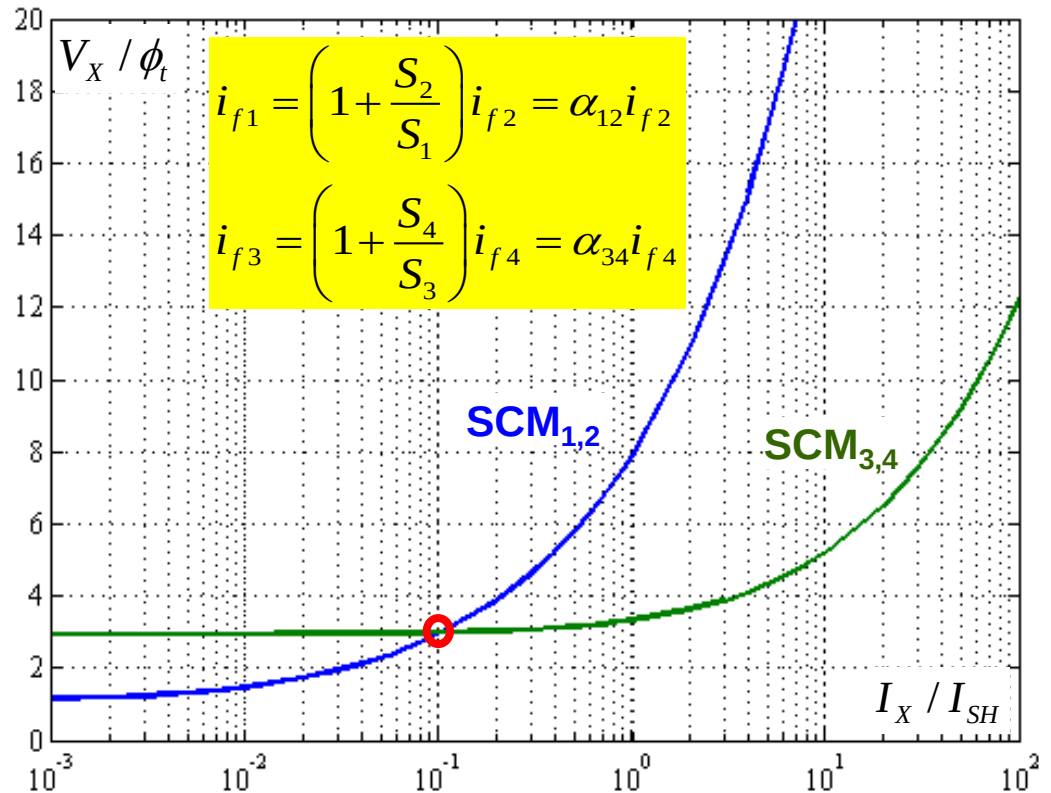
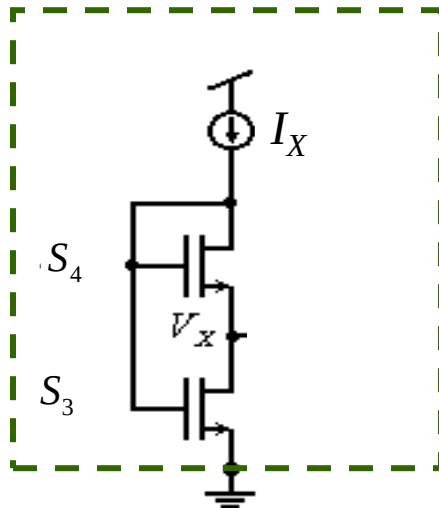
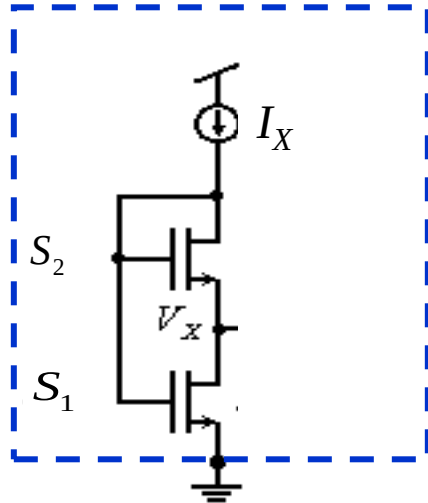
$$i_{f1} = \left(1 + \frac{I_{S2}}{I_{S1}}\right) i_{f2} = \left(1 + \frac{S_2}{S_1}\right) i_{f2} = \alpha i_{f2}$$

Applying
UICM to M1 $\Rightarrow$

$$\frac{V_X}{\phi_t} = \sqrt{1 + \alpha i_{f2}} - \sqrt{1 + i_{f2}} + \ln \left(\frac{\sqrt{1 + \alpha i_{f2}} - 1}{\sqrt{1 + i_{f2}} - 1} \right)$$

4. Current and voltage references

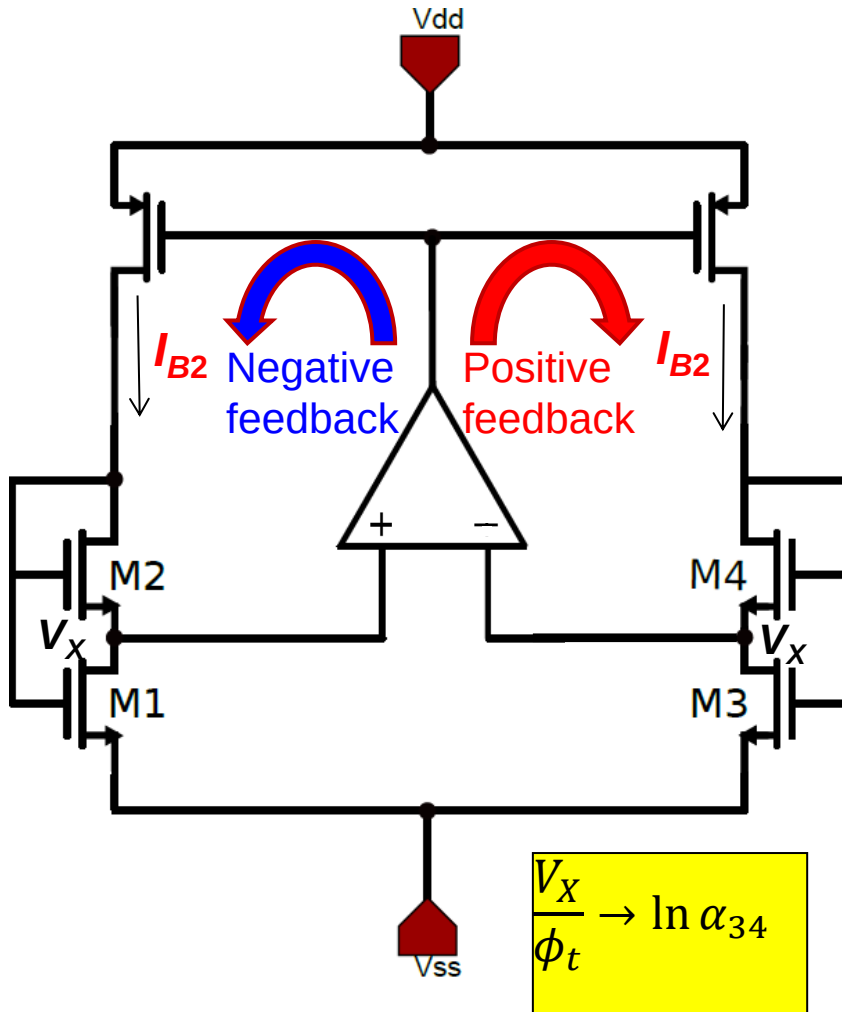
The self-cascode MOSFET (SCM)



$$\frac{V_X}{\phi_t} = \sqrt{1 + \alpha \frac{I_X}{SI_{SH}}} - \sqrt{1 + \frac{I_X}{SI_{SH}}} + \ln \left(\frac{\sqrt{1 + \alpha \frac{I_X}{SI_{SH}}} - 1}{\sqrt{1 + \frac{I_X}{SI_{SH}}} - 1} \right)$$

4. Current and voltage references

A self-biased current source (SBCS)



If M_3 and M_4 operate in WI :

$$\ln \alpha_{34} = \sqrt{1 + \alpha_{12} i_{f2}} - \sqrt{1 + i_{f2}} + \ln \left(\frac{\sqrt{1 + \alpha_{12} i_{f2}} - 1}{\sqrt{1 + i_{f2}} - 1} \right);$$

$$\alpha_{12} = 1 + \frac{S_2}{S_1} \quad \alpha_{34} = 1 + \frac{S_4}{S_3}$$

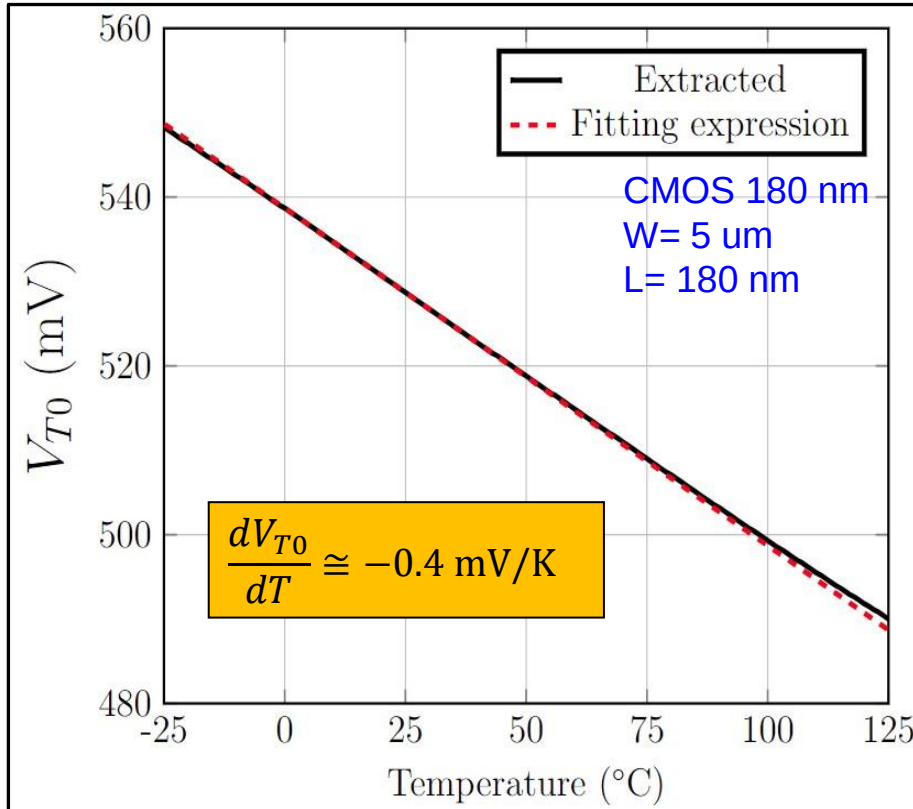
Thus, i_{f2} (the inversion level of M_2) is constant (it depends only on the geometrical ratios α_{12} and α_{34})

The reference current $I_{B2} = I_{S2}i_{f2}$ is proportional to the specific current of M_2 . $I_{S2} = I_{SH}(W/L)_2 = I_{SH}S_2$

- **Stability at the operating point:**
Negative feedback > positive feedback

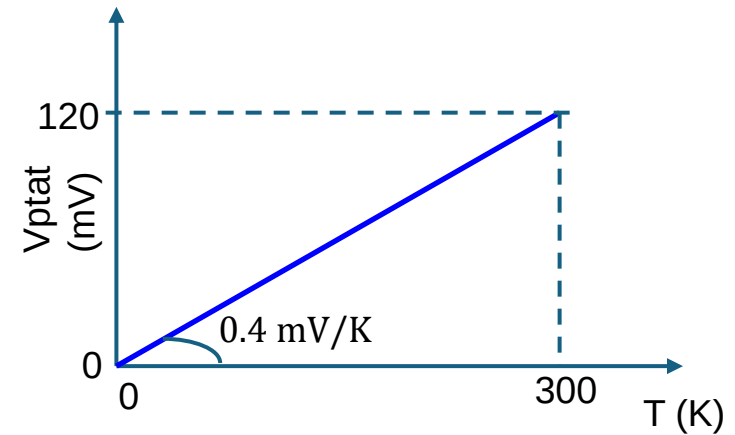
4. Current and voltage references

MOS-only voltage reference



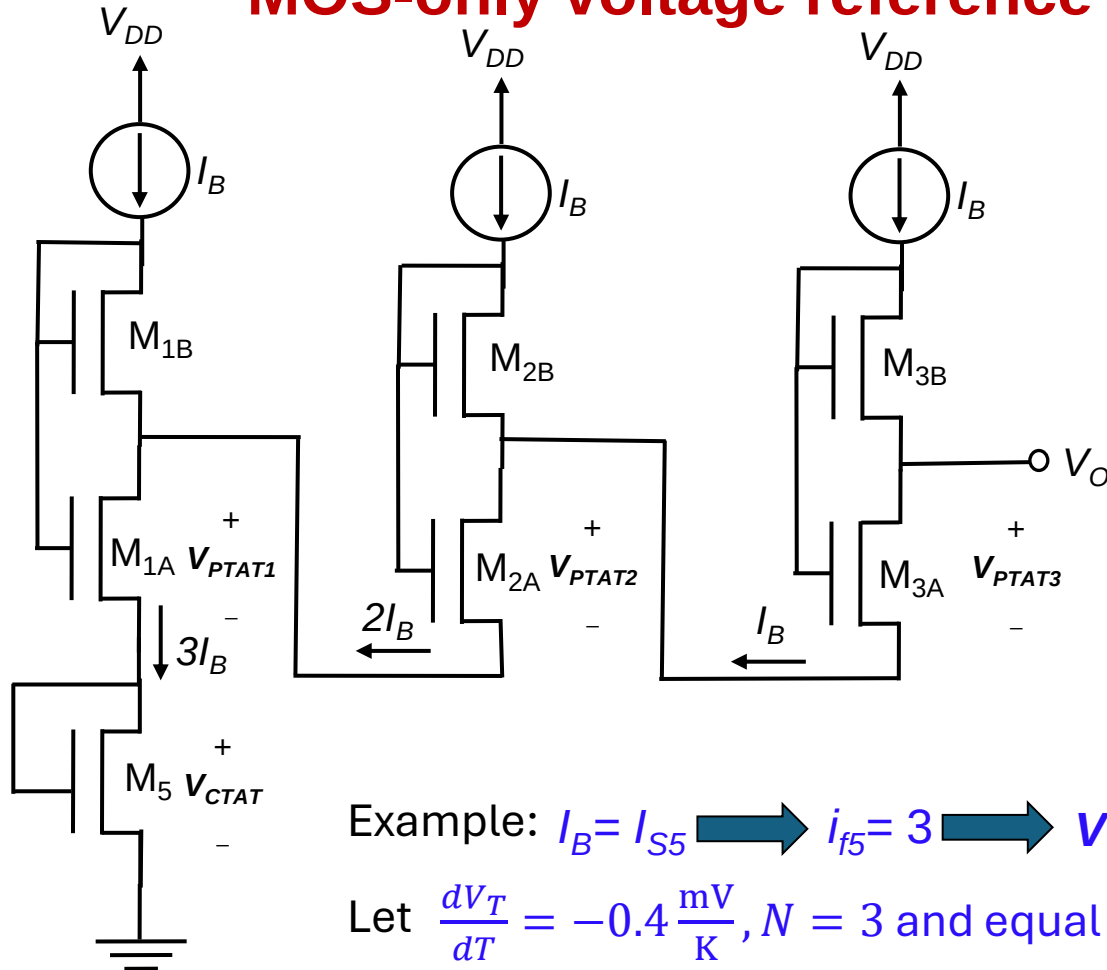
Temperature dependence of the threshold voltage (CTAT)

+



4. Current and voltage references

MOS-only voltage reference



$$V_O = V_{CTAT} + \sum_{i=1}^N V_{PTATi}$$

	W/L ($\mu\text{m}/\mu\text{m}$)	i_f	i_r
M5	1/1	3	$\ll 3$
M1A	20/1	19/100	1/25
M1B	25/1	1/25	$\ll 1/25$
M2A	20/1	4.7/37	1/37
M2B	37/1	1/37	$\ll 1/37$
M3A	20/1	4.7/74	1/74
M3B	74/1	1/74	$\ll 1/74$

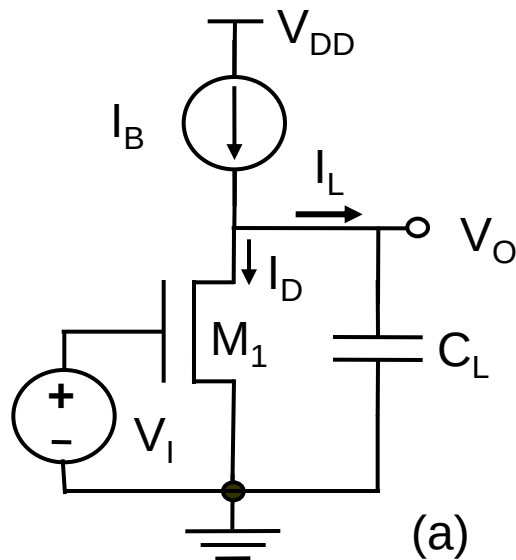
$$\frac{dV_O}{dT} = 0 \rightarrow 3 \frac{dV_{PTAT}}{dT} = -\frac{dV_{CTAT}}{dT}$$

$$\frac{dV_O}{dT} = 0 \rightarrow \frac{dV_{PTAT}}{dT} = 133 \mu\text{V/K}$$

5. Single-stage amplifiers

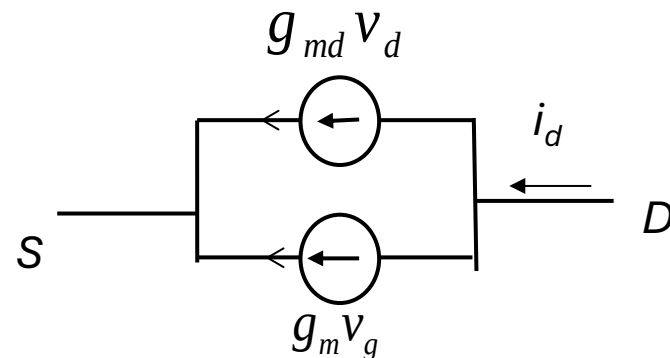
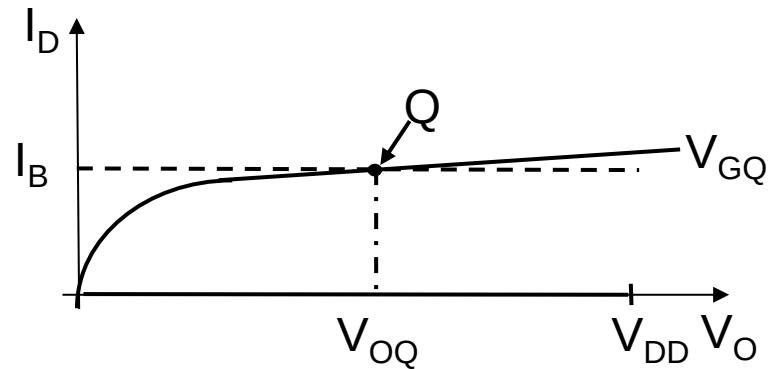
Common-source amplifier

The intrinsic gain



$$A_V = \frac{-g_m}{g_{md}}$$

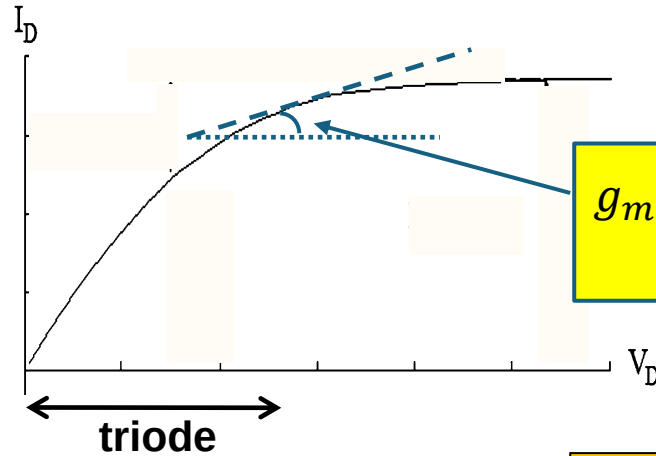
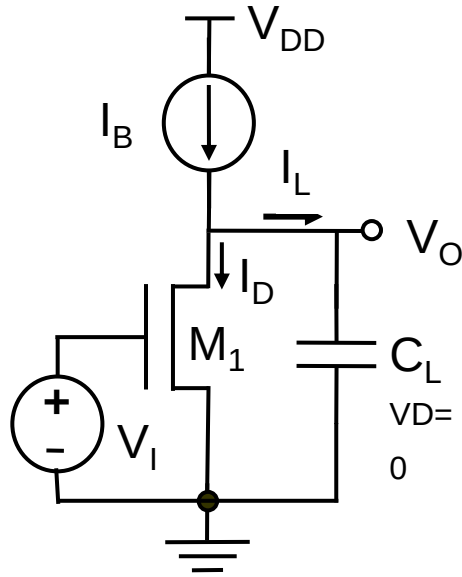
The gain is a function of technology, transistor channel length, bias current, and output voltage



5. Single-stage amplifiers

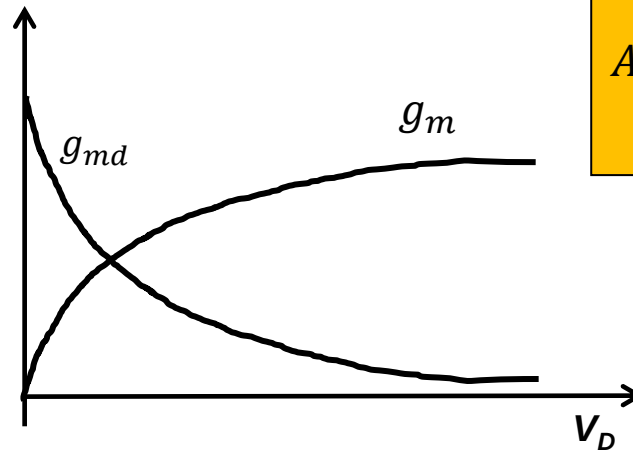
Common-source amplifier

Velocity saturation
not accounted for :



$$g_m = \frac{2I_S}{n\phi_t} (q_s - q_d)$$

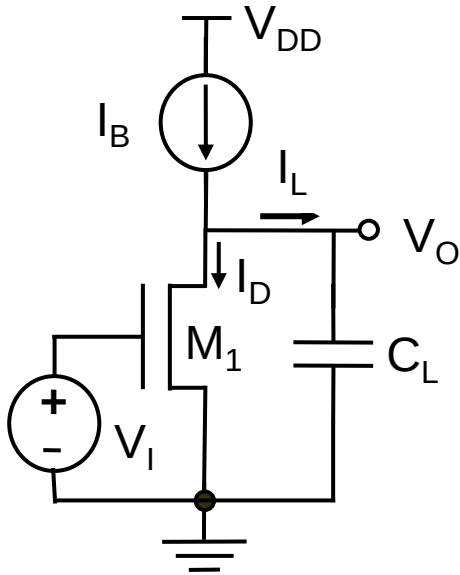
$$g_{md} = \frac{2I_S}{n\phi_t} (q_s - q_d)\sigma + \frac{2I_S}{\phi_t} q_d$$



$$A_V = -\frac{g_m}{g_{md}} = -\frac{\frac{q_s}{q_d} - 1}{\sigma \left(\frac{q_s}{q_d} - 1 \right) + n}$$

5. Single-stage amplifiers

Common-source amplifier

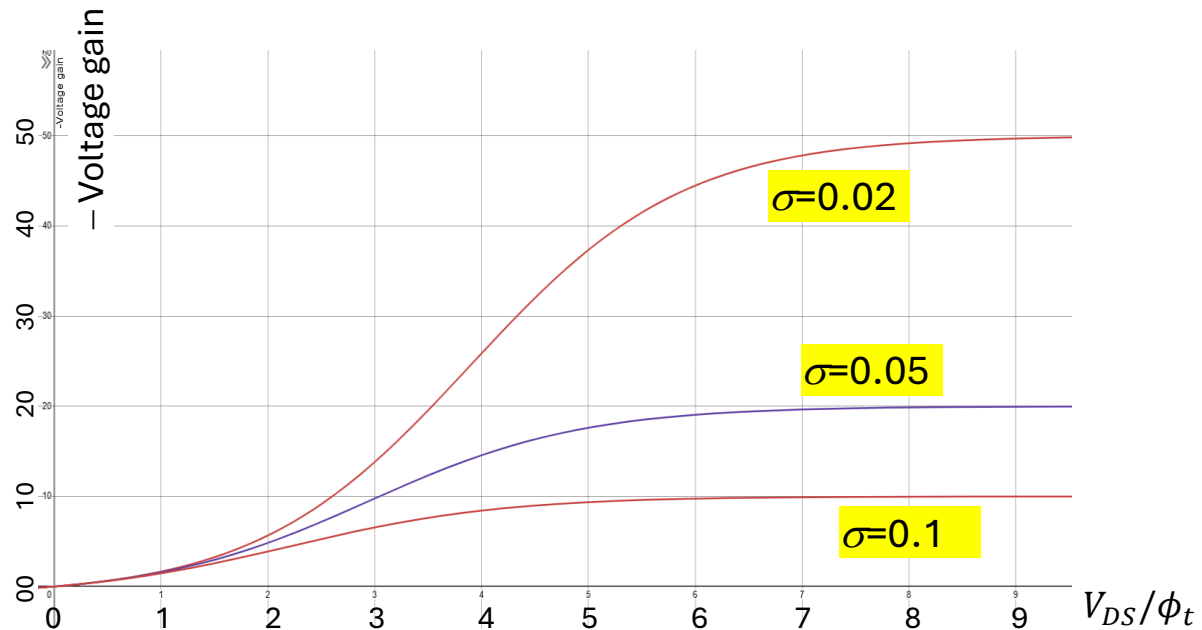


$$A_V = -\frac{g_m}{g_{md}} = -\frac{\frac{q_s}{q_d} - 1}{\sigma \left(\frac{q_s}{q_d} - 1 \right) + n}$$

Weak inversion

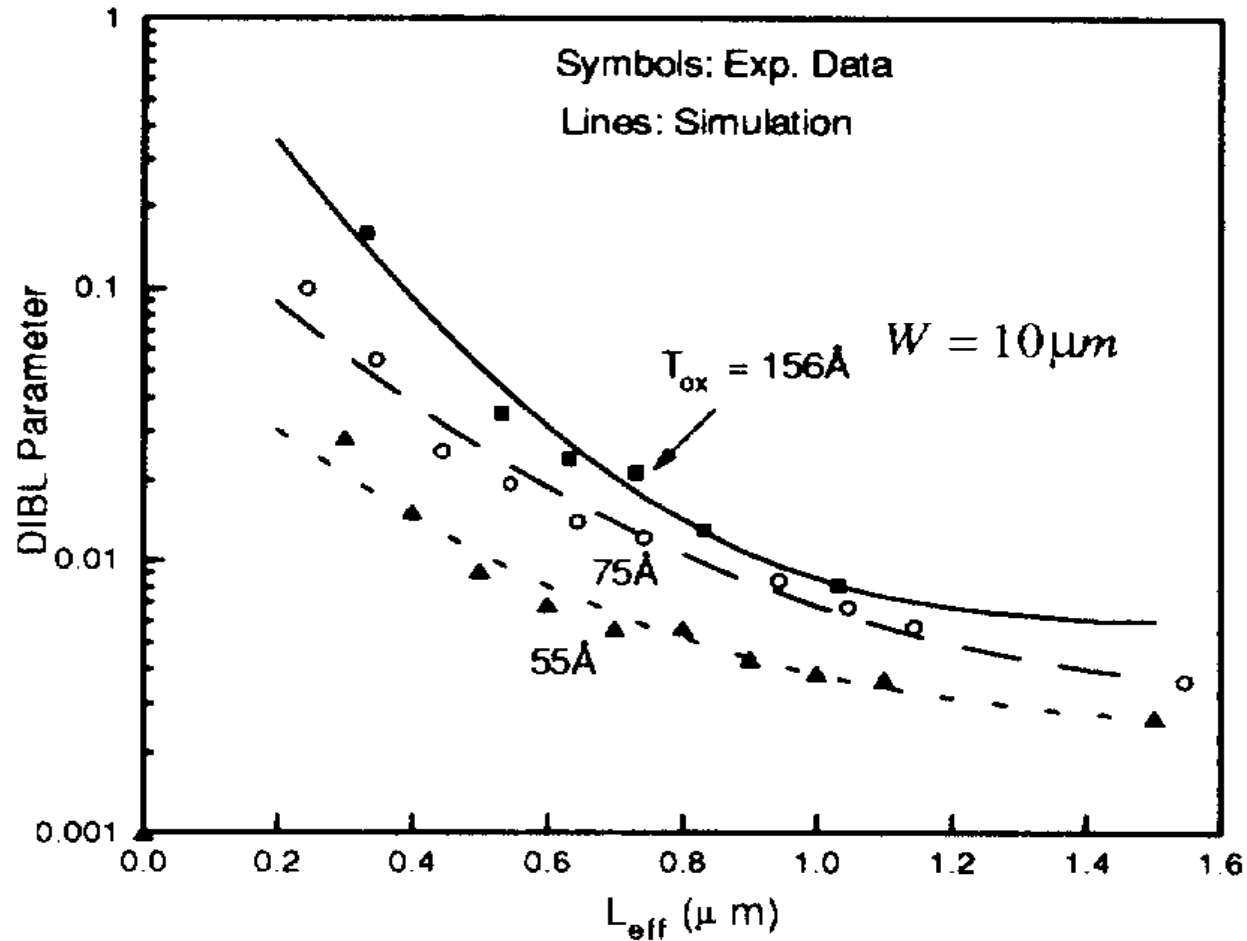


$$A_V = -\frac{e^{\frac{V_{DS}}{\phi_t}} - 1}{\sigma \left(e^{\frac{V_{DS}}{\phi_t}} - 1 \right) + n}$$



DIBL coefficient versus channel length

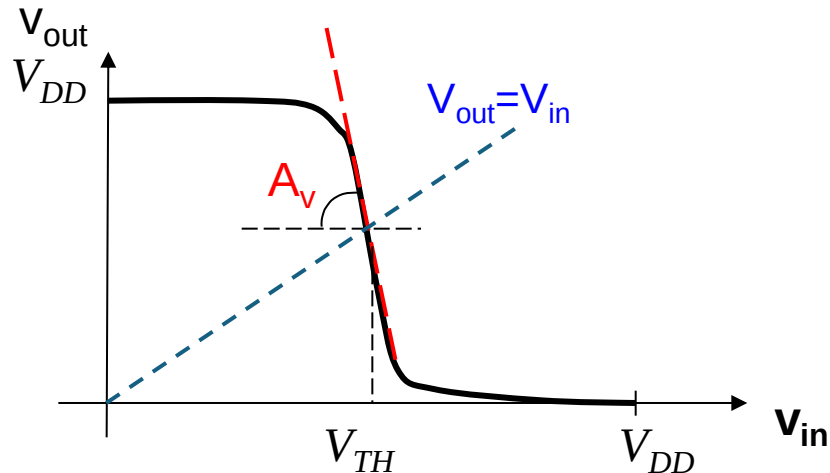
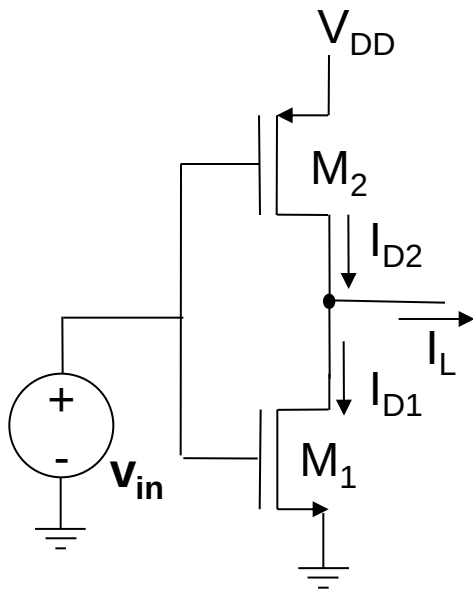
σ : dibl parameter



J. H. Huang et al., A physical model for MOSFET output resistance, IEDM 1992

5. Single-stage amplifiers

Push-pull amplifier (static CMOS inverter)



Voltage gain: $A_V = -\frac{g_{m1} + g_{m2}}{g_{md1} + g_{md2}}$ $g_{m1(2)} = \frac{I_{TH}}{n_{1(2)}\phi_t} \frac{2}{1 + \sqrt{1 + i_{f1(2)}}}$ $g_{md1(2)} = \sigma_{1(2)}g_{m1(2)}$

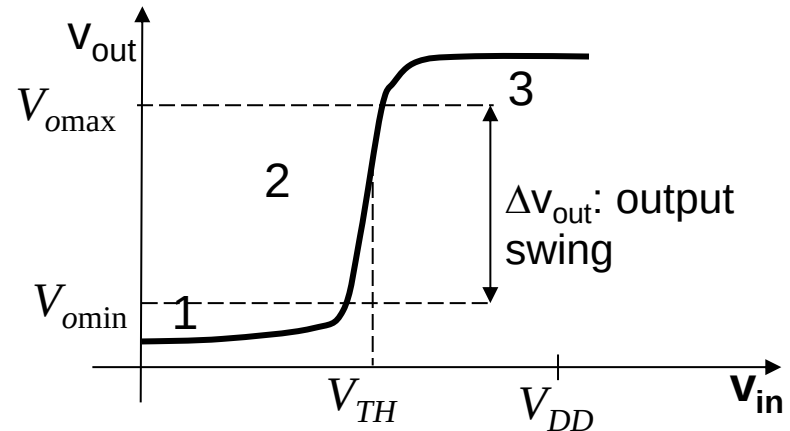
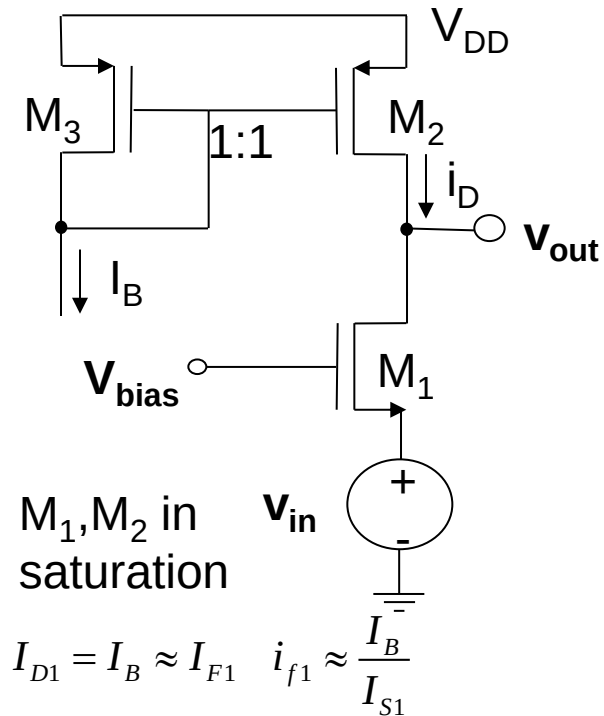
For “deep” saturation:

$$\frac{1}{A_V} = -\left(\frac{g_{m1}\sigma_1}{g_{m1} + g_{m2}} + \frac{g_{m2}\sigma_2}{g_{m1} + g_{m2}}\right)$$

$$A_V = -\frac{2}{\sigma_1 + \sigma_2} \text{ for } g_{m1} = g_{m2}$$

5. Single-stage amplifiers

Common-gate amplifier

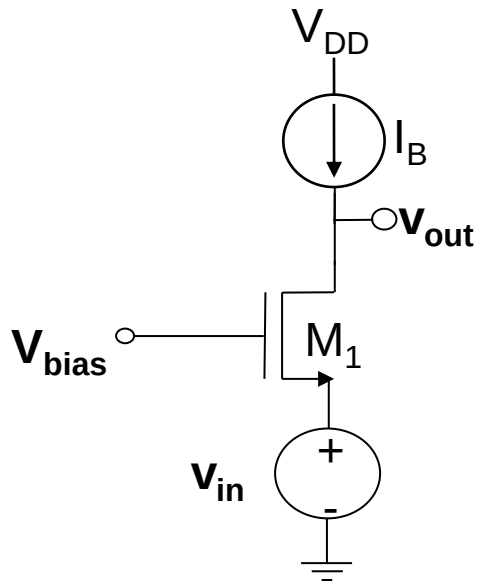


UICM

$$V_{TH} = \frac{V_{bias} - V_{T0N}}{n_N} - \phi_t \left[\sqrt{1 + i_{f1}} - 2 + \ln(\sqrt{1 + i_{f1}} - 1) \right]$$

5. Single-stage amplifiers

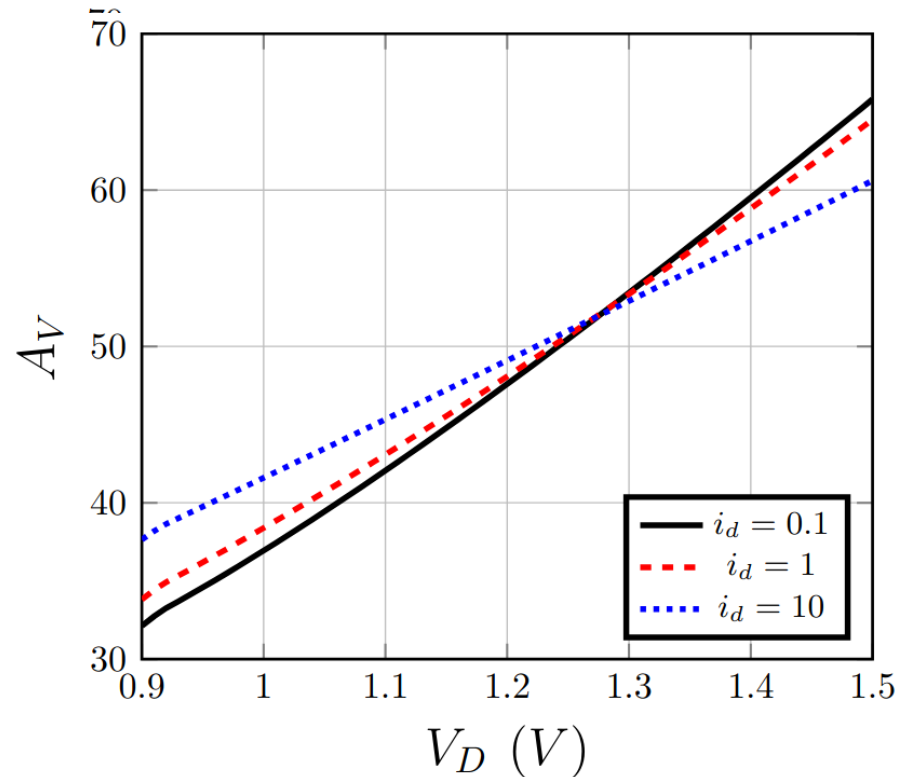
Common-gate amplifier



Voltage gain – weak inversion

$$A_V = \frac{g_{ms}}{g_{md}} = \frac{\frac{1}{\frac{-V_{DS}}{\phi_t}} - \frac{\sigma}{n}}{\frac{\sigma}{n} + \frac{1}{\frac{V_{DS}}{\phi_t} - 1}}$$

$$\frac{V_{DS}}{e \phi_t} \gg 1 \Rightarrow A_V = \frac{g_{ms}}{g_{md}} = \frac{n}{\sigma} - 1$$



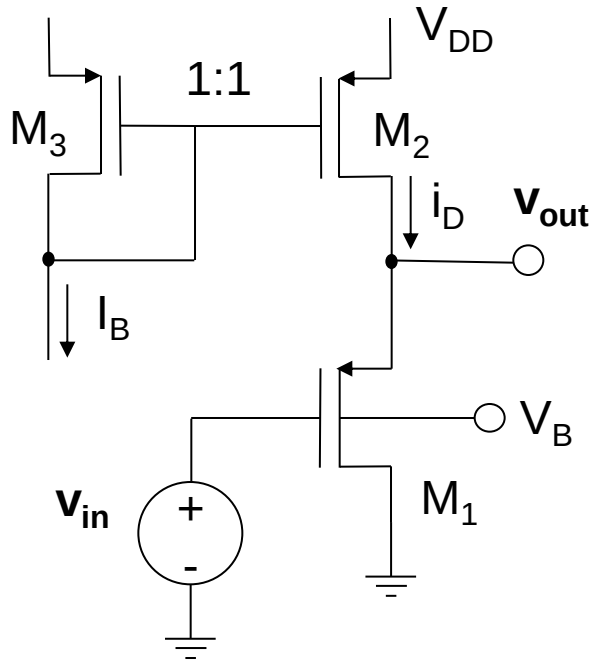
$I_B \approx 0.5 \text{ uA}, 5 \text{ uA}, 50 \text{ uA}$

For $n=1.25$,
 $\sigma = 0.025$

$$A_V = \frac{n}{\sigma} - 1 = 49$$

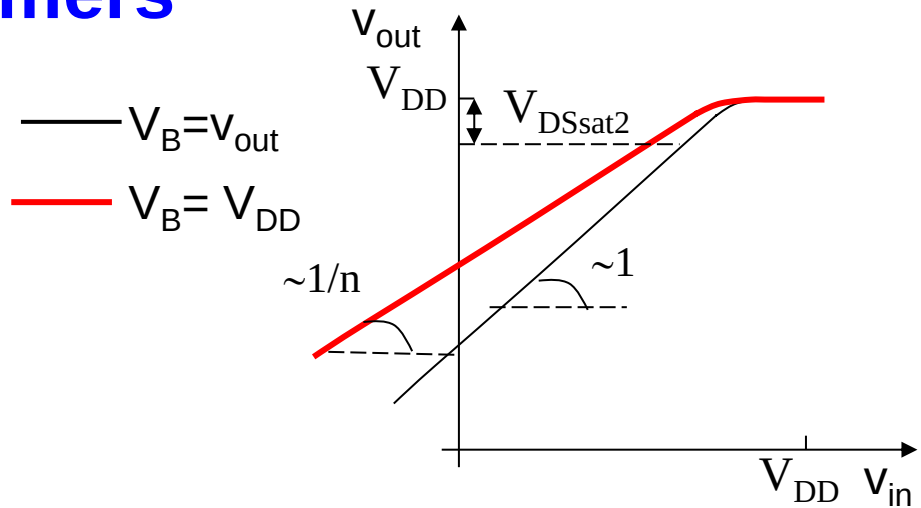
5. Single-stage amplifiers

Source follower



$$V_{out,max} = V_{DD} - |V_{DS,sat2}|$$

$$V_{out,min} = |V_{DS,sat1}|$$



Weak inversion
Saturation

$$V_B = V_{out}$$

$$\frac{v_{out}}{v_{in}} = \frac{1}{1 + \frac{g_{md1} + g_{md2}}{g_{m1}}} = \frac{1}{1 + \sigma_1 + \frac{g_{m2}}{g_{m1}} \sigma_2}$$

$$V_B = V_{DD}$$

$$\frac{v_{out}}{v_{in}} = \frac{1}{1 + \frac{g_{mb1} + g_{md1} + g_{md2}}{g_{m1}}} = \frac{1}{n + \sigma_1 + \frac{g_{m2}}{g_{m1}} \sigma_2}$$

**THANKS FOR ATTENDING THE
ANALOG SECTION OF THE
TUTORIAL**

QUESTIONS?

Part 4

Advanced Compact MOSFET Model: Design Methodology-Application to Low Noise Amplifier



"Scan me"

Sylvain Bourdel

Agenda

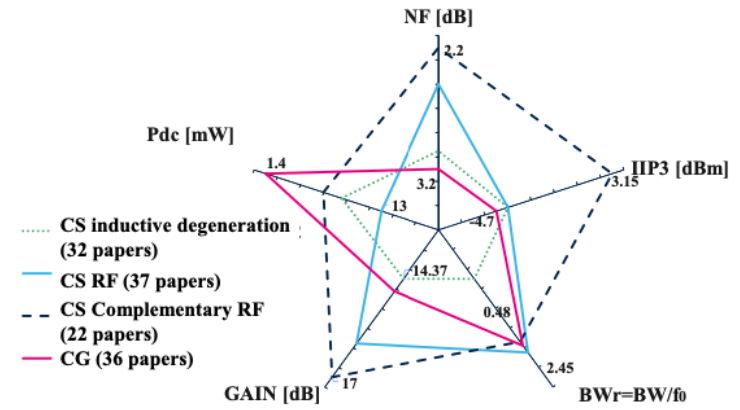
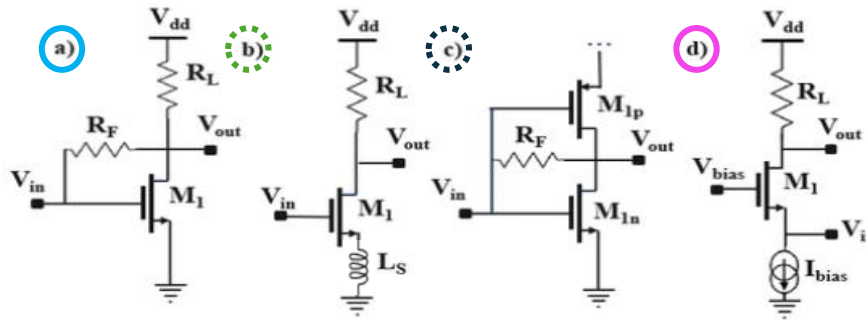
- Overview of Design Methods for RFIC
- LNA Design Considerations
- Resistive Feedback LNA
- What we need in ACM-2
- Inversion Level Based Method for R-Feedback LNA with ACM-2

Agenda

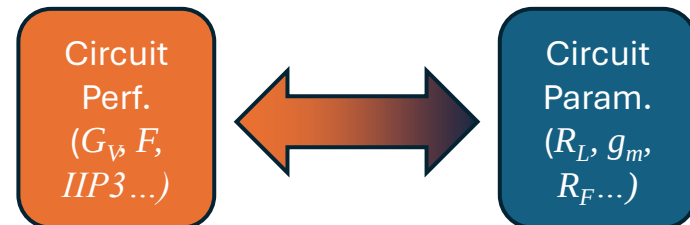
- Overview of Design Methods for RFIC
- LNA Design Considerations
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- What we need in ACM-2
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Overview of Design Methods for RFIC General Principle

• 1st Step: Architecture Choice



• 2nd Step: Circuit Analysis

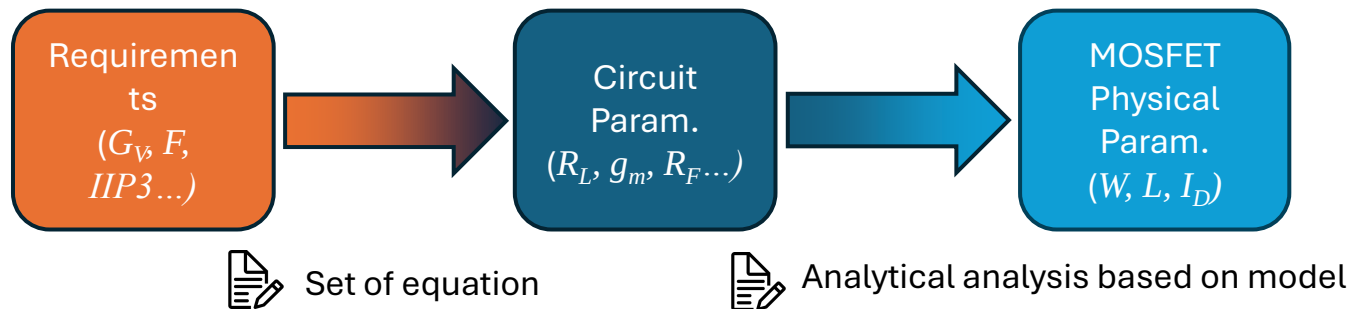


$$|G_T| = |G_v|Q_{IN} = \frac{(G_m R_F - 1)R_O}{(R_O + R_F)} \sqrt{1 + Q_P^2}$$

Overview of Design Methods for RFIC

General Principle

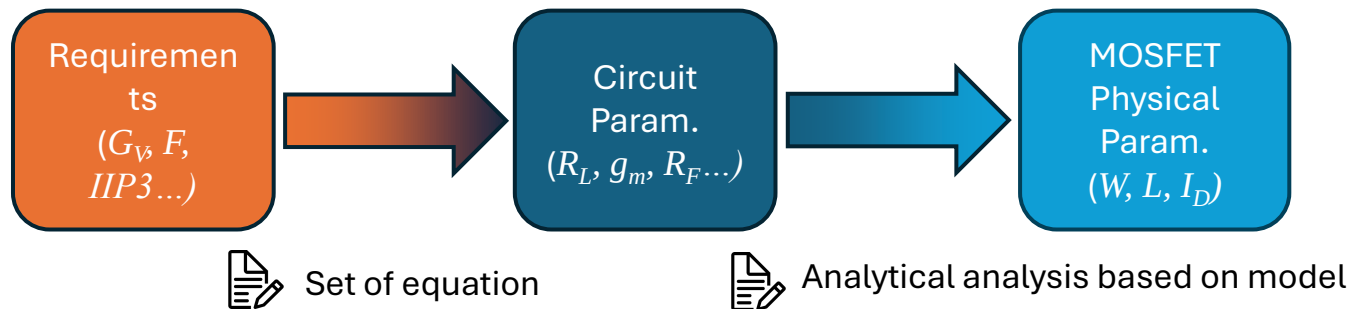
- 3rd Step: Circuit Sizing



Overview of Design Methods for RFIC

General Principle

- 3rd Step: Circuit Sizing



Region based model

$$sat. \quad I_d = 2.K_n \frac{W}{L} \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

$$lin. \quad I_d = K_n \frac{W}{L} (V_{gs} - V_t)^2 \left(1 + \frac{k_{en}}{L} V_{ds} \right)$$

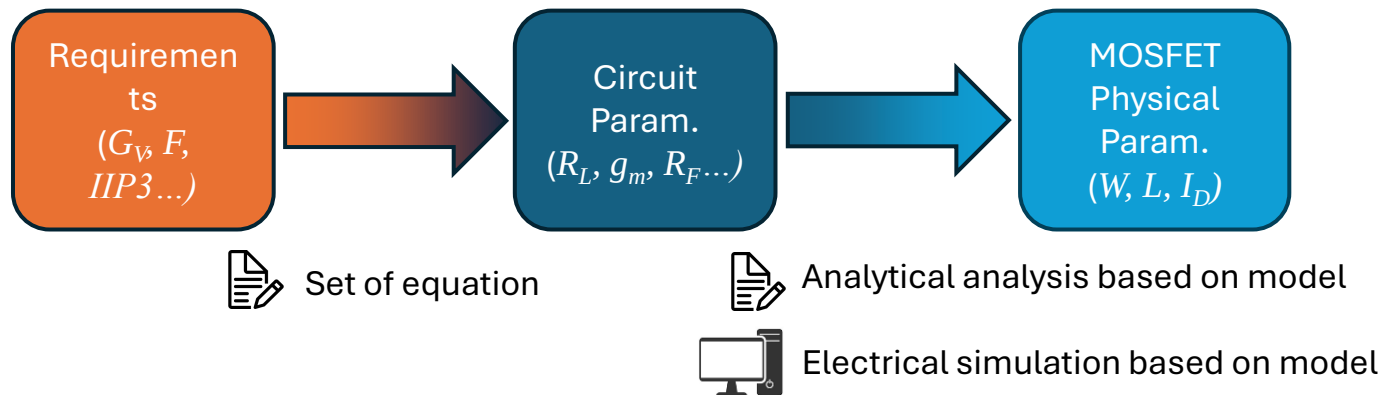
PROS: simple

CONS: Inaccurate with short channel MOS

Overview of Design Methods for RFIC

General Principle

- 3rd Step: Circuit Sizing



BSIM3 / UTSOI compact model

PROS:

- *Accurate*
- *Direct relationship between requirements and physical parameters*

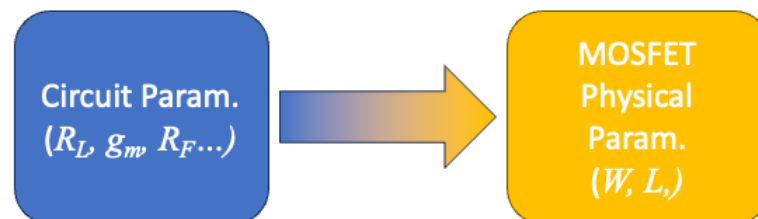
CONS:

- *Increases the gap between Physic and design*
- *Long optimization*

Overview of Design Methods for RFIC

New Trends

- **Issue in the sizing step**



- How to maintain simplicity **and** accuracy with the scaling and especially Short Channel Effects (SCE) at the sizing step.

- **A possible solution**

- g_m/I_d design approaches
 - LUT based
 - ACM or EKV based

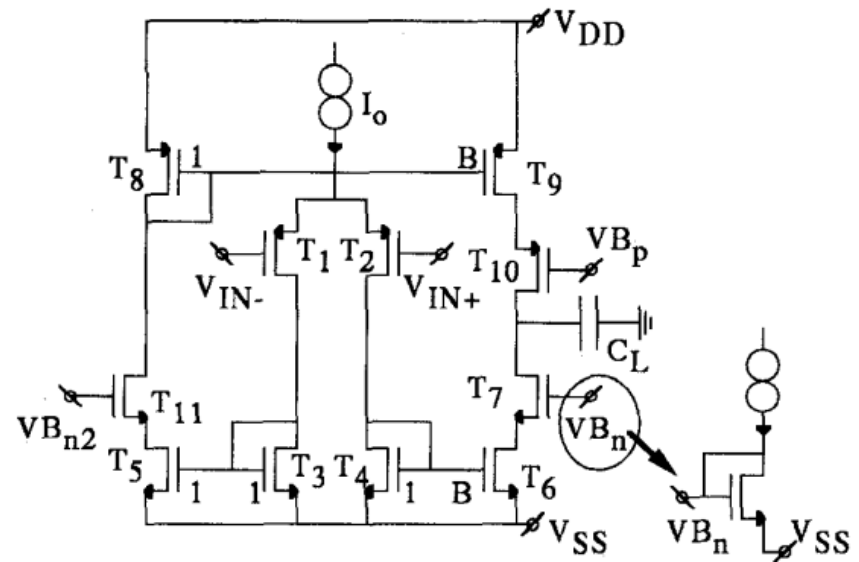
Overview of Design Methods for RFIC

g_m/I_d approach based on LUT

[Silveira, Flandre, Jespers – JSSC 1996]

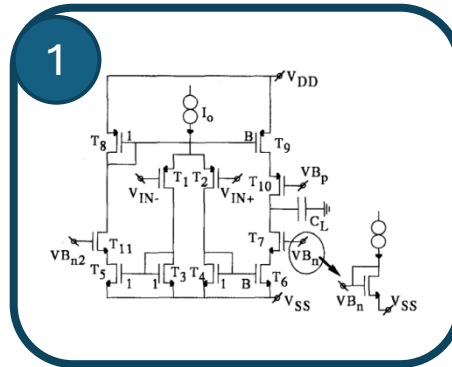
General Principle

- For a given structure

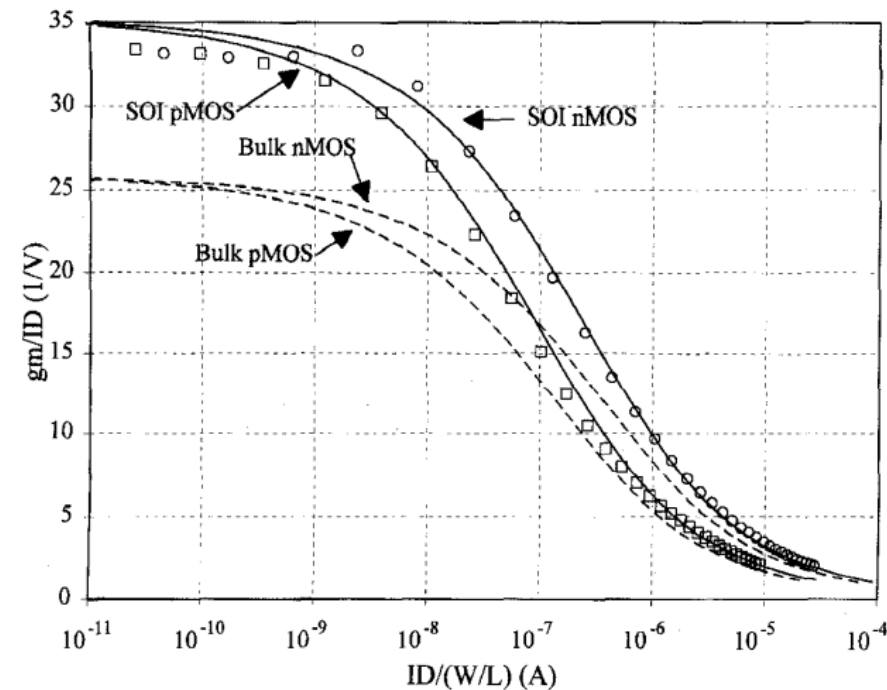


Overview of Design Methods for RFIC g_m/I_D approach based on LUT

General Principle



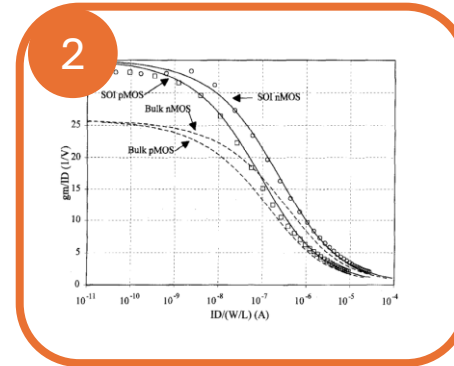
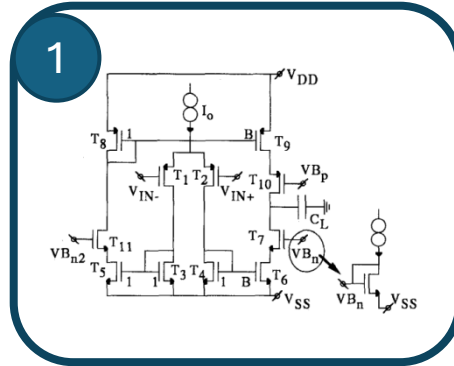
- For a given structure
- Methods are based on
 - ⇒ Extraction of MOS parameters (LUT)
 - $I_D/(W/L)$ vs g_m/I_D
 - g_m/I_D vs V_{gs0}



Overview of Design Methods for RFIC

g_m/I_D approach based on LUT

General Principle



- For a given structure
- Methods are based on
 - ⇒ Extraction of MOS parameters (LUT)
 - $I_D/(W/L)$ vs g_m/I_D
 - g_m/I_D vs V_{gs0}

⇒ Set of equations

$$SR = \frac{B \cdot I_{D1}}{C_L}$$

$$f_T = \frac{B \cdot g_{m1}}{2\pi \cdot C_L} \quad (5)$$

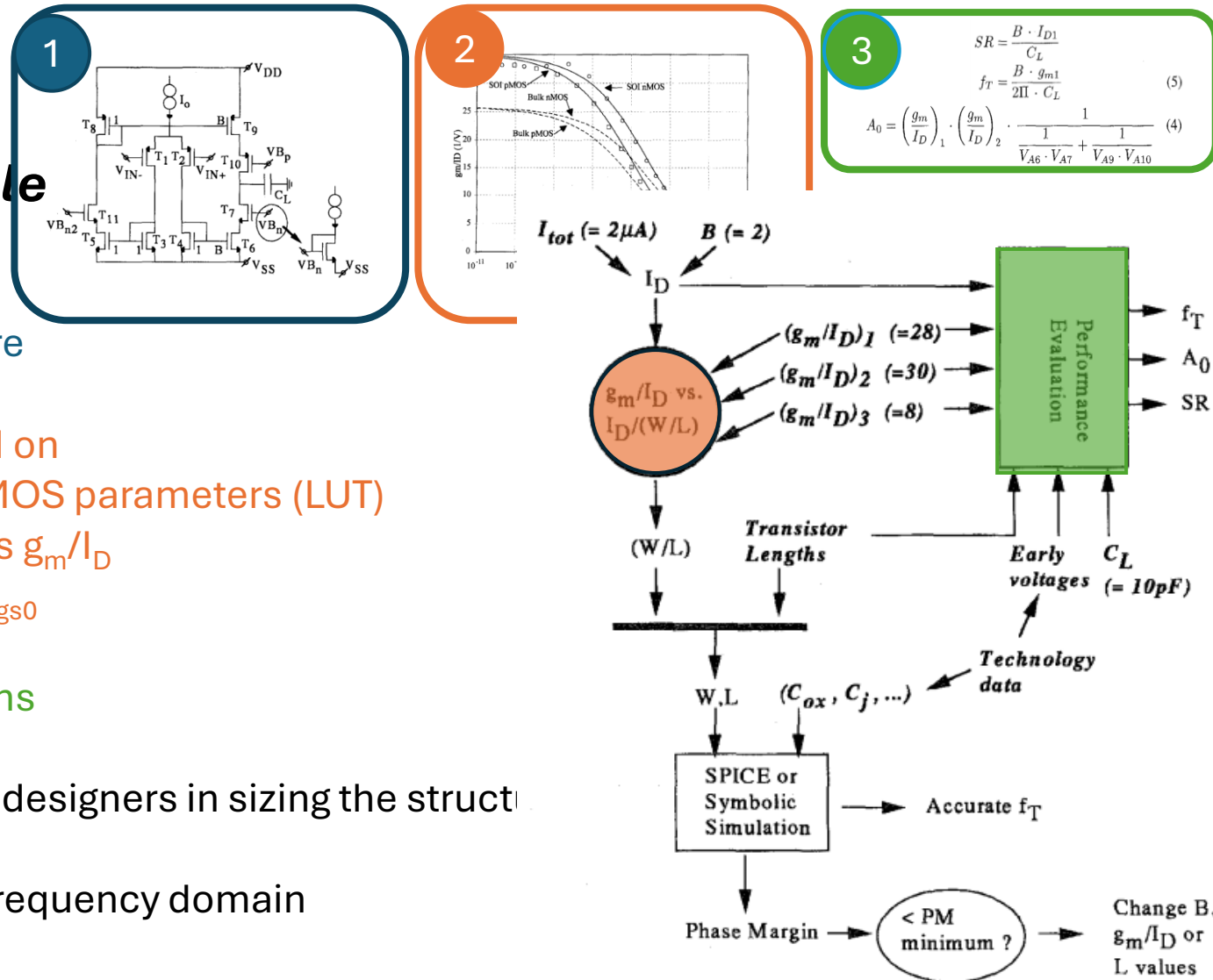
$$A_0 = \left(\frac{g_m}{I_D} \right)_1 \cdot \left(\frac{g_m}{I_D} \right)_2 \cdot \frac{1}{\frac{1}{V_{A6} \cdot V_{A7}} + \frac{1}{V_{A9} \cdot V_{A10}}} \quad (4)$$

Overview of Design Methods for RFIC

g_m/I_D approach based on LUT

General Principle

- For a given structure
- Methods are based on
 - Extraction of MOS parameters (LUT)
 - $I_D/(W/L)$ vs g_m/I_D
 - g_m/I_D vs V_{gs0}
 - Set of equations
- Methods helps the designers in sizing the structure
- Quite used in low frequency domain



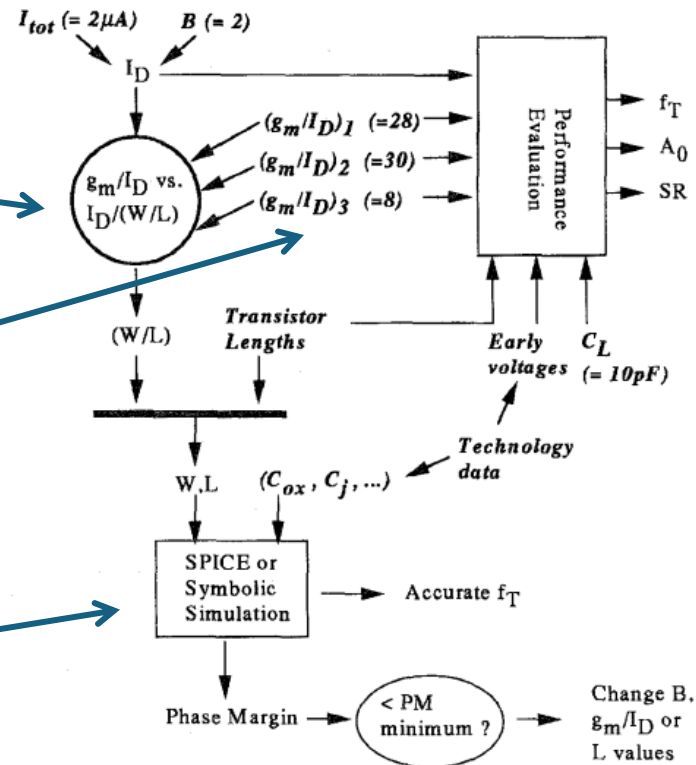
Overview of Design Methods for RFIC

g_m/I_D approach based on LUT

LIMITATIONS ?

- Time Consuming
- Lack of precision (parametric set of characteristics)
- Long optimization sequence

Need for a model based approach



Overview of Design Methods for RFIC

g_m/I_D approach based All Region Models (ACM/EKV)

All Region 3PM model

- EKV – [Enz, Krummenacher, Vittoz]
- ACM – [Schneider, Galup]



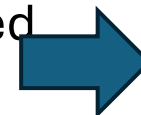
- With a 3PM model we have :

$$\frac{g_m}{I_D} = \frac{2}{n\phi_t \left(1 + \sqrt{1 + i_f}\right)} \quad g_m = \frac{2I_S}{\phi_t} \left(\sqrt{1 + i_f} - 1\right) \quad V_P - V_{S(D)} = \phi_t \left[\sqrt{1 + i_{f(r)}} - 2 + \ln \left(\sqrt{1 + i_{f(r)}} - 1 \right) \right]$$

$$V_P \cong \frac{V_G - V_{T0}}{n}$$

- PROS: The sizing is straightforward
- CONS: Inaccurate with SCE

Non-linearities can't be captured
gds effect can't be captured



Can we design RFICs

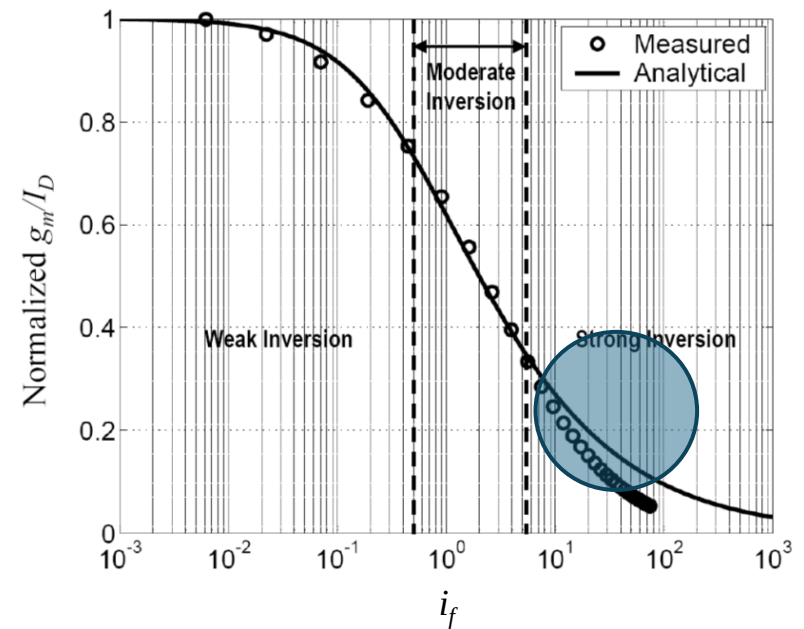
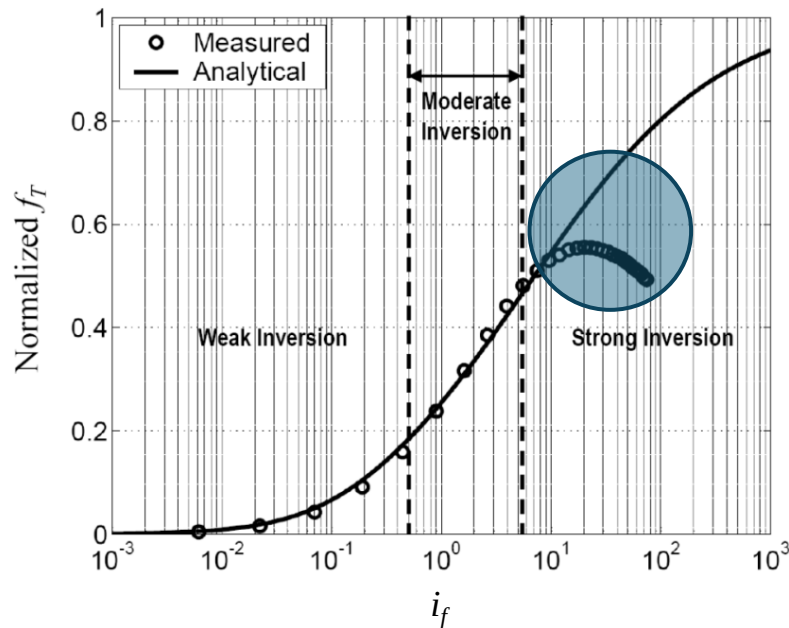
Overview of Design Methods for RFIC

g_m/I_D in RFIC design

Design Regions

- f_t grows with i_f

● Year 2000 (RF-180nm)

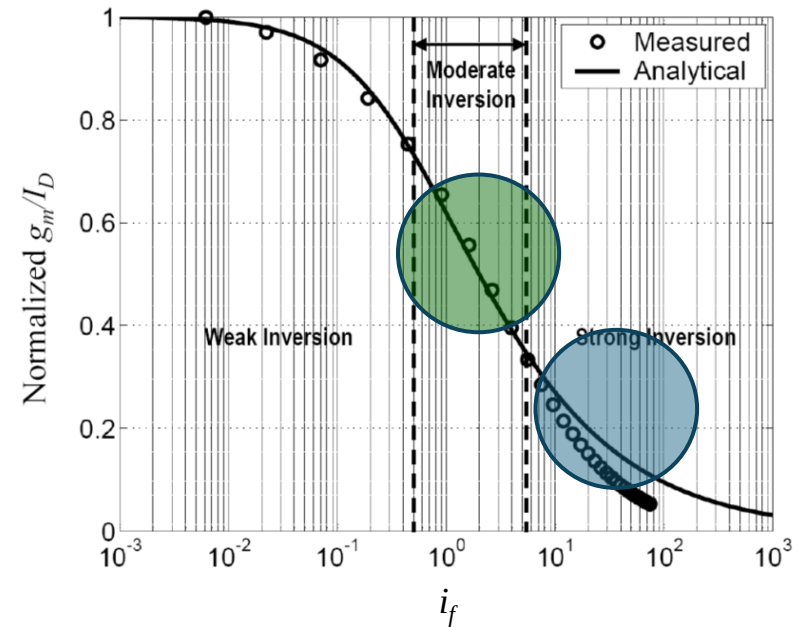
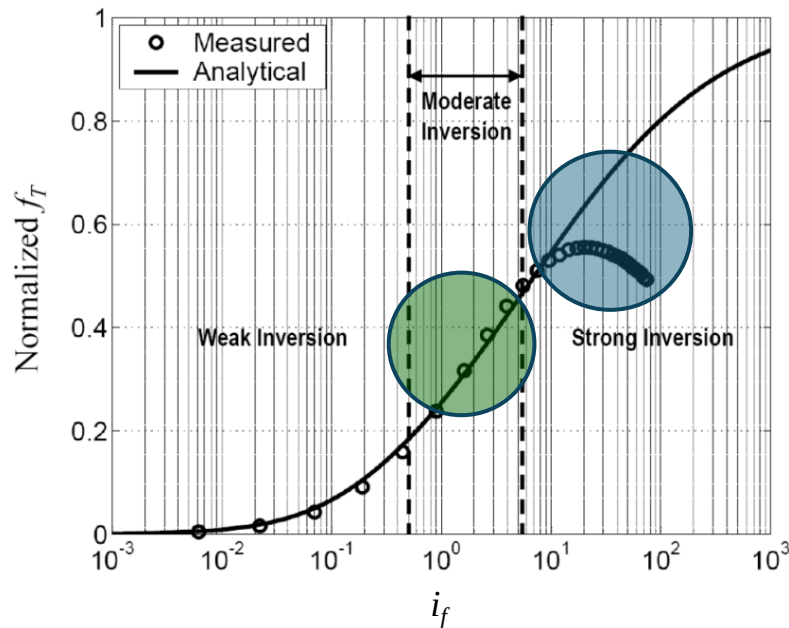
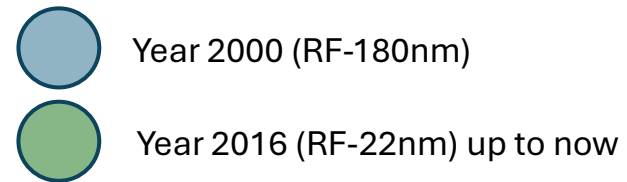


Overview of Design Methods for RFIC

g_m/I_D in RFIC design

Design Regions

- f_t grows with i_f **but** g_m/I_D reduces with i_f



Overview of Design Methods for RFIC

g_m/I_D in RFIC design

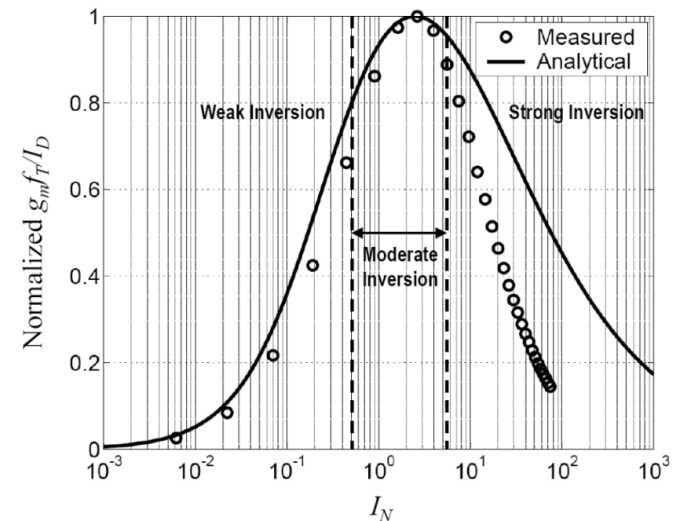
$g_m f_T / I_D$: A First Approach

A. Shameli et P. Heydari, « Ultra-Low Power RFIC Design Using Moderately Inverted MOSFETs: An Analytical/Experimental Study », *RFIC*, 2004.

- For RF design
- A FOM that maximize the gain bandwidth product

$$\frac{g_m f_T}{I_D}$$

- Gives the i_f that produces the best GBW product
- Not optimal :
 - the gain or the bandwidth might be oversized.
 - Do not depends on the topology



Overview of Design Methods for RFIC

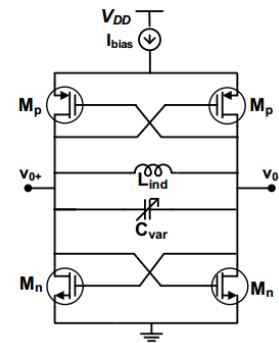
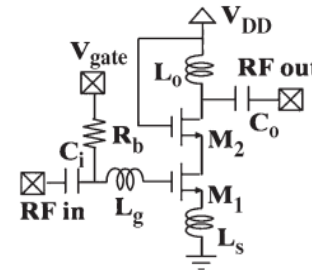
g_m/I_D in RFIC design

LN
A

LO

Function Based FoM

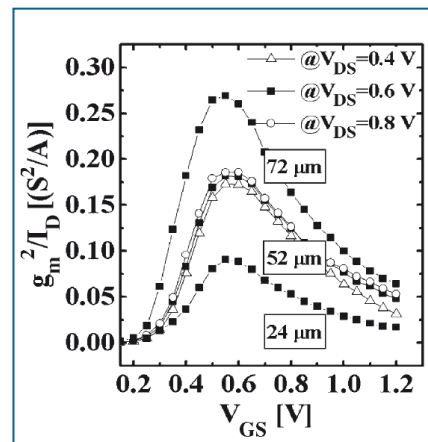
- For a given Function
- Define a FoM for the function



$$\text{FoM}_{\text{LNA}} = \frac{G}{(F-1) \cdot P} \propto \frac{g_m^2}{\left(\frac{I_D^2}{g_m^2}\right) \cdot I_D} \propto \left(\frac{g_m^2}{I_D}\right)^2$$

- Find the i_f that maximizes the FoM

I. Song et B.-G. Park, « A Simple Figure of Merit of RF MOSFET for Low-Noise Amplifier Design », *Electron Device Lett. IEEE*, vol. 29(12), 2008.



$$PN(\Delta f) = 10 \log \left(\frac{k_B T \pi^2}{64 Q^2} \frac{g_m}{I_D} \frac{1}{I_D} \frac{f_0}{(\Delta f)^2} \lambda \right)$$

R. Fiorelli ; J. Núñez ; F. Silveira; “All-Rnversion region gm/ID methodology for RF circuits in FinFET technologies”
2018 NEWCAS

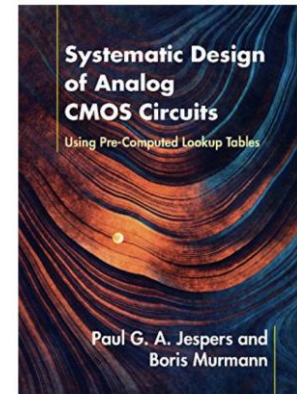
Overview of Design Methods for RFIC

g_m/I_d in RFIC design

Circuit Based Method

Circuit Sizing Based on LUT

- For a given Circuit
- Similar to the one introduced by Silveira, Flandre, Jespers in 1996
- Capacitance and noise effects can be captured in LUT for bandwidth and noise analysis
- Well known and well referenced
- But ...
 - Still complicated
 - The gap between physics and design remains
 - No analytical relationships to size « by hand »



Circuit Sizing Based on ACM

- Equations can be derived for a given circuit
- Quasi analytical approach
 - R-F LNA designed with 3PM-ACM [Bourdel - ICECS 2019]
- G_{ds} and g_{m3} (NL) are taken into account
 - R-F LNA designed with 7PM-ACM [Bouchoucha – ISCAS 2023]
 - R-F LNA designed with 5PM-ACM2 [Alves Neto – ACCESS 2024]



Resistive Feedback LNA design using a 7-parameter design-oriented model for advanced technologies

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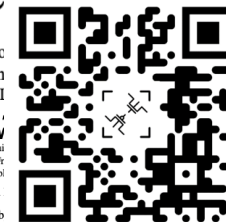
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Laboratory, Grenoble, France; and in part by

https://github.com/ACMmodel/MOSFET_model

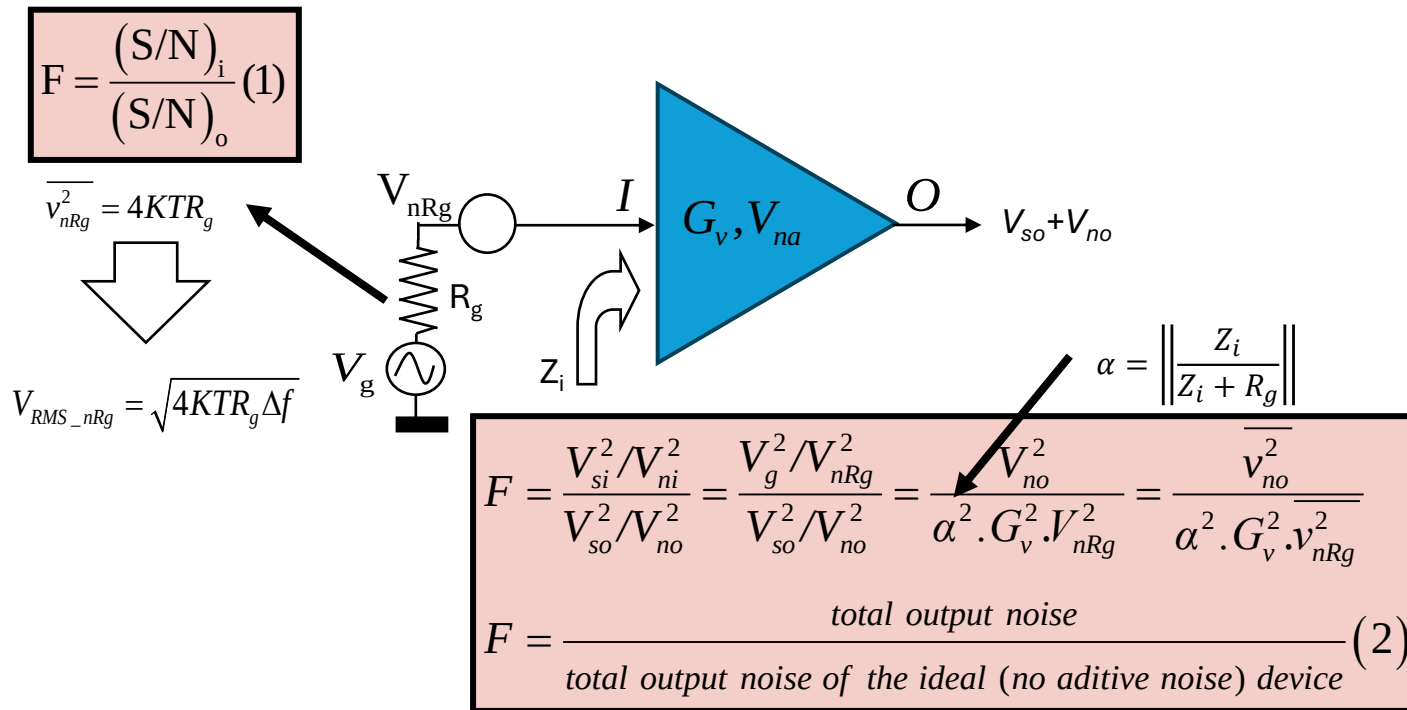
Agenda

- Overview of Design Methods for RFIC
- LNA Design Considerations
- Resistive Feedback LNA
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LNA Design considerations

Noise Factor

- NF is the signal to noise ratio (SNR) degradation (Eq. 1)

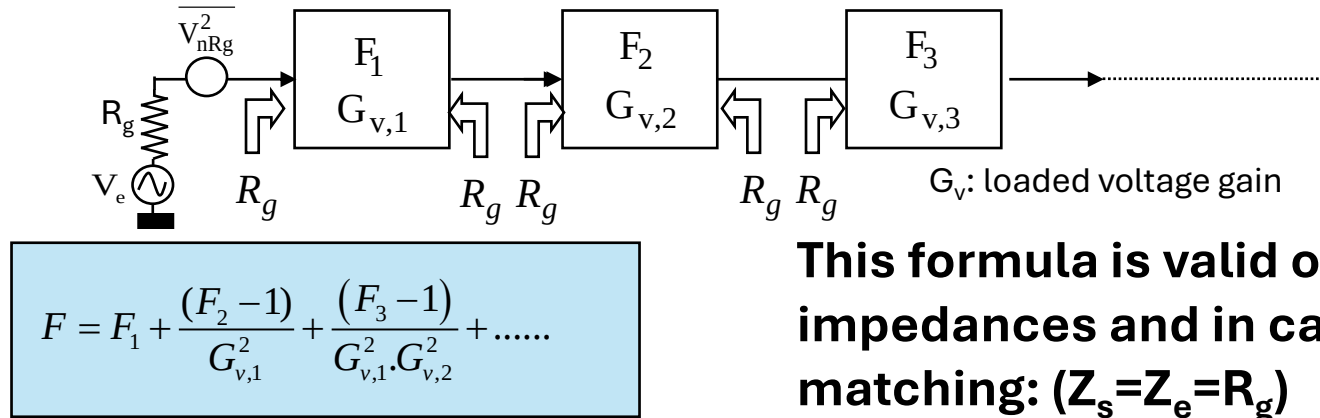


- NF also express the quantity of noise added by the stage regarding the noise delivered by the source (R_g) on Δf . (Eq. 2)

LNA Design considerations

Friss Formula

➤ RF Basis



This formula is valid only for real impedances and in case of power matching: ($Z_s = Z_e = R_g$)

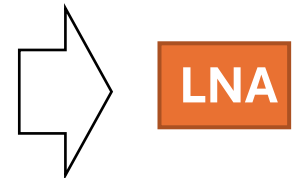
In RFIC, impedances are complex and never equal. This formula cannot be used. However, the following statements remain valid in RFIC.

The gain of the first stages reduces the NF of the following ones.

=> The receiver chain must start by amplifiers

=> The NF of the first stage must be as low as possible

Note that lossy stages ($G_v < 1$) increase the NF, especially when they are in front of the receiver chain (***antenna filter, image rejection filter, antenna switch***).

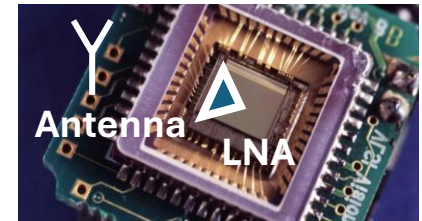
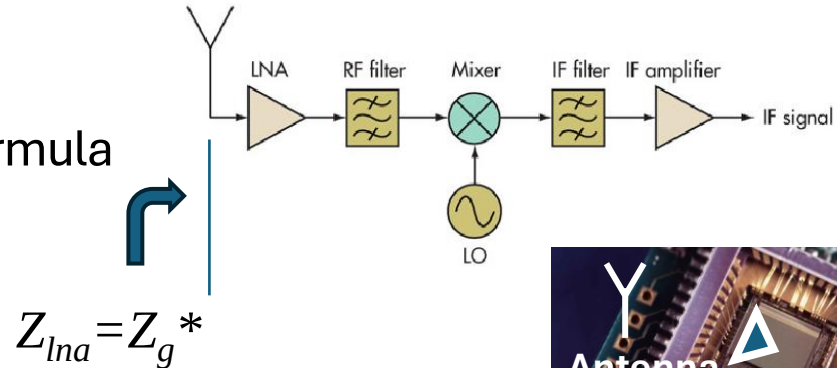


LNA Design considerations

Input patching

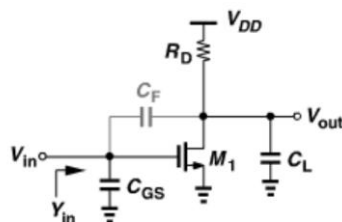
System point of view

- LNA is the first device due to Friss formula
- Antenna is mainly $50\ \Omega$.



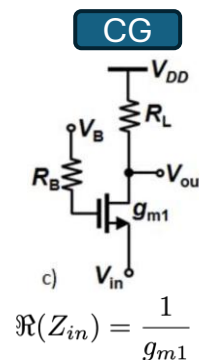
In CMOS

- Generally, the input impedance is capacitive $\Re(Y_{11})$ is very low
- For exemple : CS amplifier

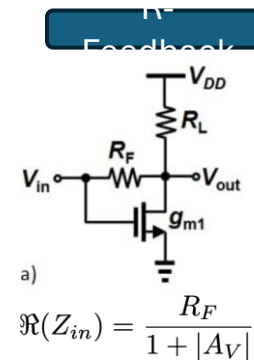


$$\Re(Y_{in}) = R_D \parallel 1/g_m$$

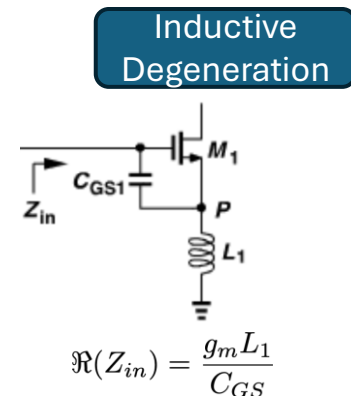
$$\Im(Y_{in}) = \frac{1}{C_{GS}C_F\omega}$$



$$\Re(Z_{in}) = \frac{1}{g_{m1}}$$



$$\Re(Z_{in}) = \frac{R_F}{1 + |A_V|}$$

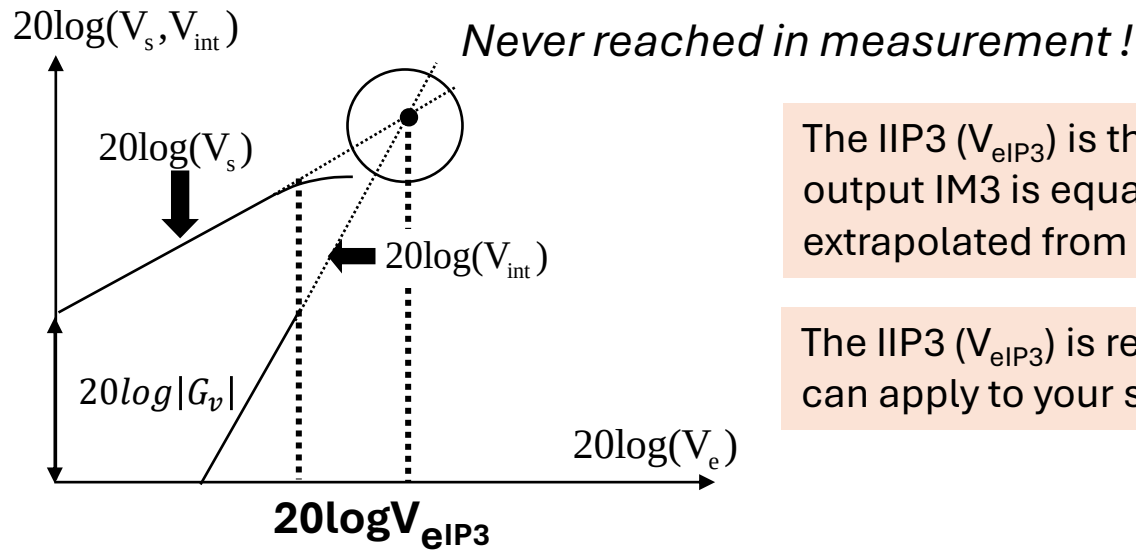
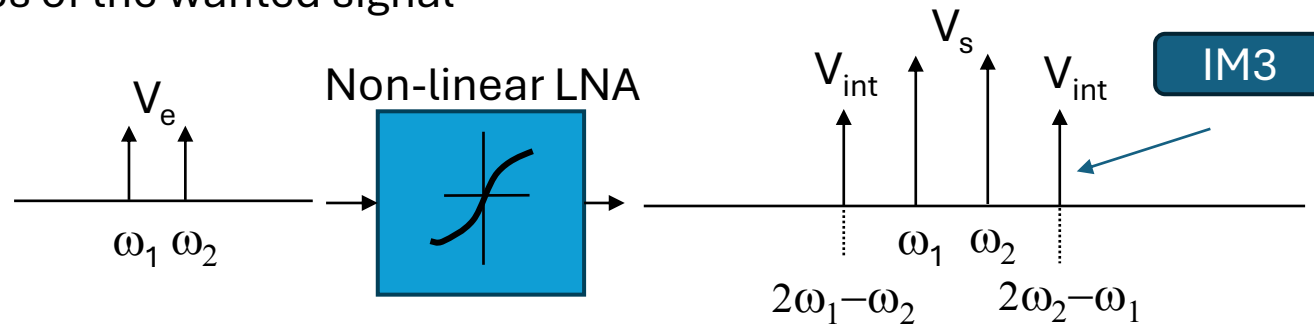


$$\Re(Z_{in}) = \frac{g_m L_1}{C_{GS}}$$

LNA Design considerations

Non-Linearities

ω_1 and ω_2 are 2 harmonics of the wanted signal



The IIP3 (V_{eIP3}) is the input signal level for which the output IM3 is equaling the fundamental harmonic (V_s) extrapolated from small signal (AC).

The IIP3 (V_{eIP3}) is related to the maximum power you can apply to your system

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Resistive Feedback LNA

Topology

R-Feedback general considerations

- Simple
- Compact (No-inductors)
- Wideband
- RF allows to synthesize a real part in Z_{in}
- L_G and C_{GS}' will help in cancelling the imaginary part
- while controlling Q_{in}

Input impedance

$$Z_{IN} = L_G s + \left(\frac{1}{(C_{GS} + C'_{GS})s} // Z_P \right)$$

$$\Re(Z_P) = R_P = \frac{R_O + R_F}{1 + G_m R_O},$$

$$C_P = \frac{R_O^2 C_L (G_m R_F - 1)}{(R_O + R_F)^2}$$

$$\Re(Z_{IN}) = R_S = \frac{R_P}{(1 + Q_P^2)} = 50 \Omega$$

$$Q_P = R_P C_T \omega_0$$

$$C_T = C_{GS} + C'_{GS} + C_P$$

$$\Im(Z_{IN}) = 0 = L_G s + \frac{Q_P^2}{C_T s (1 + Q_P^2)}$$

Note :

{

$$\Re(Z_P) = \frac{1 + \left(\frac{f}{f_c}\right)^2}{\frac{1 + G_m R_O}{R_O + R_F} + \frac{1}{R_F} \left(\frac{f}{f_c}\right)^2},$$

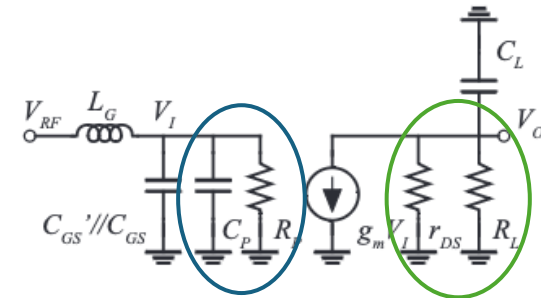
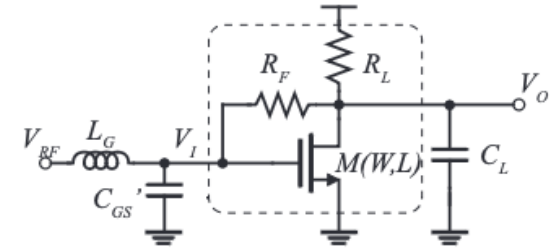
$$\Im(Z_P) = - \left(1 + \frac{R_F}{R_O}\right)^2 \frac{1 + \left(\frac{f}{f_c}\right)^2}{2\pi f C_L (G_m R_F - 1)}$$

Reduces

For $\frac{f}{f_c} = N >$

3

102



$$R_O = \frac{R_L r_{DS}}{R_L + r_{DS}}$$

Resistive Feedback LNA Topology

Voltage Gain

$$|G_o| = \left| \frac{V_o}{V_{RF}} \right| = \frac{|G_T|}{\sqrt{\left(1 + \left(\frac{f_0}{f_c}\right)^2\right)}}$$

$$|G_T| = |G_v| Q_{IN} = \frac{(G_m R_F - 1) R_O}{(R_O + R_F)} \sqrt{1 + Q_P^2}$$

$$Q_{IN} = V_I / V_{RF} \quad Q_{IN} = \sqrt{1 + Q_P^2} = \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}}$$

$$\frac{f_0}{f_c} = N > 3$$

BW

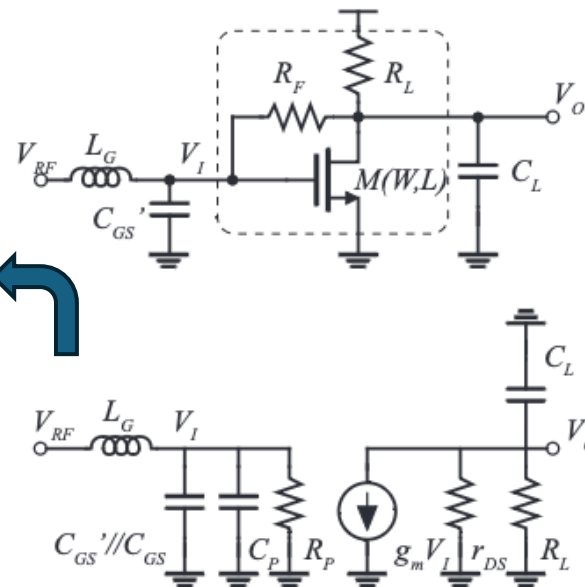
$$f_c = \frac{R_O + R_F}{2\pi R_O R_F C_L} = \frac{f_0}{N}$$

R_S
(the source resistance seen
at the 50 Ω input)

Noise Figure

$$F = 1 + \frac{4 \left(\frac{R_F}{Q_{IN}^2} + R_S \right)^2}{Q_{IN}^2 G_m R_S \left[\frac{R_F}{Q_{IN}^2} + R_S + \frac{G_m R_S R_F R_L}{R_F + R_L} \right]^2} \left[\gamma + \frac{1}{G_m R_L} + \frac{\left(1 + \frac{Q_{IN}^2 G_m R_S R_F}{R_F + Q_{IN}^2 R_S} \right)^2}{G_m R_F} \right]$$

$$\left(F_{min} = 1 + \frac{(1 + G_m) R_S^2 R_F}{R_S (1 - G_m R_F)^2} + \frac{\gamma g_m (R_S + R_F)^2}{R_S (1 - G_m R_F)^2} + \frac{(R_S + R_F)^2}{R_S R_L (1 - G_m R_F)^2} \right)$$



IIP₃

$$V_{IIP3} = \frac{2}{Q_{IN}} \sqrt{\frac{2G_m}{G_{m3}}}$$

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What we need in ACM-2

Small signal parameters

Small signal (AC) parameters

- G_T , NF, IIP3 vs (G_m , G_{m3} , G_{DS})

$$g_m = \frac{\partial I_D}{\partial V_G} = \frac{I_S}{\phi_t} g^G = \frac{I_S}{n\phi_t} \frac{2(q_s - q_d) - i_d \zeta \left(\frac{q_s}{1+q_s} - \frac{q_d}{1+q_d} \right)}{1 + \zeta (q_s - q_d)}$$

$$g_{ds} = \frac{I_S}{\phi_t} g^D = \frac{I_S}{n\phi_t} \frac{2(q_s \sigma - q_d (\sigma - n)) - i_d \zeta \left(\sigma \frac{q_s}{1+q_s} - (\sigma - n) \frac{q_d}{1+q_d} \right)}{1 + \zeta (q_s - q_d)}$$

$$g_{m3} = \frac{I_S}{(n\phi_t)^3} \left\{ \frac{\frac{2q_s}{(1+q_s)^3} - \frac{2q_d}{(1+q_d)^3} - \zeta n^2 g^G \left(\frac{q_s}{1+q_s} - \frac{q_d}{1+q_d} \right) - \zeta \left[2n g^G \left(\frac{q_s}{(1+q_s)^3} - \frac{q_d}{(1+q_d)^3} \right) + i_d \left(\frac{q_d(1-2q_d)}{(1+q_d)^5} - \frac{q_d(1-2q_d)}{(1+q_d)^5} \right) \right]}{1 + \zeta (q_s - q_d)} \right\}$$

- Valid in all region (q_s and q_d shall be explored)
- We consider only saturation

$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

$$g_{msat} = \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}$$

$$g_{dsat} = \frac{\sigma}{n} \frac{2I_S}{\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}$$

$$g_{msat3} = \frac{16I_S}{(n\phi_t)^3} \frac{q_s}{(q_s + 1)^3} \frac{2 - 2\zeta q_s - 3\zeta q_s^2}{(q_s + 1)^4}$$

- It reduces the exploration to only q_s .

What we need in ACM-2

Small signal parameters

Large signal (DC) parameters

- To compute the final voltages

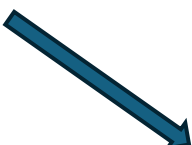
$$V_T = V_{T0} - \sigma(V_{SB} + V_{DB})$$

$$V_P = \frac{V_{GB} - V_T}{n}$$

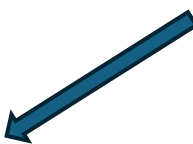
$$\frac{V_P - V_{S(D)B}}{\phi_t} = q_{s(d)} - 1 + \ln q_{s(d)}$$

- Sometime we'll use i_f in the code

$$i_d = i_f - i_r \quad \longleftrightarrow \quad q_{S(D)} = \sqrt{1 + i_{f(r)}} - 1 \quad \longleftrightarrow \quad i_d = \frac{(q_s + q_d + 2)}{1 + \zeta|q_s - q_d|} (q_s - q_d)$$



$$I_D = I_S \cdot i_d$$



$$I_S = \frac{\mu C'_{ox} n (U_T)^2 W}{2 L}$$

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Inversion Level Based Method for R-Feedback LNA with ACM-2

General Approach :

Requirements

- GT, NF, IIP3, IDC  Possible Tradeoff
- fo, CL, BW  No Tradeoff

Design parameters

- ACM parameters for a fixed L
- $V_{T0}, I_S, n, \sigma, \zeta$

Design Variables (parameters)

- W, q_s are first order design variables (q_s = inversion level sets the energy efficiency and the voltages)
- R_L, R_F are second order design variables

Approach :

- Explore the different tradeoff (the design space) on GT, NF, IIP3 and IDC by playing with W, q_s .
- In saturation region (only q_s is needed)

$$g_{msat} = \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \quad g_{msat3} = \frac{16I_S}{(n\phi_t)^3} \frac{q_s}{(q_s + 1)^3} \frac{2 - 2\zeta q_s - 3\zeta q_s^2}{(\zeta q_s + 2)^4} \quad g_{dsat} = \sigma \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}$$

- Circuit equations depend on G_m and ACM gives g_m (normalized)...

$$|G_T| = \frac{(G_m R_F - 1) R_O}{(R_O + R_F)} \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}}$$

$$G_{[m;DS]x} = \mu C'_{ox} \frac{(U_T)^{2-x}}{2} \frac{W}{L} g_{[m;DS]x}$$

Where we introduce W in the design space

$$I_S = \mu C'_{ox} n \frac{(U_T)^2}{2} \frac{W}{L} \quad I_{SL} = \frac{I_S}{W} = \mu C'_{ox} n \frac{(U_T)^2}{2L}$$

Inversion Level Based Method for R-Feedback LNA with ACM-2

Reducing the variables :

Starting from : $|G_T| = \frac{(G_m R_F - 1) R_O}{(R_O + R_F)} \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}}$ $G_T(G_m; R_0; R_F; R_S) = G_T(\underbrace{W; q_s; I_D; V_{DSAT}}_{\text{variables}}, \underbrace{V_{T0}; I_S; n; \sigma; \zeta; f_c; C_L}_{\text{parameters}})$

$$G_{[m;DS]x} = \mu C'_{ox} \frac{(U_T)^{2-x}}{2} \frac{W}{L} g_{[m;DS]x} \rightarrow g_{msat} = \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \rightarrow \boxed{G_m(q_s; W)}$$

$$\rightarrow g_{dsat} = \frac{\sigma}{n} \frac{2I_S}{\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \rightarrow \boxed{G_{DS}(q_s; W)}$$

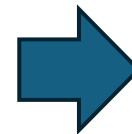
$$R_O = \frac{R_L r_{DS}}{R_L + r_{DS}} \rightarrow \boxed{G_{DS}(q_s; W)}$$

$$R_L(I_D) = \frac{V_{DD} - V_{DSAT}}{I_D} \rightarrow V_{DSAT} = \phi_t(\sqrt{1 + i_f} + 3)$$

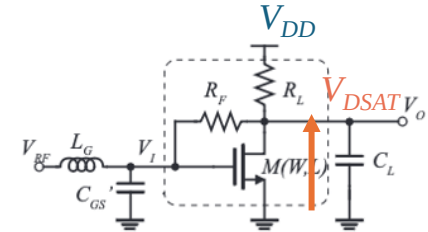
$$I_D = I_{SL} * W * i_f \quad i_f = (1 + q_s)^2 \rightarrow \boxed{R_O(q_s; W)}$$

$$R_F = \frac{R_0}{2\pi R_0 C_L f_c - 1} \rightarrow \boxed{R_F(q_s; W) \Big|_{f_c}}$$

$$\left(f_c = \frac{R_O + R_F}{2\pi R_O R_F C_L} \right)$$



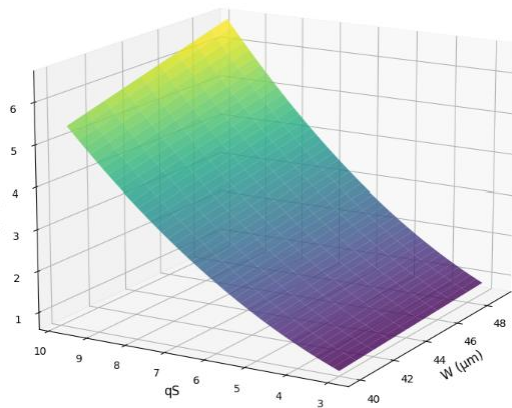
$$G_T(G_m; R_0; R_F; R_S) = G_T(W; q_s)$$



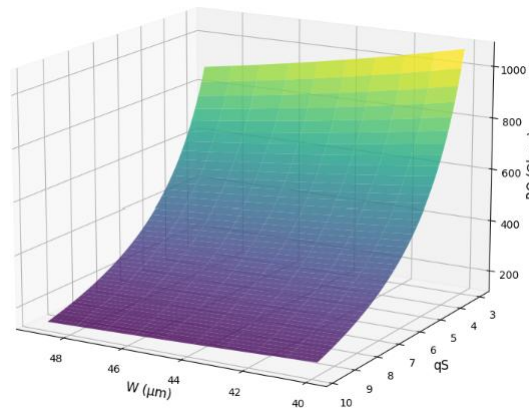
Inversion Level Based Method for R-Feedback LNA with ACM-2

Finally : $I_D(q_s; W)$; $R_F(q_s; W)$; $R_0(q_s; W)$; $G_T(q_s; W)$

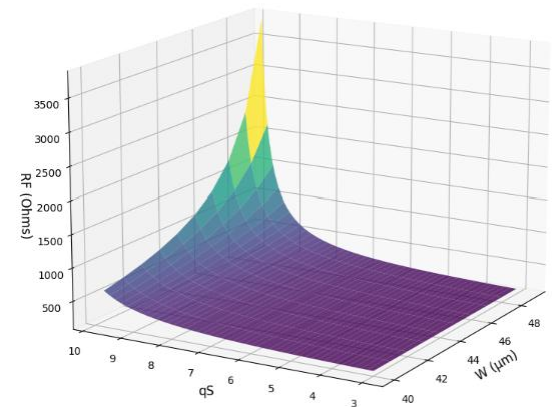
$I_D(q_s; W)$



$R_0(q_s; W)$



$R_F(q_s; W)$



Inversion Level Based Method for R-Feedback LNA with ACM-2

Computing G_T , F and $IIP3$:

Finally: $|G_T| = \frac{(G_m R_F - 1) R_O}{(R_O + R_F)} \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}}$

↳ $G_T(q_s; W)|_{f_c; C_L}$

$V_{IIP3} = \frac{2}{Q_{IN}} \sqrt{\frac{2G_m}{G_{m3}}}$

↳ $IIP_3(q_s; W)|_{f_c; C_L; R_S}$

$$F = 1 + \frac{4 \left(\frac{R_F}{Q_{IN}^2} + R_S \right)^2}{Q_{IN}^2 G_m R_S \left[\frac{R_F}{Q_{IN}^2} + R_S + \frac{G_m R_S R_F R_L}{R_F + R_L} \right]^2} \left[\gamma + \frac{1}{G_m R_L} + \frac{\left(1 + \frac{Q_{IN}^2 G_m R_S R_F}{R_F + Q_{IN}^2 R_S} \right)^2}{G_m R_F} \right]$$

↳ $F(q_s; W)|_{f_c; C_L; R_S}$

with $G_T(q_s; W)|_{f_c; C_L}$

↳ $F(G_T; W)$

Remember :

$$Q_{IN} = V_I / V_{RF} = \sqrt{1 + Q_P^2} = \sqrt{\frac{R_O + R_F}{R_S(1 + G_m R_O)}}$$

Inversion Level Based Method for R-Feedback LNA with ACM-2

Exploring the design space:

Setting GT (or NF, or IIP3) helps to build 2D plots.

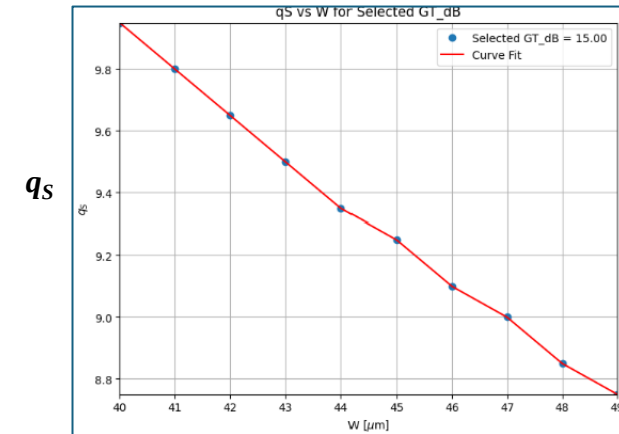
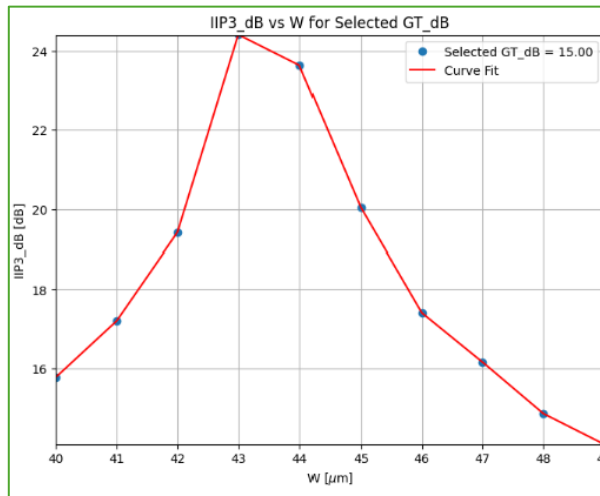
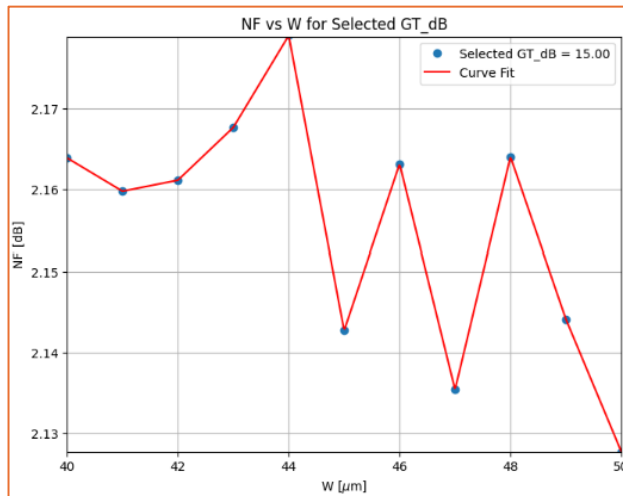
For a particular G_T

=> We get the relationship $q_s \Leftrightarrow W$

=> We plot

=> We plot $IIP3(W)$

And We already have I_D



$W(\mu m)$

Inversion Level Based Method for R-Feedback LNA with ACM-2

Setting the final value:

- We choose W , that gives q_S for a given G_T .
- $(W; q_S)$ gives $R_O, R_L, I_D, G_m, V_{DSAT}$ and V_G

Input Matching

- With $R_O, R_F, G_m \Rightarrow R_P$ $\Re(Z_P) = R_P = \frac{R_O + R_F}{1 + G_m R_O},$
- With R_S and R_P calculate Q_P $\Re(Z_{IN}) = R_S = \frac{R_P}{(1 + Q_P^2)} = 50 \Omega$
- With Q_P calculate C_T and C_{GS} , $Q_P = R_P C_T \omega_0$ $C_T = C_{GS} + C'_{GS} + C_P$ $C_P = \frac{R_O^2 C_L (G_m R_F - 1)}{(R_O + R_F)^2}$
- With C_T calculate L_G $\Im(Z_{IN}) = 0 = L_G s + \frac{Q_P^2}{C_T s (1 + Q_P^2)}$

Validation

Now, let's simulate with a real PDK

